Chapter 12

Conic Sections-II

Solutions

SECTION - A

Objective Type Questions (One Option is correct)

[General Problems on Parabola]

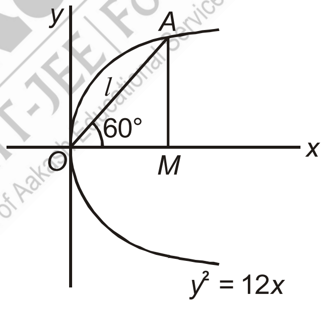
1. The length of the chord of the parabola $y^2 = 12x$ passing through the vertex and making an angle of 60° with the axis of x is

(1) $\frac{8}{3}$ (2) 8 (3) $\frac{16}{3}$ (4) 4

Sol. Answer (2)

$OM = l\cos 60° = \frac{l}{2}$

$AM = l\sin 60° = \frac{l\sqrt{3}}{2}$



Point $A\left(\frac{l}{2}, \frac{l\sqrt{3}}{2}\right)$ lies on the parabola $y^2 = 12x$

$\frac{l^2.3}{4} = 12.\frac{l}{2}$

$\Rightarrow \quad l = 8$

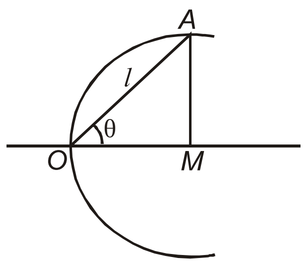
2. For the parabola $y^2 = 4ax$, the length of the chord passing through the vertex and inclined to the x-axis at angle θ, is

(1) $\frac{4a\cos\theta}{\sin^2\theta}$ (2) $\frac{4a\sin\theta}{\cos^2\theta}$ (3) $a\sec^2\theta$ (4) $a\,\text{cosec}^2\,\theta$

Sol. Answer (1)

$OM = l\cos\theta$

$AM = l\sin\theta$



$A(l\cos\theta, l\sin\theta)$ lies on the curve $y^2 = 4ax$

$\Rightarrow \quad l^2\sin^2\theta = 4a\,.\,l\cos\theta$

$l = \frac{4a\cos\theta}{\sin^2\theta}$

3. The length of the latus rectum of the parabola $x^2 - 6x + 5y = 0$ is

(1) 3 (2) 5 (3) 7 (4) 1

Sol. Answer (2)

$x^2 - 6x = -5y$

$\Rightarrow (x-3)^2 = -5\left(y - \frac{9}{5}\right)$

Comparing with $x^2 = 4ay$

$4a = -5$

Length of latus rectum = $|4a| = 5$

4. Length of the chord intercepted by the parabola $y = x^2 + 3x$ on the line $x + y = 5$ is

(1) $3\sqrt{26}$ (2) $2\sqrt{26}$ (3) $\sqrt{26}$ (4) $6\sqrt{2}$

Sol. Answer (4)

$y = x^2 + 3x$ … (i)

$y = 5 - x$ … (ii)

Solve these equations $A(1, 4)$, $B(-5, 10)$

$AB = \sqrt{36+36} = 6\sqrt{2}$

5. If b and c are the lengths of the segments of any focal chord of a parabola $y^2 = 4ax$, then length of the semi-latus rectum is

(1) $\frac{b+c}{2}$ (2) $\frac{bc}{b+c}$ (3) $\frac{2bc}{b+c}$ (4) $\sqrt{bc}$

Sol. Answer (3)

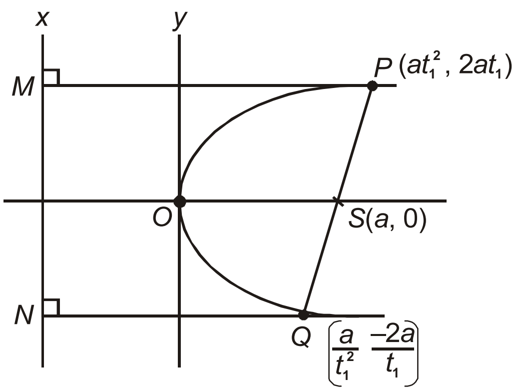
Let $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ are end points of a chord

PQ is a focal chord

$\therefore \quad t_1t_2 = -1$

$t_2 = -\frac{1}{t_1}$

$Q\left(\frac{a}{t_1^2}, \frac{-2a}{t_1}\right)$



$b = SP = PM = a + at_1^2$

$b - a = at_1^2$ … (i)

$c = SQ = QN = a + \frac{a}{t_1^2}$

$c - a = \frac{a}{t_1^2}$ … (ii)

Equation (i), (ii) $(b-a)(c-a) = a^2$

$\Rightarrow$ $bc = a(b+c)$

$\Rightarrow$ $a = \dfrac{bc}{b+c}$

$\Rightarrow$ $2a = \dfrac{2bc}{b+c}$

Semi latusrectum = $\dfrac{2bc}{b+c}$

6. If the vertex and the focus of the parabola are (–1, 1) & (2, 3) respectively, then the equation of the directrix is

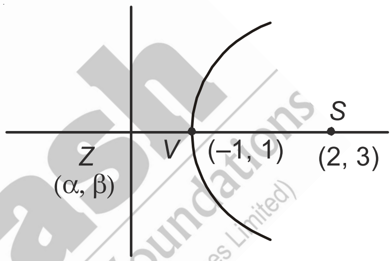
(1) $3x + 2y + 14 = 0$ (2) $3x + 2y - 25 = 0$ (3) $2x - 3y + 10 = 0$ (4) $x - y + 5 = 0$

Sol. Answer (1)

As we know that, vertex is the mid-point of the perpendicular drop from the focus to the directrix.

$\therefore$ $\dfrac{\alpha+2}{2} = -1 \Rightarrow \alpha + 2 = -2 \Rightarrow \alpha = -4$

$\dfrac{\beta+3}{2} = 1 \Rightarrow \beta = 2 - 3 = -1$



Slope of axis of the parabola = $\dfrac{3-\beta}{2-\alpha} = \dfrac{3+1}{2+4} = \dfrac{2}{3}$

$\Rightarrow$ Slope of the directrix = $-\dfrac{3}{2}$

Equation of directrix is

$y + 1 = -\dfrac{3}{2}(x+4)$

$\Rightarrow$ $2y + 2 = -3x - 12$

$\Rightarrow$ $3x + 2y + 14 = 0$

7. If the parabola $y^2 = 4ax$ passes through (3, 2), then the length of its latus rectum is

(1) $\dfrac{2}{3}$ (2) $\dfrac{4}{3}$ (3) $\dfrac{1}{3}$ (4) 4

Sol. Answer (2)

Given equation of parabola is $y^2 = 4ax$, which is passes through (3, 2), so we can write

$4 = 4a \times 3 = 12a$

$\Rightarrow$ $a = \dfrac{4}{12} = \dfrac{1}{3}$

$\Rightarrow$ $4a = \dfrac{4}{3}$

$\therefore$ Length of latus rectum = $4a = \dfrac{4}{3}$

8. The curve is given by $x = \cos 2t$, $y = \sin t$ represents

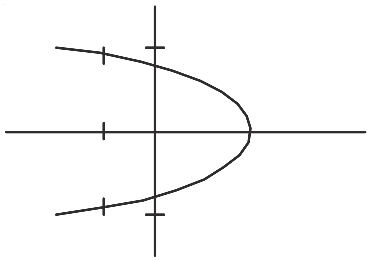
(1) A parabola (2) Circle

(3) Part of a parabola (4) A pair of straight lines

Sol. Answer (3)

Given parabolic equations are

$x = \cos 2t$, $y = \sin t$



$\Rightarrow$ Obviously, $x, y \in [-1, 1]$

$\therefore$ $x = 1 - 2\sin^2 t = 1 - 2y^2$

$\Rightarrow$ $2y^2 = 1 - x = -(x - 1)$

$\Rightarrow$ $y^2 = -\frac{1}{2}(x-1)$

$\Rightarrow$ It represents a part of a parabola

9. The vertex of the parabola $y^2 = (a - b)(x - a)$ is

(1) (b, a) (2) (a, b) (3) $(a, 0)$ (4) $(b, 0)$

Sol. Answer (3)

Given equation of a parabola is

$y^2 = (a - b)(x - a)$

$\Rightarrow$ $Y^2 = (a - b)X$, where $Y = y$ & $X = x - a$

$\therefore$ Vertex $\equiv (0, 0)$

$\Rightarrow$ $X = 0$, $Y = 0$

$\Rightarrow$ $x - a = 0$, $y = 0$

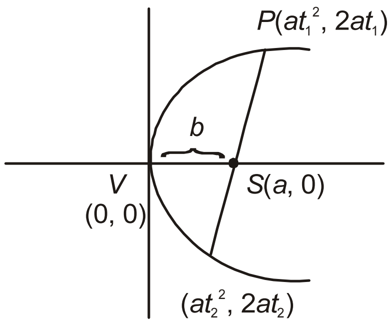
$\Rightarrow$ $x = a$, $y = 0$

Co-ordinates of vertex are $(a, 0)$

10. The length of a focal chord of the parabola $y^2 = 4ax$ at a distance b from the vertex is c. Then

(1) $a^2 = bc$ (2) $a^3 = b^2c$ (3) $b^2 = ac$ (4) $b^2c = 4a^3$

Sol. Answer (4)



$PQ : y(t_1 + t_2) = 2(x + at_1t_2)$

$\Rightarrow$ $2x - y(t_1 + t_2) + 2at_1t_2 = 0$

$\therefore$ $SV = b$.

$\Rightarrow$ $b = \frac{2at_1t_2}{\sqrt{4+(t_1+t_2)^2}} = \frac{-2a}{\sqrt{t_1^2+t_2^2+2}}$ $(\because t_1t_2 = -1)$

Now, $c = \sqrt{a(t_1^2-t_2^2)^2 + 4a^2(t_1-t_2)^2}$

$\Rightarrow$ $c^2 = a^2(t_1-t_2)^2\{(t_1+t_2)^2+4\}$

$= a^2(t_1^2+t_2^2+2)(t_1^2+t_2^2+2)$

$\Rightarrow$ $c^2 = a^2(t_1^2 + t_2^2 + 2)^2$

$\Rightarrow$ $c = a(t_1^2 + t_2^2 + 2)$

$\therefore$ $b = \dfrac{-2a}{\sqrt{t_1^2 + t_2^2 + 2}}$

$\Rightarrow$ $b^2 = \dfrac{4a^2}{t_1^2 + t_2^2 + 2} = \dfrac{4a^2}{c/a} = \dfrac{4a^3}{c}$

$\Rightarrow$ $b^2c = 4a^3$

11. The number of points with integral co-ordinates that lie in the interior of the region common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 4x$ is

(1) 13 (2) 10 (3) 18 (4) 20

Sol. Answer (1)

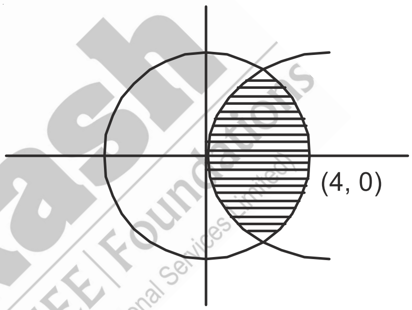
Let (α, β) lies in the shaded part

$\Rightarrow$ $\alpha^2 + \beta^2 - 16 < 0$

and $\beta^2 - 4\alpha < 0$

$\Rightarrow$ $\alpha > \dfrac{\beta^2}{4}$

$\Rightarrow$ $\alpha > \left(\dfrac{\beta}{2}\right)^2$



If $\beta = 0, \alpha = 1, 2, 3$

$\beta = 1, \alpha = 1, 2, 3,$

$\beta = 2, \alpha = 2, 3,$

$\beta = 3$, α has no integral value

$\beta = -1, \alpha = 1, 2, 3$

$\beta = -2, \alpha = 2, 3$

Also, if $\alpha^2 < 16 - \beta^2$

If $\beta = 1, \alpha = 1, 2, 3$

$\beta = 2, \alpha = 1, 2, 3$

$\beta = 3, \alpha = 2, 3$

$\beta = 4$, α has no integral value

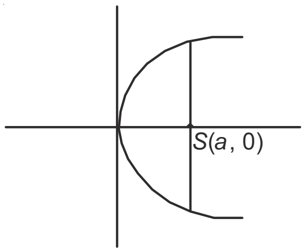
Thus, the points are

(1, 0) (2, 0) (3, 0), (1, 1), (2, 1), (3, 1) (2, 2), (3, 2), (1, –1), (2, –1), (3, –1), (2, –2), (3, –2)

12. The point $(a, 2a)$ is an interior point of the region bounded by the parabola $y^2 = 16x$ and the double ordinate through the focus. Then

(1) $a < 4$ (2) $0 < a < 4$ (3) $0 < a < 2$ (4) $a > 4$

Sol. Answer (2)



Since $(a, 2a)$ is an interior point of $y^2 = 16x$, then

$4a^2 - 16a < 0$

$\Rightarrow a^2 - 4a < 0$

$\Rightarrow a(a-4) < 0$

$\Rightarrow 0 < a < 4$ ...(i)

Also, the point $(a, 2a)$ and $(0, 0)$ are lie on the same side of the line $x = 4$

$\therefore 0 - 4 = -4 < 0$

$\Rightarrow a - 4 < 0$

$\Rightarrow a < 4$ ...(ii)

From (i) & (ii), we get

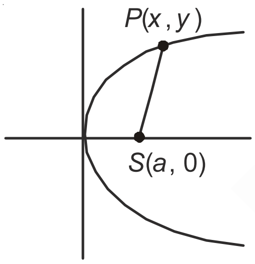
$0 < a < 4$

[Tangents to Parabola]

13. A particle moves on a parabolic path $y^2 = 4ax$. If its distance from the focus is minimum, then x is

(1) –1 (2) 0 (3) $-a$ (4) a

Sol. Answer (2)



From figure minimum distance is of (0, 0).

14. The equation of the tangent to the parabola $y^2 = 9x$, which passes through the point (4, 10) is

(1) $x + 4y + 1 = 0$ (2) $9x + 4y + 4 = 0$ (3) $x - 4y + 36 = 0$ (4) $19x - 14y + 4 = 0$

Sol. Answer (3)

Equation of tangent to the parabola

$y^2 = 9x$

$$y = mx + \frac{9}{4m} \quad \text{... (i)}$$

Point (4, 10) lies on this tangent $10 = 4m + \dfrac{9}{4m}$

$\Rightarrow 16m^2 - 40m + 9 = 0$

$\Rightarrow m = \dfrac{1}{4}$ and $m = \dfrac{9}{4}$

$\therefore$ Tangents are

$x - 4y + 36 = 0$ and $9x - 4y + 4 = 0$

15. The line 4x – 7y + 10 = 0 intersects the parabola, $y^2 = 4x$ at the points P and Q. The coordinates of the point of intersection of the tangents drawn at the points P and Q are

(1) $\left(-\frac{7}{2}, -\frac{5}{2}\right)$ (2) $\left(\frac{5}{2}, \frac{7}{2}\right)$

(3) $\left(-\frac{5}{2}, -\frac{7}{2}\right)$ (4) $\left(\frac{7}{2}, \frac{5}{2}\right)$

Sol. Answer (2)

Let point of intersection of tangents at P and Q be $A(h, k)$

$\therefore$ Equation of chord of contact (PQ) is

$T = 0$

$yk = 2(x + h)$

$2x - yk + 2h = 0$ … (i)

$4x - 7y + 10 = 0$ … (ii)

Lines (i) and (ii) are coincident lines

$$\frac{2}{4} = \frac{-k}{-7} = \frac{2h}{10} \quad \Rightarrow \quad k = \frac{7}{2}$$

$$\Rightarrow \quad h = \frac{5}{2}$$

Point of intersection is $\left(\frac{5}{2}, \frac{7}{2}\right)$

16. The coordinates of the point at which the line

$x \cos \alpha + y \sin \alpha + a \sin \alpha \tan \alpha = 0$ touches the parabola $y^2 = 4ax$ are

(1) $(a \tan \alpha, 2a \tan \alpha)$ (2) $(2a \tan \alpha, a \tan \alpha)$

(3) $(a \tan^2 \alpha, -2a \tan \alpha)$ (4) $(a \cos \alpha, 2a \cos \alpha)$

Sol. Answer (3)

Slope of tangent = $-\frac{\cos\alpha}{\sin\alpha}$

$$\therefore \quad \frac{dy}{dx} = -\frac{\cos\alpha}{\sin\alpha}$$

Let $P(h, k)$ lies on the curve

$y^2 = 4ax$

$$\Rightarrow \quad 2y\frac{dy}{dx} = 4a$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{2a}{y}$$

Slope of $P(h, k)$

$$-\frac{\cos\alpha}{\sin\alpha} = \frac{2a}{k}$$

$k = -2a \tan\alpha$

$P(h, k)$ lie on the curve

$k^2 = 4ah$

$\Rightarrow\ 4a^2 \tan^2\alpha = 4ah$

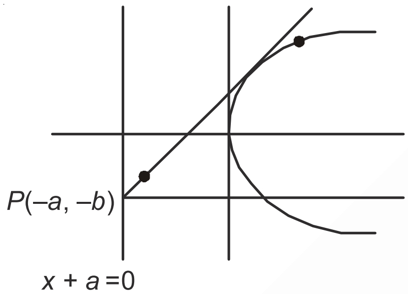
$\Rightarrow\ h = a \tan^2\alpha$

Point of contact is $(a \tan^2\alpha, -2a \tan\alpha)$

17. If $y + b = m_1(x + a)$ and $y + b = m_2(x + a)$ are two tangents to the parabola $y^2 = 4ax$, then

(1) $m_1 + m_2 = 0$ (2) $m_1 m_2 = 1$ (3) $m_1 m_2 = -1$ (4) $m_1 = m_2$

Sol. Answer (3)



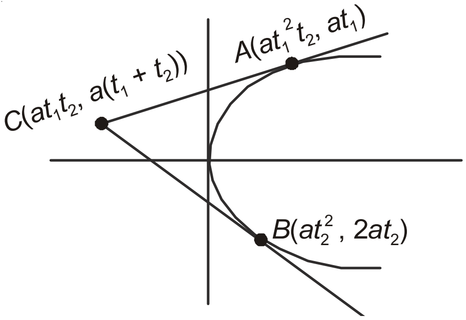
As we know that, tangent are drawn any point on the directrix arc at right angle

$\Rightarrow\ m_1m_2 = -1$

18. If tangent at A and B on the parabola intersect at point C, then the ordinates of A, C and B are in

(1) A.P. (2) G.P. (3) H.P. (4) A.G.P

Sol. Answer (1)



Let $y_1 = 2at_1$, $y_2 = 2at_2$, $y_3 = a(t_1 + t_2)$

$\Rightarrow\ y_1 + y_2 = 2(at_1 + at_2)$

$$\Rightarrow\ \frac{y_1 + y_2}{2} = a(t_1 + t_2) = y_3$$

$\Rightarrow\ y_1, y_3, y_2, \in$ A.P.

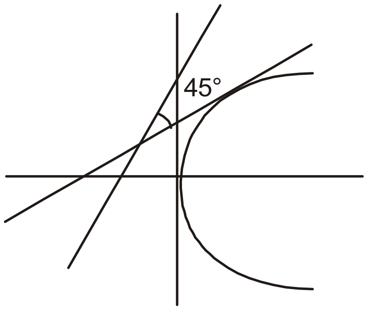
$\Rightarrow\ A, C, B$ are A.P.

19. A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the straight line $y = 3x + 5$. The equation of tangent is

(1) $x + 2y + 1 = 0$ (2) $2x + y + 1 = 0$

(3) $x + y + 2 = 0$ (4) $x + y + 1 = 0$

Sol. Answer (2)



Equation of any tangent to the parabola $y^2 = 8x$ is $y = mx + \frac{a}{m} = mx + \frac{2}{m}$

$L : y = 3x + 5$

$\therefore \quad \tan 45° = \left|\frac{m-3}{1+3m}\right|$

$\Rightarrow \quad \frac{m-3}{1+3m} = \pm 1$

$\Rightarrow \quad m - 3 = 1 + 3\text{ m} \qquad \Rightarrow \quad m - 3 = -1 - 3\text{m}$

$\Rightarrow \quad m - 3m = 4 \qquad \Rightarrow \quad 4m = 2$

$\Rightarrow \quad m = \frac{4}{-2} = -2 \qquad \Rightarrow \quad m = \frac{1}{2}$

Equation of tangent is

$y = -2x - 1 \qquad y = \frac{1}{2}x + 4$

$\Rightarrow \quad y + 2x + 1 = 0 \qquad \Rightarrow \quad 2y - x - 8 = 0$

[Normals to Parabola]

20. If the normals drawn at the points t_1 and t_2 on the parabola meet the parabola again at its point t_3, then t_1t_2 equals

(1) 2 (2) –1 (3) –2 (4) $t_3 - \frac{2}{t_3}$

Sol. Answer (1)

Let normal at t meets the parabola again at t_3

$\therefore \quad t_3 = -t - \frac{2}{t}$

$\therefore \quad t^2 + t_3.t + 2 = 0$

t_1 and t_2 are the roots of the equation

$\therefore \quad t_1t_2 = 2$

21. The length of the normal chord to the parabola $y^2 = 4x$, which subtends a right angle at the vertex is

(1) $3\sqrt{3}$ (2) $6\sqrt{3}$ (3) 2 (4) 1

Sol. Answer (2)

Let equation of normal

$y + tx = 2t + t^3$

$\frac{y+tx}{2t+t^3} = 1 \qquad \ldots \text{(i)}$

$y^2 = 4x \qquad \ldots \text{(ii)}$

Equation of pair of lines through origin

$$y^2 = 4x\left(\frac{y+tx}{2t+t^3}\right)$$

$4tx^2 - (2t + t^3)y^2 + 4xy = 0$

These lines are perpendicular

$\therefore$ $4t - 2t - t^3 = 0$

$\Rightarrow$ $2t - t^3 = 0$

$\Rightarrow$ $t = 0$ and $t^2 = 2$

Equation of normal

$$y = -\sqrt{2}x + 4\sqrt{2} \qquad \ldots \text{(iii)}$$

$y^2 = 4x$

$A(x_1, y_1)$ and $B(x_2, y_2)$ are the points of intersections of normal (i) and parabola

Then $\left(4\sqrt{2} - \sqrt{2}x\right)^2 = 4x$

$x^2 - 10x + 16 = 0$

Roots are x_1 and x_2

$x_1 + x_2 = 10$

$x_1 x_2 = 16$

From (iii), $y_2 - y_1 = -\sqrt{2}\left(x_2 - x_1\right)$

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + 2(x_2 - x_1)^2}$$

$$= \sqrt{3(x_2 - x_1)^2}$$

$$= \sqrt{3}\sqrt{(x_2 + x_1)^2 - 4x_2x_1}$$

$$= \sqrt{3}\sqrt{100 - 64}$$

$$= \sqrt{3}\sqrt{36}$$

$$= 6\sqrt{3}$$

22. The locus of the point of intersection of the normals at the ends of a focal chord of the parabola $y^2 = 4ax$ is

(1) $y^2 = 4a(x - 3a)$ (2) $y^2 = 2a(x - 3a)$ (3) $y^2 = a(x - 3a)$ (4) $y^2 = 16a(x - 3a)$

Sol. Answer (3)

End points of focal chord are $(at^2, 2at)$ and $\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$

$\therefore$ Equation of normals

$y + tx = 2at + at^3$ … (i)

and $y - \dfrac{x}{t} = -\dfrac{2a}{t} - \dfrac{a}{t^3}$ … (ii)

For locus of point of intersection of normals (i) and (ii), eliminate t

Equation (i) – (ii)

$x = 2a + a\left(t^2 + \dfrac{1}{t^2} - 1\right)$ … (iii)

Equation (i) + t^2(ii)

$y = a\left(t - \dfrac{1}{t}\right)$

$\dfrac{y^2}{a^2} = t^2 + \dfrac{1}{t^2} - 2$ … (iv)

From (iii) and (iv)

$x = 2a + a\left(\dfrac{y^2}{a^2} + 1\right)$

$\dfrac{y^2}{a} = x - 3a$

$y^2 = a(x - 3a)$

23. Three normals are drawn to the curve $y^2 = x$ from a point $(c, 0)$. Out of three, one is always the x-axis. If two other normals are perpendicular to each other, then 'c' is

(1) $\dfrac{3}{4}$ (2) $\dfrac{1}{2}$ (3) $\dfrac{3}{2}$ (4) 2

Sol. Answer (1)

Equation of normal to the curve $y^2 = x$

$y = mx - 2am - am^3$ Put $a = \dfrac{1}{4}$

$y = mx - \dfrac{1}{2}m - \dfrac{1}{4}m^3$ but these normals passes through $(c, 0)$

$\Rightarrow 0 = mc - \dfrac{1}{2}m - \dfrac{1}{4}m^3$

$m\left(\dfrac{m^2}{4} + \dfrac{1}{2} - c\right) = 0$

$m = 0$ and $\dfrac{m^2}{4} + \dfrac{1}{2} - c = 0$

$m = 0$ $m^2 + 2 - 4c = 0$

$\therefore$ Normal is x-axis

Remaining normals are perpendicular

$\therefore \quad m_1m_2 = -1$

$2 - 4c = 1$

$4c = 3$

$c = \dfrac{3}{4}$

24. The number of distinct normals that be drawn from (–2, 1) to the parabola $y^2 - 4x - 2y - 3 = 0$ is

(1) 1 (2) 2 (3) 3 (4) 0

Sol. Answer (1)

Given parabola is

$y^2 - 2y = 4x + 3$

$\Rightarrow \quad y^2 - 2y + 1 = 4x + 4$

$\Rightarrow \quad (y-1)^2 = 4(x+1)$

$\Rightarrow \quad Y^2 = 4X, \quad Y = y - 1$

$X = x + 1$

Axis is $Y = 0$

$\Rightarrow \quad y - 1 = 0$

Also the point (–2, 1) lies on the axis,

And $1 + 8 - 2 - 3 = 2 > 0$

Point lies exterior to the parabola.

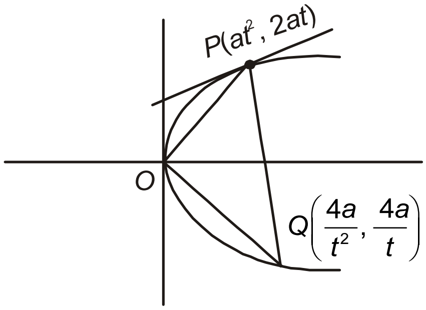
$\Rightarrow$ Thus only one normal is possible.

25. The normal at any point $P(t^2, 2t)$ on the parabola $y^2 = 4x$ meets the curve again at Q, then the ar(ΔPOQ) in m the form of $\dfrac{k}{|t|}(1 + t^2)(2 + t^2)$. The value of k is

(1) $k > 2$ (2) $k = 2$ (3) $k < 2$ (4) $k = 1$

Sol. Answer (2)

$$ar(\Delta POQ) = \left| \frac{1}{2}\begin{vmatrix} at^2 & 2at & 1 \\ 0 & 0 & 1 \\ \frac{4a}{t^2} & \frac{4a}{t} & 1 \end{vmatrix} \right|$$



$$= \left| -\frac{1}{2}\begin{vmatrix} at^2 & 2at \\ \frac{4a}{t^2} & \frac{4a}{t} \end{vmatrix} \right|$$

$$= \frac{1}{2}\left(4a^2t - \frac{8a^2}{t}\right)$$

$$= \frac{1}{2}4a^2\left(t - \frac{2}{t}\right)$$

$$= 2a^2\left(t - \frac{2}{t}\right)$$

$\Rightarrow k = 2$

[Locus and Miscellaneous Problems]

26. If the segment intercepted by the parabola $y^2 = 4ax$ with the line $\ell x + my + n = 0$ subtends a right angle at the vertex, then

(1) $4a\ell + n = 0$ (2) $4a\ell + 4am + n = 0$

(3) $4am + n = 0$ (4) $a\ell + n = 0$

Sol. Answer (1)

Equation of pair of lines passes through origin

$y^2 = 4ax\left(\frac{lx+my}{-n}\right)$ [Homogeneous equation of second degree]

$4alx^2 + ny^2 + 4amxy = 0$

These lines are perpendicular

$\therefore$ $4al + n = 0$

27. Let $P(1, 0)$ and Q be any point on $y^2 = 8x$. The locus of mid-points of PQ must be

(1) $y^2 + 2 = 4x$ (2) $y^2 + 4x + 2 = 0$ (3) $x^2 + 2 = 4y$ (4) $x^2 + 4y + 2 = 0$

Sol. Answer (1)

$P(1, 0)$, $Q(x, y)$ lie on the curve

Let mid-point of PQ be (h, k)

$\therefore$ $h = \frac{1+x}{2}$ $\Rightarrow$ $x = 2h - 1$

$k = \frac{y}{2}$ $\Rightarrow$ $y = 2k$

$\therefore$ Puts the values of x and y in $y^2 = 8x$

$4k^2 = 8(2h - 1)$

$k^2 = 2(2h - 1)$

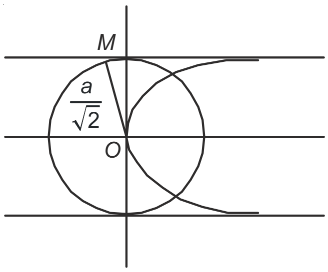
Locus of (h, k)

$y^2 + 2 = 4x$

28. Two common tangents to the circle $x^2 + y^2 = \frac{a^2}{2}$ and the parabola $y^2 = 4ax$ are

(1) $x = \pm (y + 2a)$ (2) $y = \pm (x + 2a)$ (3) $x = \pm (y + a)$ (4) $y = \pm (x + a)$

Sol. Answer (4)



Equation of any tangent to the parabola $y^2 = 4ax$ is $y = mx + \frac{a}{m}$

$\Rightarrow$ $my = m^2x + a$

$\Rightarrow$ $m^2x - my + a = 0$

Length of perpendicular from centre = radius of a circle to the tangent

$\Rightarrow$ $\frac{a}{\sqrt{m^4 + m^2}} = \frac{a}{\sqrt{2}}$

$\Rightarrow$ $m^4 + m^2 = 2$

$\Rightarrow$ $m^4 + m^2 - 2 = 0$

$\Rightarrow$ $m^4 + 2m^2 - m^2 - 2 = 0$

$\Rightarrow$ $m^2 (m^2 + 2) - 1 (m^2 + 2) = 0$

$\Rightarrow$ $(m^2 + 2) (m^2 - 1) = 0$

$\Rightarrow$ $m = \pm 1$

$\therefore$ Equation of common tangents are

$y = \pm x \pm a.$

$\Rightarrow$ $y = \pm (x + a)$

29. The line $y = a - x$ touches the parabola $y = x - x^2$ and $f(x) = \sin^2 x + \sin^2\left(x+\frac{\pi}{3}\right)+\cos x\cdot\cos\left(x+\frac{\pi}{3}\right)$

$g\left(\frac{5}{4}\right)=1$ and $b = gof$, then

(1) $a = b$ (2) $a = 2b$ (3) $a = -b$ (4) $2a = b$

Sol. Answer (1)

$\therefore$ $y = a - x$ & $y = x - h^2$

$\Rightarrow$ $a - x = x - h^2$

$\Rightarrow$ $h^2 - 2x + a = 0$

$D = 0 \Rightarrow 4 - 4a = 0$

$\Rightarrow$ $a = 1$

$$\therefore\quad f(x)=\sin^2 x+\sin^2\left(x+\frac{\pi}{3}\right)+\cos x\cdot\cos\left(x+\frac{\pi}{3}\right)$$

$$=\frac{1}{2}\left[2\sin^2 x+2\sin^2\left(x+\frac{\pi}{3}\right)+2\cos x\cos\left(x+\frac{\pi}{3}\right)\right]$$

$$=\frac{1}{2}\left[1-\cos 2x+1-\cos\left(2x+\frac{2\pi}{3}\right)+\cos\left(2x+\frac{\pi}{3}\right)+\frac{1}{2}\right]$$

$$=\frac{1}{2}\left[\frac{5}{2}-\cos 2x+\cos\left(2x+\frac{\pi}{3}\right)-\cos\left(2x+\frac{2\pi}{3}\right)\right]$$

$$=\frac{1}{2}\left[\frac{5}{2}-\cos 2x+\cos 2x\right]$$

$$=\frac{5}{4}$$

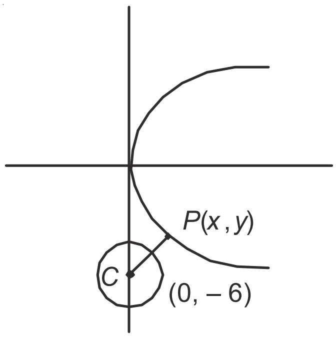
$$\therefore\quad b=(gof)(x)=g(f(x))=g\left(\frac{5}{4}\right)=1$$

$\Rightarrow$ $a = b$

30. The co-ordinates of the points on the parabola $y^2 = 8x$, which is at minimum distance from the circle $x^2 + (y + 6)^2 = 1$ are

(1) (2, 4) (2) (–2, 4) (3) (2, –4) (4) (–2, –4)

Sol. Answer (3)



$CP = \sqrt{x^2 + (y+6)^2}$

$\Rightarrow \quad z = x^2 + (y + 6)^2$

$\quad = \dfrac{y^4}{64} + (y+6)^2$

$\Rightarrow \quad \dfrac{dz}{dy} = \dfrac{4y^3}{64} + 2(y+6) = 0$

$\Rightarrow \quad \dfrac{y^3}{16} + 2(y+6) = 0$

$\Rightarrow \quad y^3 + 32y + 192 = 0$

$\Rightarrow \quad y = -4$

$\Rightarrow \quad 8x = 16$

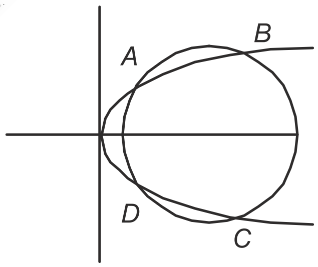
$\Rightarrow \quad x = 2$

Point is (2, – 4)

31. If a circle intersects the parabola $y^2 = 4ax$ at points $A(at_1^2, 2at_1)$, $B(at_2^2, 2at_2)$, $C(at_3^2, 2at_3)$, $D(at_4^2, 2at_4)$, then $t_1 + t_2 + t_3 + t_4$ is

(1) 1 (2) –1 (3) 0 (4) 2

Sol. Answer (3)



Let the equation of the circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$

Which is passing through a point $(at^2, 2at)$, so

$a^2t^4 + 4a^2t^2 + 2g.at^2 + 2f.2at + c = 0$

$\Rightarrow \quad a^2t^4 + (4a^2 + 2ga)t^2 + 4af\,t + c = 0 \quad$ …(i)

Let t_1, t_2, t_3 and t_4 are the roots of equation (i)

$\Rightarrow \quad t_1 + t_2 + t_3 + t_4 = 0$

32. If the line $y - \sqrt{3}x + 3 = 0$ cuts the parabola $y^2 = x + 2$ at A and B. If $P \equiv (\sqrt{3}, 0)$ then $PA.PB$ is

(1) $\dfrac{2}{3}(\sqrt{3}+2)$ (2) $2\sqrt{3}$ (3) $\dfrac{4}{3}(2-\sqrt{3})$ (4) $\dfrac{4}{3}(\sqrt{3}+2)$

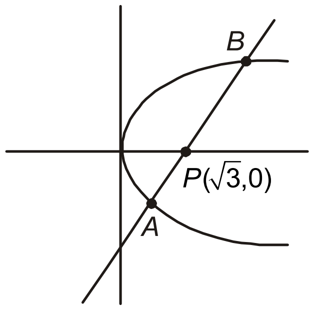
Sol. Answer (4)

$L: \; y - \sqrt{3}x + 3 = 0$

$\Rightarrow \quad \dfrac{y-0}{\frac{\sqrt{3}}{2}} = \dfrac{x-\sqrt{3}}{\frac{1}{2}} = r(\text{say})$

$\Rightarrow \quad x = \sqrt{3} + \frac{r}{2}, y = \frac{\sqrt{3}}{2}r$

$P : y^2 = x + 2$



$\Rightarrow \quad \left(\frac{\sqrt{3}}{2}r\right)^2 = \left(\sqrt{3} + \frac{r}{2}\right) + 2$

$\Rightarrow \quad \frac{3}{4}r^2 = (2 + \sqrt{3}) + \frac{r}{2}$

$\Rightarrow \quad \frac{3}{4}r^2 - \frac{r}{2} - (2 + \sqrt{3}) = 0$

$\Rightarrow \quad 3r^2 - 2r - (4\sqrt{3} + 8) = 0$

Let its roots are r_1, r_2

$\therefore \quad r_1 r_2 = \left|\frac{4\sqrt{3}+8}{3}\right|$

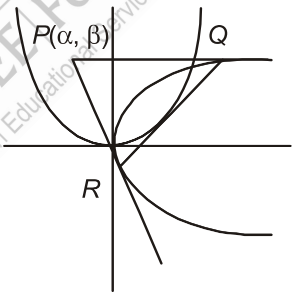
$\Rightarrow \quad PA.PB = \frac{4\sqrt{3}+8}{3}$

33. If the chord of contact of tangent from a point to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, then the locus of p is

(1) A circle (2) A parabola (3) A pair of straight lines (4) A hyperbola

Sol. Answer (4)

$\therefore \quad QR : y\beta = 2a(x + \alpha)$



$\Rightarrow \quad y = \frac{2a}{\beta}(x + \alpha)$

Parabola : $x^2 = 4by$

$\Rightarrow \quad x^2 = 4b \cdot \frac{2a}{\beta}(x + \alpha)$

$\Rightarrow \quad \beta x^2 = 8ab(x + \alpha)$

$\Rightarrow \quad Bx^2 - 8abx - 8ab\alpha = 0$

$\because \quad D = 0$

$\Rightarrow \quad 64a^2b^2 + 32ab\alpha\beta = 0$

$\Rightarrow \quad \alpha\beta = -\frac{64a^2b^2}{32ab} = -2ab$

$\Rightarrow$ Locus of (α, β) is a hyperbola

34. A ray of light moving parallel to x-axis gets reflected from a parabolic mirror whose equation is $4(x + y) - y^2 = 0$. After reflection the ray pass through the $pt(a, b)$. Then the value of $a + b$ is

(1) 2 (2) 1 (3) –2 (4) –1

Sol. Answer (1)

Given equation is $4(x + y) - y^2 = 0$

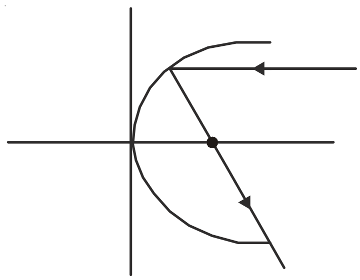
$\Rightarrow$ $y^2 - 4y = 4x$

$\Rightarrow$ $y^2 - 4y + 4 = 4x + 4$

$\Rightarrow$ $(y - 2)^2 = 4(x + 1)$

$Y^2 = 4X$, $X = x + 1$

$Y = y - 2$



The ray of light, after reflection must pass through a focus

$\therefore$ Focus = $(a, 0)$

$\therefore$ $X = a = 1$ $\quad$ $Y = 0$

$\Rightarrow$ $x + 1 = 1$ $\quad$ $\Rightarrow$ $y - 2 = 0$

$\therefore$ $x = 0$ $\quad$ $\Rightarrow$ $y = 2$

Focus $\equiv (0, 2) \equiv (a, b)$

$\Rightarrow$ $a + b = 0 + 2 = 2$

35. If the parabola $y = (a - b)x^2 + (b - c)x + (c - a)$ touches the x-axis then the line $ax + by + c = 0$ always passes through a fixed point is

(1) (1, 2) (2) (–2, 1) (3) (2, 1) (4) (2, –1)

Sol. Answer (2)

Put $y = 0$

$\therefore$ $(a - b)x^2 + (b - c)x + (c - a) = 0$

Since the parabola touches the x-axis, so

$D = 0$

$\Rightarrow$ $(b - c)^2 - 4(a - b)(c - a) = 0$

$\Rightarrow$ $b^2 + c^2 - 2bc - 4(ac - a^2 - bc + ab) = 0$

$\Rightarrow$ $4a^2 + b^2 + c^2 + 2bc - 4ac - 4ab = 0$

$\Rightarrow$ $(b + c)^2 + (2a)^2 - 2.2a(b + c) = 0$

$\Rightarrow$ $(b + c - 2a)^2 = 0$

$\Rightarrow$ $b + c - 2a = 0$

$\Rightarrow$ $-2a + b + c = 0$

$\Rightarrow$ $ax + by - c = 0$

$\Rightarrow$ Point is (–2, 1)

36. The vertex of the parabola $y^2 = 8x$ is at the centre of a circle and the parabola cuts the circle of the ends of the latus rectum. Then the equation of the circle is

(1) $x^2 + y^2 = 4$ (2) $x^2 + y^2 = 20$ (3) $x^2 + y^2 = 80$ (4) $x^2 + y^2 = 1$

Sol. Answer (2)

$\therefore \quad y^2 = 8x \Rightarrow 4a = 8$

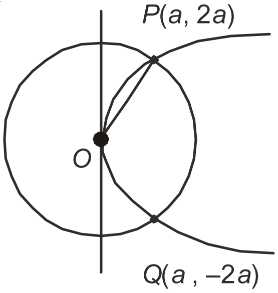
$\Rightarrow a = 2$

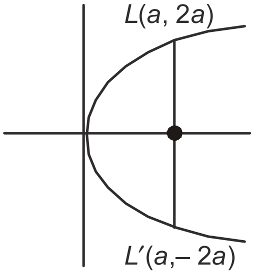
$\therefore \quad P = (a, 2a) = (2, 4)$

$\therefore \quad$ Radius = $OP = \sqrt{a^2 + 4a^2} = a\sqrt{5} = 2\sqrt{5}$

Equation of circle is $x^2 + y^2 = OP^2 = 20$

$\Rightarrow \quad x^2 + y^2 = 20$





37. The mirror image of the focus to the parabola $4(x + y) = y^2$ w.r.t. the directrix is

(1) (0, 2) (2) (2, 2) (3) (–4, 2) (4) (–2, 2)

Sol. Answer (3)

Given parabola is

$y^2 = 4x + 4y$

$\Rightarrow \quad y^2 - 4y = 4x$

$\Rightarrow \quad y^2 - 4y + 4 = 4x + 4$

$\Rightarrow \quad (y - 2)^2 = 4(x + 1)$

$\Rightarrow \quad Y^2 = 4x, \qquad X = x + 1$

$\qquad\qquad Y = y - 2$

$\therefore \quad$ Focus : $(a, 0)$

$X = a = 1 \qquad Y = 0$

$\Rightarrow \quad x + 1 = 1 \qquad \Rightarrow \quad y - 2 = 0$

$\Rightarrow \quad x = 0 \qquad \Rightarrow \quad y = 2$

$F : (0, 2)$

Directrix : $X = -a$

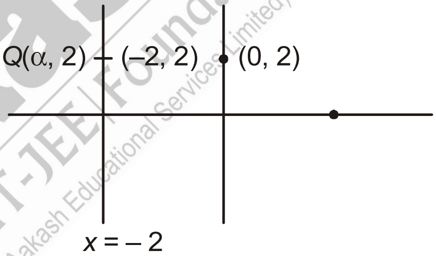
$\Rightarrow \quad x + 1 = -1$

$\Rightarrow \quad x + 2 = 0$

$\therefore \quad \dfrac{\alpha + 0}{2} = -2$

$\Rightarrow \quad \alpha = -4$

Image of $(0, 2) \equiv (-4, 2)$



38. Let m_1, m_2 are slopes of two tangents that are drawn from (2, 3) to the parabola $y^2 = 4x$ and α is the harmonic mean of m_1 & m_2. If β is the value of (1 + tan23°) (1 + tan22°), then $\frac{3}{2}\alpha + \beta$ is

(1) 2 (2) 3 (3) 1 (4) 5

Sol. Answer (2)

Equation of any tangent to the parabola $y^2 = 4x$ is

$$y = mx + \frac{1}{m}$$

Which is passes through (2, 3)

$\therefore \quad 3 = 2m + \dfrac{1}{m}$

$\Rightarrow \quad 2m^2 - 3m + 1 = 0$

$\Rightarrow \quad 2m^2 - 2m - m + 1 = 0$

$\Rightarrow \quad 2m(m - 1) - 1(2m - 1) = 0$

$\Rightarrow \quad m = 1, \dfrac{1}{2}$

$\therefore \quad m_1 = 1, m_2 = \dfrac{1}{2}$

$\therefore \quad \alpha = \dfrac{2}{\dfrac{1}{m_1} + \dfrac{1}{m_2}} = \dfrac{2}{1+2} = \dfrac{2}{3}$

Also, $\beta = 2$

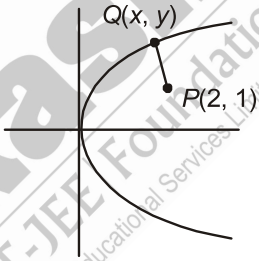
$\therefore \quad \dfrac{3}{2}\alpha + \beta = \dfrac{3}{2} \times \dfrac{2}{3} + 2 = 3$

39. A point on the curve $y^2 = 4x$, which is nearest to the point (2, 1) is

(1) (1, –2) (2) (–2, 1) (3) $(1, 2\sqrt{2})$ (4) (1, 2)

Sol. Answer (4)

$\therefore \quad PQ = (x - 2)^2 + (y - 1)^2$



$\Rightarrow \quad z = \left(\dfrac{y^2}{4} - 2\right)^2 + (y - 1)^2$

$\Rightarrow \quad \dfrac{dz}{dy} = 2\left(\dfrac{y^2}{4} - 2\right) \times \dfrac{y}{2} + 2(y - 1) = 0$

$\Rightarrow \quad \dfrac{y^3}{4} - 2y + 2y - 2 = 0$

$\Rightarrow \quad y^3 = 8$

$\Rightarrow \quad y = 2$

When $y = 2$, $4x = 4 \Rightarrow x = 1$

$\therefore$ Point is (1, 2)

40. From a point P, two tangents are drawn to the the parabola $y^2 = 4ax$. If the slope of one tangent is twice the slope of other, the locus of P is

(1) Circle (2) Straight line (3) Parabola (4) Ellipse

Sol. Answer (3)

Let $P(h, k)$ is point

Equation of line through $P(h, k)$

$y - k = m(x - h)$

$y = mx + (k - mh)$

Condition of tangency $c = \dfrac{a}{m}$

$k - mh = \dfrac{a}{m}$

$m^2h - km + a = 0$

Roots are m_1 and $2m_1$

$m_1 + 2m_1 = \dfrac{k}{h}$ $\quad\quad$ $m_1 2m_1 = \dfrac{a}{h}$

$m_1 = \dfrac{k}{3h}$ $\quad\quad$ and $\quad 2m_1^2 = \dfrac{a}{h}$

Eliminate m_1

$\therefore \quad k^2 = \dfrac{9a}{2}h$

Locus of $P(h, k)$ is $y^2 = \dfrac{9a}{2}x$ which is a parabola

[General Problems to Ellipse]

41. The eccentricity of the ellipse, which passes through the points (2, –2) and (–3, 1) is

(1) $\dfrac{1}{\sqrt{5}}$ $\quad$ (2) $\sqrt{\dfrac{2}{5}}$ $\quad$ (3) $\sqrt{\dfrac{3}{5}}$ $\quad$ (4) $\sqrt{\dfrac{4}{5}}$

Sol. Answer (2)

Ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

Passes through $A(2, -2)$ and $B(-3, 1)$

$\dfrac{4}{a^2} + \dfrac{4}{b^2} = 1$ ... (i)

$\dfrac{9}{a^2} + \dfrac{1}{b^2} = 1$ ... (ii)

By (ii) – (i), $\dfrac{5}{a^2} - \dfrac{3}{b^2} = 0$

$\dfrac{b^2}{a^2} = \dfrac{3}{5}$

$\Rightarrow \quad 1 - e^2 = \dfrac{3}{5}$

$\Rightarrow \quad e^2 = \dfrac{2}{5}$

$\Rightarrow \quad e = \sqrt{\dfrac{2}{5}}$

42. The parametric coordinates of a point on the ellipse, whose foci are (–3, 0) and (9, 0) and eccentricity $\dfrac{1}{3}$, are

(1) $(-3 + 9\cos\theta, 9\sin\theta)$ $\quad$ (2) $(4 - 3\cos\theta, 4 + 9\sin\theta)$

(3) $(3 + 18\cos\theta, 4 + 9\sin\theta)$ $\quad$ (4) $(3 + 18\cos\theta, 12\sqrt{2}\sin\theta)$

Sol. Answer (4)

Distance between foci

$2ae = 12$ Centre (3, 0)

$\Rightarrow$ $ae = 6$

$\Rightarrow$ $a = 18$

$b^2 = a^2(1 - e^2) = (18)^2 \left[1-\frac{1}{9}\right]$

$= 18 \times 16 \Rightarrow b = 12\sqrt{2}$

Equation of ellipse $\frac{(x-3)^2}{(18)^2}+\frac{(y-0)^2}{18\times16}=1$

Parametric coordinate

$x - 3 = a \cos$

$\Rightarrow$ $x = 3 + 18 \cos\theta$

$\Rightarrow$ $y - 0 = \sqrt{18\times16} . \sin\theta$

$\Rightarrow$ $y = 12\sqrt{2} \sin\theta$

$(3 + 18 \cos\theta, 12\sqrt{2} \sin\theta)$

43. The radius of the circle passing through the foci of $\frac{x^2}{16}+\frac{y^2}{9}=1$, and having centre (0, 3) is

(1) 4 (2) 3 (3) $\sqrt{12}$ (4) $\frac{7}{2}$

Sol. Answer (1)

$a^2 = 16; b^2 = 9$

$\Rightarrow$ $e^2 = 1 - \frac{b^2}{a^2} = 1-\frac{9}{16}=\frac{7}{16}$

$\Rightarrow$ $e=\frac{\sqrt{7}}{4}$

foci $(\pm ae, 0) = (\pm\sqrt{7}, 0)$

Centre (0, 3) $\therefore$ $r=\sqrt{7+9}=\sqrt{16}=4$

44. The centre of the ellipse $4x^2 + 9y^2 + 16x - 18y - 11 = 0$ is

(1) (–2, –1) (2) (–2, 1) (3) (2, –1) (4) (2, 1)

Sol. Answer (2)

The given equation is

$4x^2 + 9y^2 + 16x - 18y - 11 = 0$

$\Rightarrow$ $4(x^2 + 4x) + 9(y^2 - 2y) = 11$

$\Rightarrow$ $4(x^2 + 4x + 4) + 9(y^2 - 2y + 1) = 11 + 16 + 9$

$\Rightarrow$ $4(x + 2)^2 + 9(y - 1)^2 = 36$

$\Rightarrow$ $\frac{(x+2)^2}{9}+\frac{(y-1)^2}{4}=1$

$\Rightarrow$ Centre : (–2, 1)

45. The length of the latus rectum of the ellipse

$2x^2 + 3y^2 - 4x - 6y - 13 = 0$ is

(1) 5 (2) 4 (3) 8 (4) 12

Sol. Answer (2)

The given equation can be written as

$2(x^2 - 2x) + 3(y^2 - 2y) = 13$

$\Rightarrow 2(x^2 - 2x + 1) + 3(y^2 - 2y + 1) = 13 + 2 + 3$

$\Rightarrow 2(x-1)^2 + 3(y-1)^2 = 18$

$\Rightarrow \dfrac{(x-1)^2}{9} + \dfrac{(y-1)^2}{6} = 1$

Length of the latus rectum

$= \dfrac{2b^2}{a}$

$= \dfrac{2\times 6}{3} = 4$ units

46. The co-ordinates of foci of an ellipse

$3x^2 + 4y^2 + 12x + 16y - 8 = 0$ is

(1) $(\pm\sqrt{3}-2, -2)$ (2) $(-2, -2)$ (3) $(2-\sqrt{3}, \pm 2)$ (4) $(2+\sqrt{3}, -2)$

Sol. Answer (1)

The given equation reduces to

$3(x^2 + 4x) + 4(y^2 + 4y) = 8$

$\Rightarrow 3(x^2 + 4x + 4) + 4(y^2 + 4y + 4) = 8 + 12 + 16$

$\Rightarrow 3(x + 2)^2 + 4(y + 2)^2 = 36$

$\Rightarrow \dfrac{(x+2)^2}{12} + \dfrac{(y+2)^2}{9} = 1$

$\therefore\ e = \sqrt{1-\dfrac{b^2}{a^2}} = \sqrt{1-\dfrac{9}{12}} = \sqrt{1-\dfrac{3}{4}} = \sqrt{\dfrac{1}{4}} = \dfrac{1}{2}$

Foci : $(\pm ae, 0)$

$\therefore\ x + 2 = \pm ae = \pm 2\sqrt{3}\times\dfrac{1}{2} = \pm\sqrt{3}$

$\Rightarrow\ x = \pm\sqrt{3} - 2$

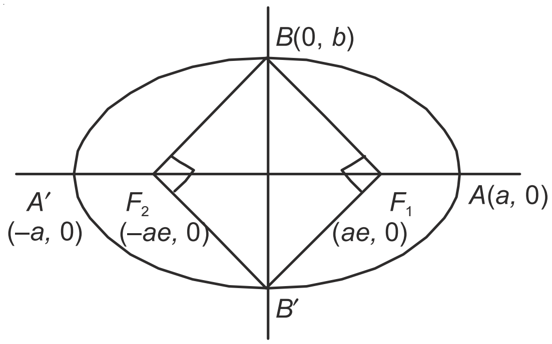
$y + 2 = 0 \Rightarrow y = -2$

Foci $\equiv (\pm\sqrt{3} - 2, -2)$

47. If the line joining foci subtends an angle of 90° at an extremity of minor axis, then the eccentricity e is

(1) $\dfrac{1}{\sqrt{6}}$ (2) $\dfrac{1}{\sqrt{3}}$ (3) $\dfrac{1}{\sqrt{2}}$ (4) $\dfrac{1}{2\sqrt{2}}$

Sol. Answer (3)



$\therefore\quad F_1B \perp F_2B$

$\Rightarrow\quad m_1m_2 = -1$

$$\Rightarrow\quad \frac{b-0}{0-ae}\times\frac{b-0}{0+ae} = -1$$

$$\Rightarrow\quad \frac{b^2}{a^2e^2} = 1$$

$$\Rightarrow\quad \frac{a^2(1-e^2)}{a^2e^2} = 1$$

$\Rightarrow\quad e^2 = 1 - e^2$

$\Rightarrow\quad 2e^2 = 1$

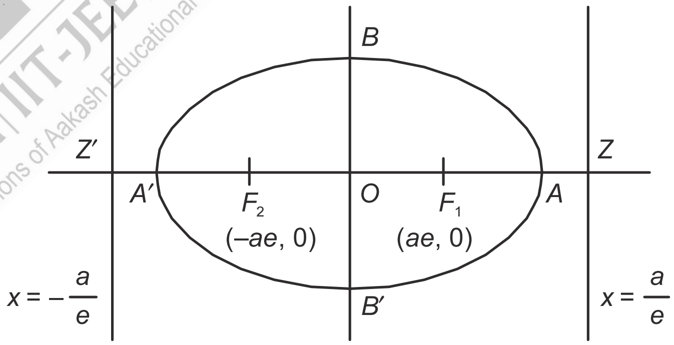
$$\Rightarrow\quad e^2 = \frac{1}{2}$$

$$\Rightarrow\quad e = \frac{1}{\sqrt{2}}$$

48. In an ellipse the distance between the foci is 8 and the distance between the directrices is 25, then the ratio of the length of major and minor axis is

(1) $\frac{5}{\sqrt{17}}$ (2) $\frac{3}{\sqrt{17}}$ (3) $\frac{4}{\sqrt{17}}$ (4) $\frac{6}{\sqrt{17}}$

Sol. Answer (1)



Given $F_1F_2 = 8 \Rightarrow 2ae = 8 \Rightarrow ae = 4$

Also, $ZZ' = 25$

$$\Rightarrow\quad \frac{2a}{e} = 25$$

$$\Rightarrow\quad \frac{2a}{\frac{4}{a}} = 25 \qquad e = \frac{4}{a}$$

$$\Rightarrow\quad 2a^2 = 25\times 4 = 100 \qquad = \frac{4}{5\sqrt{2}}$$

$\Rightarrow\quad a^2 = 50$

$\Rightarrow\quad a = 5\sqrt{2}$

$$\therefore\quad b^2 = a^2(1-e^2) \qquad \text{Ratio} = \frac{5\sqrt{2}}{\sqrt{34}}$$

$$= 50\left(1-\frac{16}{50}\right) \qquad = \frac{5}{\sqrt{17}}$$

$$= \frac{50}{50}\times 34$$

$\Rightarrow\quad b = \sqrt{34}$

49. If the equation $\dfrac{x^2}{2-r}+\dfrac{y^2}{r-5}+1=0$ represents an ellipse, then r lies in

(1) $(-\infty, 2) \cup (5, \infty)$ (2) $(2, 5)$ (3) $(5, 10)$ (4) $(-\infty, 1)$

Sol. Answer (2)

The given equation can reduces to

$$\frac{x^2}{2-r}+\frac{y^2}{r-5}=-1$$

$$\Rightarrow \quad \frac{x^2}{r-2}+\frac{y^2}{5-r}=1$$

For ellipse, $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1, a^2, b^2>0$

$\Rightarrow \quad r-2>0,\ 5-r>0$

$\Rightarrow \quad r>2,\ r<5$

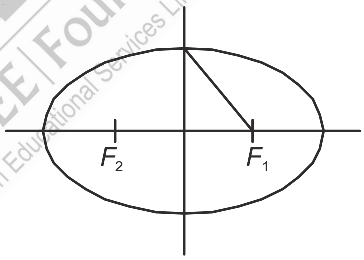
$\Rightarrow \quad 2<r<5$

50. If the focal distance of an end of minor axis of any ellipse, (whose axes along the x and y axes respectively is k and the distance between the foci is $2h$. Then the equation of the ellipse is

(1) $\dfrac{x^2}{h^2}+\dfrac{y^2}{k^2}=1$ (2) $\dfrac{x^2}{k^2}+\dfrac{y^2}{k^2-h^2}=1$ (3) $x^2+\dfrac{y^2}{k^2}=1$ (4) $\dfrac{x^2}{h^2}+y^2=1$

Sol. Answer (2)

Let the equation of the ellipse be $\dfrac{x^2}{a^2}=\dfrac{y^2}{b^2}=1$



Given

$F_1F_2 = 2h$ Also, $a + ex = k$

$\Rightarrow \quad 2ae = 2h \qquad \Rightarrow \quad a + e(0) = k$

$\Rightarrow \quad ae = h \qquad \Rightarrow \quad a = k$

$\therefore \quad b^2 = a^2(1-e^2) = a^2 - a^2e^2$

$\qquad = k^2 - h^2$

Equation of the ellipse is

$$\frac{x^2}{k^2}+\frac{y^2}{k^2-h^2}=1$$

51. Distance between the foci of the curve represented by the equation $x = 3 + 4\cos\theta$, $y = 2 + 3\sin\theta$, is

(1) $3\sqrt{7}$ (2) $2\sqrt{7}$ (3) $\sqrt{7}$ (4) $\dfrac{\sqrt{7}}{2}$

Sol. Answer (2)

The given equations can be written as

$$\frac{(x-3)^2}{16}+\frac{(y-2)^2}{9}=1$$

$$\Rightarrow \quad a=4, e=\sqrt{1-\frac{9}{16}}=\frac{\sqrt{7}}{4}$$

Distance between foci = $2ae$

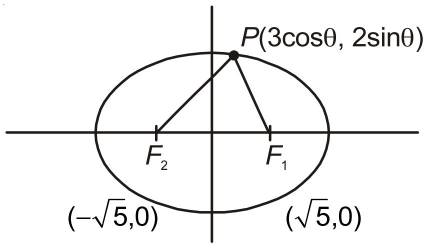
$$= 2\times4\times\frac{\sqrt{7}}{4} = 2\sqrt{7}$$

52. P is any variable point on the ellipse $4x^2 + 9y^2 = 36$ and F_1, F_2 are its foci. maximum area of ΔPF_1F_2 is (e is eccentricity of ellipse)

(1) $9e$ (2) $4e$ (3) $6e$ (4) $10e$

Sol. Answer (3)

The given equation can be reduces to $\frac{x^2}{9}+\frac{y^2}{4}=1$



$$\therefore \quad e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{4}{9}}=\frac{\sqrt{5}}{3}$$

$$F_1:(ae=0)\equiv(\sqrt{5},0)$$

$$ar(\Delta PF_1F_2)=\left|\frac{1}{2}\begin{vmatrix}3\cos\theta & 2\sin\theta & 1\\ \sqrt{5} & 0 & 1\\ -\sqrt{5} & 0 & 1\end{vmatrix}\right|$$

$$=\left|\frac{1}{2}\left\{-2\sin\theta(\sqrt{5}+\sqrt{5})\right\}\right|$$

$$=\left|\frac{1}{2}\times-2\sin\theta\times2\sqrt{5}\right|$$

$$=|-2\sqrt{5}\sin\theta|$$

$$=2\sqrt{5}\sin\theta$$

Max. area of $(\Delta PF_1F_2)=2\sqrt{5} = 2\times3\times\frac{\sqrt{5}}{3} = 6e$

[Tangents and Normals to Ellipse]

53. The number of values of c, such that $y = 4x + c$ touches $\frac{x^2}{4}+y^2=1$, is

(1) 0 (2) 1 (3) 2 (4) More than 2

Sol. Answer (3)

$c^2 = a^2m^2 + b^2$

$\Rightarrow \quad c=\pm\sqrt{a^2m^2+b^2}$

$\therefore$ Two values of c will satisfy.

54. If the line $\frac{x}{a}+\frac{y}{b}=\sqrt{2}$ touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at P, then eccentric angle of the point is equal to

(1) 0 (2) 90° (3) 45° (4) 60°

Sol. Answer (3)

Let point of contact be $P(a\cos\alpha, b\sin\alpha)$

$\therefore$ Equation of tangent

$$\frac{x\cos\alpha}{a}+\frac{y\sin\alpha}{b}=1 \qquad \ldots \text{(i)}$$

$$\frac{x}{a\sqrt{2}}+\frac{y}{b\sqrt{2}}=1 \qquad \ldots \text{(ii)}$$

These lines are coincident lines

$$\therefore \quad \frac{\cos\alpha}{\frac{1}{\sqrt{2}}}=\frac{\sin\alpha}{\frac{1}{\sqrt{2}}}=1$$

$$\cos\alpha = \frac{1}{\sqrt{2}} \text{ and } \sin\alpha = \frac{1}{\sqrt{2}}$$

$\therefore \quad \alpha = 45°$

55. The sum of the squares of the perpendiculars on any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from two points on the minor axis each at the distance $\sqrt{a^2-b^2}$ from the centre is

(1) a^2 (2) b^2 (3) $2a^2$ (4) $2b^2$

Sol. Answer (3)

Equation of tangent

$$\frac{x\cos\alpha}{a}+\frac{y\sin\alpha}{b}=1$$

Two points on minor axis

$$A\left(0, \sqrt{a^2-b^2}\right)$$

$$B\left(0, -\sqrt{a^2-b^2}\right)$$

$$AM = \frac{\frac{\sqrt{a^2-b^2}}{b}.\sin\alpha - 1}{\sqrt{\frac{\cos^2\alpha}{a^2}+\frac{\sin^2\alpha}{b^2}}}$$

$$PN = \frac{-\frac{\sqrt{a^2-b^2}}{b}.\sin\alpha - 1}{\sqrt{\frac{\cos^2\alpha}{a^2}+\frac{\sin^2\alpha}{b^2}}}$$

$$AM^2 + PN^2 = \frac{2.\left(\frac{a^2-b^2}{b^2}\right)\sin^2\alpha + 2}{\frac{\cos^2\alpha}{a^2}+\frac{\sin^2\alpha}{b^2}}$$

$$= \frac{2a^2[a^2\sin^2\alpha + b^2(1-\sin^2\alpha)]}{b^2\cos^2\alpha + a^2\sin^2\alpha} = 2a^2$$

56. The line $2x + y = 3$ cuts the ellipse $4x^2 + y^2 = 5$ at P and Q. If θ be the angle between the normals at these points, then $\tan\theta$ is equal to

(1) $\frac{1}{2}$ (2) $\frac{3}{4}$ (3) $\frac{3}{5}$ (4) 5

Sol. Answer (3)

Point of intersection of line

$2x + y = 3$ and ellipse $4x^2 + y^2 = 5$ are $P(1, 1)$ and $Q\left(\frac{1}{2}, 2\right)$

Differentiating $4x^2 + y^2 = 5$

We get, $8x + 2y\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{4x}{y}$

Slope of normal = $\frac{y}{4x}$

At $P(1, 1)$ $m_1 = \frac{1}{4}$

At $Q\left(\frac{1}{2}, 2\right)$ $m_2 = 1$

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{1 - \frac{1}{4}}{1 + 1.\frac{1}{4}}\right| = \left|\frac{\frac{3}{4}}{\frac{5}{4}}\right| = \frac{3}{5} \text{ (for acute angle)}$$

57. Two common tangents to $x^2 + y^2 = 2a^2$ and $y^2 = 8ax$ are

(1) $x = \pm (y + 2a)$ (2) $x = \pm (y + a)$ (3) $y = \pm (x + 2a)$ (4) $y = \pm (x + a)$

Sol. Answer (3)

Equation of tangent of $y^2 = 8ax$ be

$$y = mx + \frac{2a}{m} \quad \dots \text{(i)}$$

Line (i) is a tangent to the ellipse $x^2 + \frac{y^2}{3} = a^2$

$\therefore$ Condition of tangency

$c^2 = a^2m^2 + b^2$

$\Rightarrow \frac{4a^2}{m^2} = a^2m^2 + 3a^2$

$\Rightarrow \frac{4}{m^2} = m^2 + 3$

$\Rightarrow m^4 + 3m^2 - 4 = 0$ $\therefore$ $m^2 = -4$ or $m^2 = 1$

$m = \pm 1$ $\therefore$ Common tangent

$y = \pm x \pm 2a$

$\Rightarrow y = \pm (x + 2a)$

58. The line $\ell x + my + n = 0$ is a normal to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, provided

(1) $\frac{a^2}{m^2}+\frac{b^2}{\ell^2}=\frac{(a^2-b^2)^2}{n^2}$ (2) $\frac{a^2}{\ell^2}+\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$

(3) $\frac{a^2}{\ell^2}-\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$ (4) None of these

Sol. Answer (2)

Equation of normal to the ellipse

$ax\sec\theta - by\,\mathrm{cosec}\theta = a^2 - b^2$ … (i)

$lx + my + n = 0$ … (ii)

These lines are coincident lines

$$\frac{a\sec\theta}{l}=\frac{-b\,\mathrm{cosec}\theta}{m}=\frac{-(a^2-b^2)}{n}$$

$\therefore$ $\cos\theta=\frac{-an}{l(a^2-b^2)}$ and $\sin\theta=\frac{bn}{m(a^2-b^2)}$

$\sin^2\theta+\cos^2\theta=1$

$$\Rightarrow \frac{n^2}{(a^2-b^2)^2}\left[\frac{a^2}{l^2}+\frac{b^2}{m^2}\right]=1$$

$$\Rightarrow \frac{a^2}{l^2}+\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$$

59. If the tangent at a point $(a\cos\alpha, b\sin\alpha)$ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the auxiliary circle in two points and the chord joining them subtends a right angle at the centre, then the eccentricity of the ellipse is given by

(1) $1+\cos^2\alpha$ (2) $\frac{1}{\sqrt{1+\sin^2\alpha}}$ (3) $\frac{1}{\sqrt{1+\cos^2\alpha}}$ (4) $1+\sin^2\alpha$

Sol. Answer (2)

Equation of tangent

$$\frac{x\cos\alpha}{a}+\frac{y\sin\alpha}{b}=1 \quad \text{… (i)}$$

Equation of auxiliary circle

$$x^2+y^2=a^2 \quad \text{… (ii)}$$

Line (i) cuts the circle at two points. Combined equation of pair of lines passing through origin and intersection points.

$$x^2 + y^2 = a^2\left(\frac{x\cos\alpha}{a} + \frac{y\sin\alpha}{b}\right)^2$$

$$(\cos^2\alpha - 1)x^2 + \left(\frac{a^2\sin^2\alpha}{b^2} - 1\right)y^2 + \frac{2a\sin\alpha\cos\alpha}{b}xy = 0$$

Angle between pair of lines $\frac{\pi}{2}$

$\Rightarrow \quad \cos^2\alpha - 1 + \frac{a^2}{b^2}\sin^2\alpha - 1 = 0$

$\Rightarrow \quad \frac{a^2}{b^2}\sin^2\alpha = 1 + \sin^2\alpha$

$\Rightarrow \quad b^2 = a^2(1 - e^2)$

$\Rightarrow \quad \frac{1}{(1-e^2)}\sin^2\alpha = 1 + \sin^2\alpha$

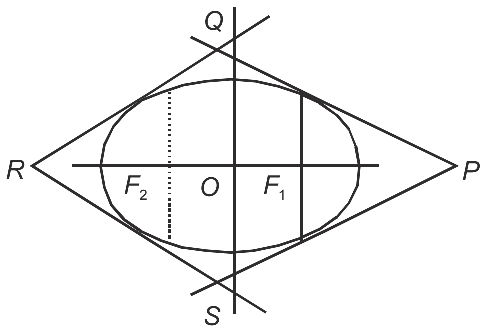
$\Rightarrow \quad e^2 = \frac{1}{1+\sin^2\alpha}$

$\Rightarrow \quad e = \frac{1}{\sqrt{1+\sin^2\alpha}}$

60. Tangents are drawn to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ at ends of latus rectum. The area of quadrilateral so formed is

(1) 27 (2) $\frac{27}{2}$ (3) $\frac{27}{4}$ (4) $\frac{27}{55}$

Sol. Answer (1)



Given $\frac{x^2}{9} + \frac{y^2}{5} = 1$

$\Rightarrow \quad e^2 = 1 - \frac{5}{9} = \frac{4}{9}$

$\Rightarrow \quad e = \frac{2}{3}$

One end of a latus rectum

$= \left(ae, \frac{b^2}{a}\right)$

$= \left(3\cdot\frac{2}{3}, \frac{5}{3}\right)$

$= \left(2, \frac{5}{3}\right)$

Equation of tangent at $\left(2, \frac{5}{3}\right)$ is

$$\frac{2x}{9} + \frac{5}{3} \cdot \frac{y}{5} = 1$$

$$\Rightarrow \frac{2x}{9} + \frac{y}{3} = 1$$

$$ar(\Delta OPQ) = \frac{1}{2} \times \frac{9}{2} \times 3 = \frac{27}{4}$$

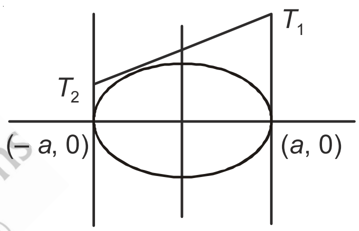
$$ar(\square PQRS) = 4 \times \frac{27}{4} = 27 \text{ sq. units}$$

61. The tangent at any point on the ellipse $16x^2 + 25y^2 = 400$ meets the tangents at the ends of the major axis at T_1 and T_2. The circle on T_1T_2 as diameter passes through

(1) (3, 0) (2) (0, 0) (3) (0, 3) (4) (4, 0)

Sol. Answer (1)

Given $16x^2 + 25y^2 = 400 \Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$



Any tangent to the ellipse is

$$\frac{x\cos\theta}{5} + \frac{y\sin\theta}{4} = 1$$

$$T_1 : \left(5, 4\tan\frac{\theta}{2}\right)$$

$$T_2 : \left(-5, 4\cot\frac{\theta}{2}\right)$$

Equation of a circle is

$$(x-5)(x+5) + \left(y - 4\tan\frac{\theta}{2}\right)\left(y - 4\cot\frac{\theta}{2}\right) = 0$$

$$\Rightarrow x^2 - 25 + y^2 + 16 - 4\left(\tan\frac{\theta}{2} + \cot\frac{\theta}{2}\right)y = 0$$

$$\Rightarrow x^2 + y^2 - 8y\operatorname{cosec}\theta - 9 = 0$$

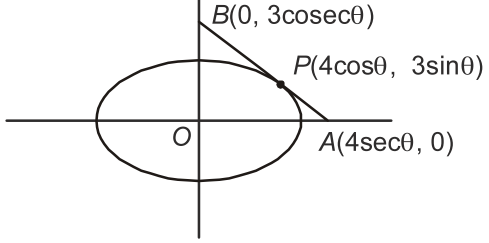
Which passes through (3, 0)

62. The minimum area of the triangle formed by the tangent to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the co-ordinate axes is

(1) 16 (2) 9 (3) 12 (4) 144

Sol. Answer (3)

Tangent at P is $\frac{x}{4}\cos\theta + \frac{y}{3}\sin\theta = 1$



$$ar(\Delta OAB) = \frac{1}{2} \times 4\sec\theta \times 3\operatorname{cosec}\theta$$

$$= \frac{12}{\sin 2\theta}$$

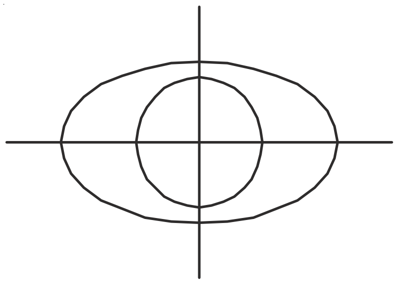
$\Rightarrow$ ar(ΔOAB) will be minimum when $\sin 2\theta$ is max.

$\Rightarrow$ ar(ΔOAB) = 12 $(\because \sin 2\theta = 1)$

63. The number of common tangents to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and the circle $x^2 + y^2 = 4$ is

(1) 2 (2) 1 (3) 0 (4) 4

Sol. Answer (3)

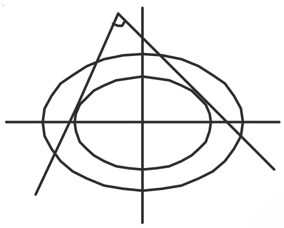


Clearly, number of common tangents = 0

64. Number of points on the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ from which pair of perpendicular tangents are drawn to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, is

(1) 2 (2) 3 (3) 4 (4) 1

Sol. Answer (1)



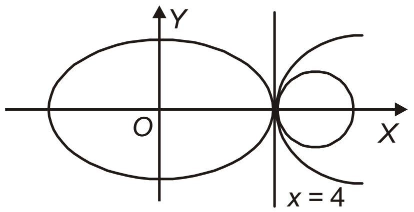
Locus of point from which mutually perpendicular tangents can be drawn to $\frac{x^2}{16}+\frac{y^2}{9}=1$ is $x^2 + y^2 = 25$

Number of points if intersection at $x^2 + y^2 = 25$ and $\frac{x^2}{25}+\frac{y^2}{16}=1$ is 2.

65. A common tangent to $9x^2 + 16y^2 = 144$, $y^2 = x - 4$ and $x^2 + y^2 - 12x + 32 = 0$ is

(1) $y = 3$ (2) $y = -3$ (3) $x = 4$ (4) $x = -4$

Sol. Answer (3)



Thus, the equation of the common tangent is $x = 4$

[Locus and Miscellaneous Problems on Ellipse]

66. The equation of the chord of the ellipse $x^2 + 4y^2 = 4$ having the middle point at $\left(-2, \frac{1}{2}\right)$ is

(1) $2x - 2y + 5 = 0$ (2) $x + 2y = 0$

(3) $3x - 2y + 4 = 0$ (4) $2x - 2y - 5 = 0$

Sol. Answer (1)

Equation of chord whose mid-point is $\left(-2, \frac{1}{2}\right)$ be

$S' = T$

$\Rightarrow \; 4 + 1 - 4 = x(-2) + 4y\left(\frac{1}{2}\right) - 4$

$\Rightarrow \; 2x - 2y + 5 = 0$

67. Pole of the line $x + 4y = 5$ with respect to the ellipse $5x^2 + 4y^2 = 10$ is

(1) $(2, 2)$ (2) $\left(\frac{2}{5}, \frac{3}{5}\right)$ (3) $\left(\frac{2}{5}, 2\right)$ (4) $\left(2, \frac{3}{5}\right)$

Sol. Answer (3)

Let pole be $P(h, k)$ of the ellipse $5x^2 + 2y^2 = 10$

$\therefore$ Equation of polar $T = 0$

$5xh + 4yk = 10$ ... (i)

$x + 4y = 5$ ... (ii)

These lines are coincident lines

$$\frac{5h}{1} = \frac{4k}{4} = \frac{10}{5}$$

$h = \frac{2}{5}$ and $k = 2$

Pole $= \left(\frac{2}{5}, 2\right)$

68. The locus of the mid points of the portion of the tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepted between the axes is

(1) $\frac{a^2}{x^2} + \frac{b^2}{y^2} = 4$ (2) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$ (3) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 16$ (4) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 25$

Sol. Answer (1)

Tangent of ellipse

$$\frac{x\cos\alpha}{a} + \frac{y\sin\alpha}{b} = 1$$

Cuts x-axis at $A\left(\frac{a}{\cos\alpha}, 0\right)$ and y-axis at $B\left(0, \frac{b}{\sin\alpha}\right)$

Let mid-point of AB be $P(h, k)$

$\therefore \; h = \frac{a}{2\cos\alpha}$ and $k = \frac{b}{2\sin\alpha}$

$\cos\alpha = \frac{a}{2h}$ and $\sin\alpha = \frac{b}{2k}$

$\cos^2\alpha + \sin^2\alpha = 1$

$$\frac{a^2}{4h^2}+\frac{b^2}{4k^2}=1$$

Locus of $P(h, k)$

$$\frac{a^2}{x^2}+\frac{b^2}{y^2}=4$$

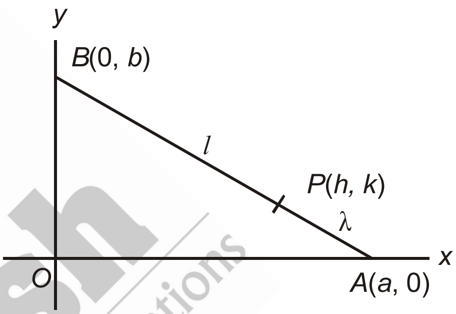
69. A rod of given length moves such that its extremities lie on two fixed perpendicular lines. Any point on the rod describes a

(1) Circle (2) Parabola (3) Ellipse (4) Straight line

Sol. Answer (3)

$a^2 + b^2 = l^2$ … (i)

Let $P(h, k)$ divide AB in the ratio $\lambda : 1$



$$\therefore \quad h=\frac{a}{\lambda+1},\ k=\frac{\lambda b}{\lambda+1}$$

From (i), $(\lambda+1)^2h^2+\frac{(\lambda+1)^2}{\lambda^2}k^2=l^2$

$\therefore$ Locus of $P(h, k)$

$(\lambda+1)^2x^2+\frac{(\lambda+1)^2}{\lambda^2}y^2=l^2$ is an ellipse

70. The locus of the feet of the perpendiculars drawn from the centre to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ on any tangent to it is

(1) $(x^2 + y^2)^2 = 9x^2 + 4y^2$ (2) $(x^2 + y^2)^2 = 4x^2 + 9y^2$

(3) $(x^2 + y^2)^2 = 3x^2 + 2y^2$ (4) $(x^2 + y^2)^2 = 2x^2 + 3y^2$

Sol. Answer (1)

$$\frac{x^2}{9}+\frac{y^2}{4}=1$$

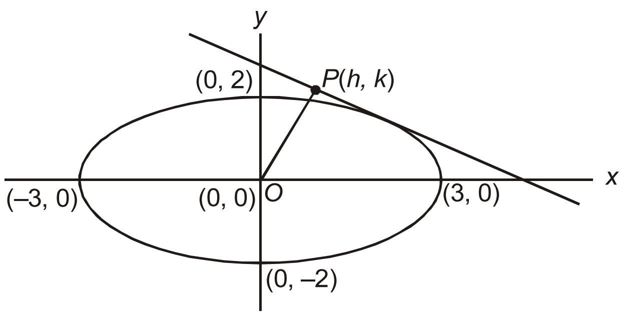
Equation of tangent of ellipse at $(3\cos\theta, 2\sin\theta)$

$$\frac{x}{3}\cos\theta+\frac{y}{2}\sin\theta=1 \quad \dots \text{(i)}$$

Line OP is perpendicular to the tangent

Equation of OP is $y = mx$

Where m is $-\frac{1}{\frac{\left(\frac{\cos\theta}{3}\right)}{\left(\frac{\sin\theta}{2}\right)}}=\frac{\left(\frac{\sin\theta}{2}\right)}{\frac{\cos\theta}{3}}=\frac{3}{2}\tan\theta$



Equation of OP, $y = \frac{3}{2}\tan\theta x$

$\Rightarrow \tan\theta = \frac{2y}{3x}$

$\sin\theta = \frac{2y}{\sqrt{4y^2+9x^2}}$

$\cos\theta = \frac{3x}{\sqrt{4y^2+9x^2}}$

Putting these values of $\sin\theta$, $\cos\theta$ in (i)

$$\frac{x}{3}\cdot\frac{3x}{\sqrt{4y^2+9x^2}}+\frac{y}{2}\cdot\frac{2y}{\sqrt{4y^2+9x^2}}=1$$

$\Rightarrow (x^2+y^2)^2 = (4y^2+9x^2)$

71. The locus of the foot of the perpendicular from any focus upon any tangent to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

(1) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (2) $x^2 + y^2 = a^2 + b^2$ (3) $x^2 + y^2 = a^2$ (4) None of these

Sol. Answer (3)

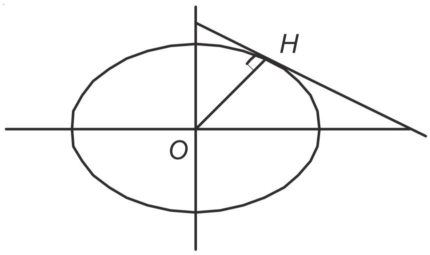
72. The locus of the foot of perpendicular from the centre on any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

(1) A circle (2) A pair of straight lines

(3) Another ellipse (4) Curve of degree 4

Sol. Answer (4)

Equation of any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is



$y = mx + \sqrt{a^2m^2+b^2}$ ...(i)

$OM : y = -\frac{1}{m}x$

$\Rightarrow x + my = 0$ ...(ii)

From (i) & (ii), we get

$$y = x\cdot\left(-\frac{x}{y}\right)+\sqrt{a^2\times\frac{x^2}{y^2}+b^2}$$

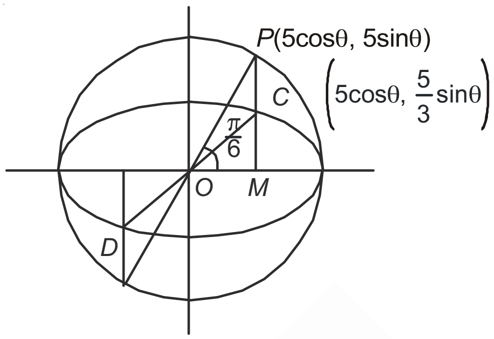
$\Rightarrow (x^2 + y^2) = (a^2x^2 + b^2y^2)^2$

$\Rightarrow$ Which represents a curve of degree 4

73. If CD is a diameter of an ellipse $x^2 + 9y^2 = 25$ and the eccentric angle of C is $\frac{\pi}{6}$, then the eccentric angle of D is

(1) $\frac{5\pi}{6}$ (2) $-\frac{5\pi}{6}$ (3) $-\frac{2\pi}{3}$ (4) $\frac{2\pi}{3}$

Sol. Answer (2)



Clearly eccentric angle of $D = \frac{7\pi}{6} - 2\pi$

$= -\frac{5\pi}{6}$

74. If θ_1, θ_2, θ_3, θ_4 be eccentric angles of the four concyclic points of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $\theta_1 + \theta_2 + \theta_3 + \theta_4$ is (where $n \in I$)

(1) $(2n+1)\frac{\pi}{2}$ (2) $(2n + 1)\pi$ (3) $2n\pi$ (4) $n\pi$

Sol. Answer (3)

The equation of chords of the ellipse are

$$\frac{x}{a}\cos\left(\frac{\theta_1+\theta_2}{2}\right)+\frac{y}{b}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=1$$

And $\frac{x}{a}\cos\left(\frac{\theta_2+\theta_3}{2}\right)+\frac{y}{b}\sin\left(\frac{\theta_3+\theta_4}{2}\right)=1$

If θ_1, θ_2, θ_3, θ_4 are concyclic points, then these lines will make equal angle with the axes

$\therefore \quad \tan\left(\frac{\theta_1+\theta_2}{2}\right)=-\tan\left(\frac{\theta_3+\theta_4}{2}\right)$

$\Rightarrow \quad \frac{\theta_1+\theta_2}{2}+\frac{\theta_3+\theta_4}{2}=n\pi$

$\Rightarrow \quad \theta_1 + \theta_2 + \theta_3 + \theta_4 = 2n\pi,\ n \in I$

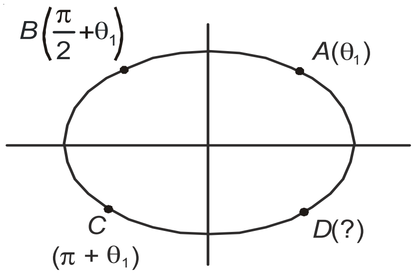
75. Let the eccentric angles of three point A, B and C on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are $\theta_1, \frac{\pi}{2}+\theta_1, \pi+\theta_1$.

A circle through A, B and C cuts the ellipse again at D. Then the eccentric angle of D is

(1) $\pi - 3\theta_1$ (2) $\frac{3\pi}{2}-3\theta_1$ (3) $\frac{\pi}{2}-3\theta_1$ (4) $3\theta_1-\frac{\pi}{2}$

Sol. Answer (3)

Let the eccentric angle of D be θ_2



Then $\theta_1 + \frac{\pi}{2} + \theta_1 + \pi + \theta_1 + \theta_2 = 2\pi$

$\Rightarrow \frac{3\pi}{2} + 3\theta_1 + \theta_2 = 2\pi$

$\Rightarrow \theta_2 = 2\pi - \frac{3\pi}{2} - 3\theta_1$

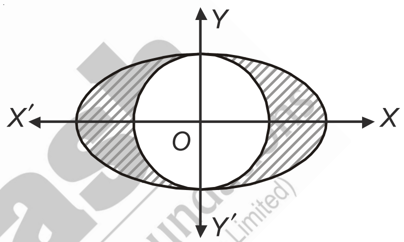
$\Rightarrow \theta_2 = \frac{\pi}{2} - 3\theta_1$

76. The area of the region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} \le 1$ and the circle $x^2 + y^2 \ge 9$ is

(1) 12π (2) 3π (3) 9π (4) 6π

Sol. Answer (2)

Area of the shaded part



= Area of an ellipse – area of a circle

$= \pi \times 3 \times 4 - \pi \times 3^2$

$= 12\pi - 9\pi$

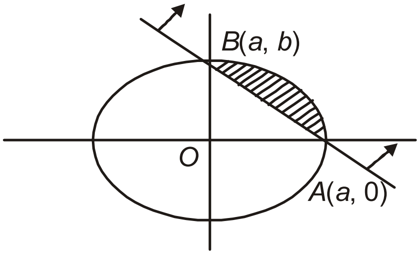
$= 3\pi$

77. Area of the region bounded by the curve $\left\{(x, y) : \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1 \le \frac{x}{a} + \frac{y}{b}\right\}$ is

(1) $\left(\frac{\pi}{4} - \frac{1}{2}\right)ab$ (2) $\left(\frac{\pi}{4} + \frac{1}{2}\right)ab$ (3) $\left(\frac{\pi}{4} - \frac{1}{3}\right)ab$ (4) $\frac{\pi}{4}ab$

Sol. Answer (1)

Area of the shaded part



$= \frac{1}{4}$th (area of the ellipse) – $ar(\Delta AOB)$

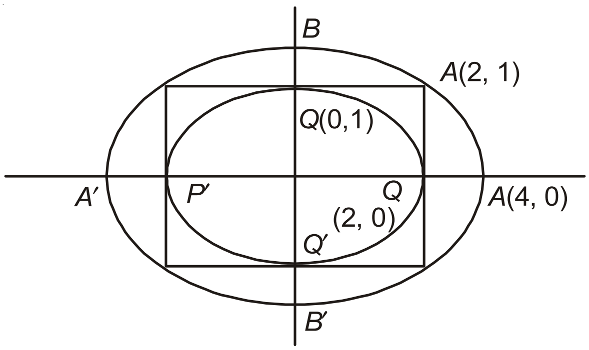
$= \frac{\pi}{4}ab - \frac{1}{2}ab$

$= \left(\frac{\pi}{4} - \frac{1}{2}\right)ab$ sq. units

78. The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the co-ordinate axes which is turn in inscribed in another ellipse that passes through the (4, 0). Then the equation of the ellipse is

(1) $x^2 + 12y^2 = 16$ (2) $4x^2 + 48y^2 = 48$ (3) $4x^2 + 64y^2 = 48$ (4) $x^2 + 16y^2 = 16$

Sol. Answer (1)



$E : x^2 + 4y^2 = 4$

$\Rightarrow \quad \dfrac{x^2}{4} + \dfrac{y^2}{1} = 1$

Let the equation of the ellipse be

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1 \qquad \ldots(i)$$

Which is passing through $A(2, 1)$, so we get,

$$\frac{4}{16} + \frac{1}{b^2} = 1$$

$\Rightarrow \quad \dfrac{1}{b^2} = 1 - \dfrac{4}{16} = \dfrac{12}{16}$

$\Rightarrow \quad b^2 = \dfrac{16}{12}$

Required equation is $\dfrac{x^2}{16} + \dfrac{y^2}{\frac{16}{12}} = 1$

$\Rightarrow \quad \dfrac{x^2}{16} + \dfrac{12y^2}{16} = 1$

$\Rightarrow \quad x^2 + 12y^2 = 16$

[General Problems on Hyperbola]

79. The curve described parametrically by $x = t^2 + t + 1$, $y = t^2 - t + 1$ represents

(1) A pair of straight lines (2) An ellipse

(3) A hyperbola (4) A parabola

Sol. Answer (4)

$x = t^2 + t + 1 \qquad \ldots(i)$

$y = t^2 - t + 1 \qquad \ldots(ii)$

(i) + (ii); $x + y = 2(t^2 + 1)$

$$\frac{x+y}{2} = t^2 + 1 \qquad \ldots(iii)$$

(i) – (ii); $x - y = 2t$

$$t = \frac{x-y}{2}$$

From (iii); $\dfrac{x+y}{2} = \left(\dfrac{x-y}{2}\right)^2 + 1$

$\Rightarrow \quad x^2 - 2xy + y^2 - 2x - 2y + 4 = 0$

Comparing with $ax^2+2hxy+by^2+2gx+2fy+c=0$,

$a = 1$, $b = 1$, $h = –1$, $g = –1$, $f = –1$, $c = 4$

$h^2 = 1$

$ab = 1$

$\therefore\quad h^2 = ab$

Also $\Delta = abc + 2fgh – af^2 – bg^2ch^2$

$= 4 + 2 \times (–1)(–1)(–1) – 1 \times (–1)^2 – 1 \times (–1)^2 – 4(–1)^2 = 4 – 2 – 1 – 1 – 4 = – 4 \neq 0$

$\therefore$ Equation represents a parabola.

80. If LM is a double ordinate of the hyperbola $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$ such that OLM is an equilateral triangle, O being the centre of the hyperbola, then the eccentricity e of the hyperbola, satisfies

(1) $e>\dfrac{2}{\sqrt{3}}$ (2) $e=\dfrac{2}{\sqrt{3}}$ (3) $e<\dfrac{2}{\sqrt{3}}$ (4) $1<e<\dfrac{2}{\sqrt{3}}$

Sol. Answer (1)

$ON^2 = l^2-\dfrac{l^2}{4}=\dfrac{3}{4}l^2$

$ON = \dfrac{\sqrt{3}}{2}l$

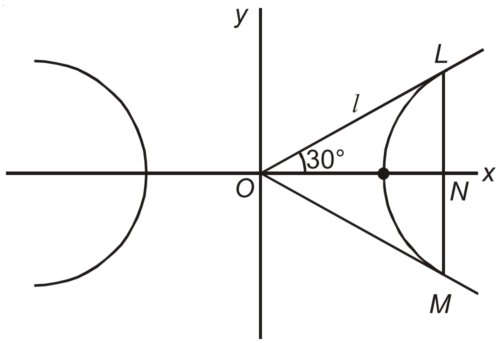
$L\left(\dfrac{\sqrt{3}}{2}l,\dfrac{l}{2}\right)$ lies on the hyperbola $\therefore\ \dfrac{3l^2}{4a^2}-\dfrac{l^2}{4b^2}=1$

But $b^2 = a^2(e^2 – 1)$

$\Rightarrow\ \dfrac{3l^2}{4a^2}-\dfrac{l^2}{4a^2(e^2-1)}=1$

$\Rightarrow\ \dfrac{l^2}{4a^2}\left[3-\dfrac{1}{e^2-1}\right]=1$

$\Rightarrow\ \dfrac{l^2}{a^2}=\dfrac{4(e^2-1)}{(3e^2-4)}$



ΔOLM is equilateral triangle

$\therefore\ l > a$ $\qquad \dfrac{l^2}{a^2}>1$

$\dfrac{4(e^2-1)}{(3e^2-4)}>1$ $\qquad \Rightarrow\ \dfrac{4(e^2-1)}{3e^2-4}-1>0$

$\Rightarrow\ \dfrac{e^2}{3e^2-4}>0$ $\qquad \therefore\ 3e^2 – 4 > 0$

$\Rightarrow\ e>\dfrac{2}{\sqrt{3}}$

81. The equation of the hyperbola, whose foci are (6, 4) and (–4, 4) and eccentricity is 2, is

(1) $12(x-1)^2 - 4(y-4)^2 = 75$ (2) $12(x+1)^2 - 4(y+4)^2 = 75$

(3) $4(x-1)^2 - 12(y-4)^2 = 75$ (4) $4(x+1)^2 - 12(y+4)^2 = 75$

Sol. Answer (1)

Distance between foci = 10

$\Rightarrow$ $2ae = 10$

$\Rightarrow$ $ae = 5$

$\Rightarrow$ $e = 2$

$\Rightarrow$ $a = \dfrac{5}{2}$

$\Rightarrow$ $b^2 = a^2(e^2 - 1) = \dfrac{25}{4}(4-1) = \dfrac{75}{4}$

Centre of hyperbola and mid-point of foci = (1, 4)

$\therefore$ Equation of hyperbola

$$\frac{4(x-1)^2}{25} - \frac{(y-4)^2.4}{75} = 1$$

$\Rightarrow$ $12(x-1)^2 - 4(y-4)^2 = 75$

82. The equation $\dfrac{x^2}{1-r} - \dfrac{y^2}{1+r} = 1$, $0 < r < 1$ represents

(1) An ellipse (2) A hyperbola (3) A circle (4) A parabola

Sol. Answer (2)

$$\frac{x^2}{1-r} - \frac{y^2}{1+r} = 1$$

Condition for hyperbola

$1 - r > 0$ and $1 + r > 0$

$r < 1$ $r > -1$

$-1 < r < 1$ for hyperbola

Thus for $0 < r < 1$, given locus is hyperbola.

83. The equation of the hyperbola with centre at (0, 0) and co-ordinate axes as its axes, distance between the directrices being $\dfrac{4}{\sqrt{3}}$ and passing through the point (2, 1), is

(1) $3x^2 - 2y^2 = 10$ (2) $3x^2 - 2y^2 = 2$ (3) $2x^2 - 3y^2 = 10$ (4) $x^2 - y^2 = 3$

Sol. Answer (1)

Distance between directrices = $\frac{4}{\sqrt{3}}$

$\Rightarrow \frac{2a}{e} = \frac{4}{\sqrt{3}}$

$\Rightarrow a = \frac{2e}{\sqrt{3}}$

Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through (2, 1)

$\Rightarrow \frac{4}{a^2} - \frac{1}{a^2(e^2-1)} = 1$

$\frac{12}{4e^2} - \frac{3}{4e^2(e^2-1)} = 1$

$4e^4 - 16e^2 + 15 = 0$

$e^2 = \frac{5}{2}$ $\qquad$ $e^2 = \frac{3}{2}$

If $e^2 = \frac{5}{2}$ $\qquad$ $a^2 = \frac{13}{3}$

$b^2 = 5$

Equation of hyperbola

$\frac{3x^2}{10} - \frac{y^2}{5} = 1$

$3x^2 - 2y^2 = 10$

84. If t is a parameter, then $x = a\left(t + \frac{1}{t}\right)$ and $y = b\left(t - \frac{1}{t}\right)$ represents

(1) An ellipse (2) A parabola (3) A hyperbola (4) A circle

Sol. Answer (3)

Given $x = a\left(t + \frac{1}{t}\right)$ and $y = b\left(t - \frac{1}{t}\right)$

$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = \left(t + \frac{1}{t}\right)^2 - \left(t - \frac{1}{t}\right)^2$

$= t^2 + \frac{1}{t^2} + 2 - t^2 - \frac{1}{t^2} + 2 = 4$

$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 4$

$\Rightarrow \frac{x^2}{4a^2} - \frac{y^2}{4b^2} = 1$ $\qquad$ Represents a hyperbola

85. If the equation $\frac{x^2}{10-\lambda}+\frac{y^2}{6-\lambda}=1$ represents a hyperbola, then λ is

(1) $\lambda > 10$ (2) $6 < \lambda < 10$ (3) $\lambda < 6$ (4) $(-\infty, 6] \cup [10, \infty)$

Sol. Answer (2)

$$H:\ \frac{x^2}{10-\lambda}+\frac{y^2}{6-\lambda}=1$$

$$\Rightarrow \frac{x^2}{10-\lambda}-\frac{y^2}{\lambda-6}=1$$

$\therefore$ $10-\lambda>0$ and $\lambda-6>0$

$\Rightarrow$ $\lambda<10, \lambda>6 \Rightarrow 6<\lambda<10$

86. If $3x^2 - 5y^2 - 6x + 20y - 32 = 0$ represents a hyperbola, then the co-ordinates of foci are

(1) $(\pm 2\sqrt{2}, 0)$ (2) $(1 \pm 2\sqrt{2}, 2)$ (3) $(0, \pm 2\sqrt{2})$ (4) (1, 2)

Sol. Answer (2)

Hyperbola : $3x^2 - 5y^2 - 6x + 20y - 32 = 0$

$\Rightarrow$ $3(x^2-2x)-5(y^2-4y)=32$

$3(x^2-2x+1)-5(y^2-4y+4) = 32 + 3 - 20$

$3(x-1)^2-5(y-2)^2 = 15$

$$\Rightarrow \frac{(x-1)^2}{5}-\frac{(y-2)^2}{3}=1$$

$$\therefore\ e=\sqrt{1+\frac{b^2}{x^2}}=\sqrt{1+\frac{3}{5}}=\frac{2\sqrt{2}}{\sqrt{5}}$$

$$\therefore\ ae=\sqrt{5}\times\frac{2\sqrt{2}}{\sqrt{5}}=2\sqrt{2}$$

Foci : $(\pm ae, 0)=\left(\pm 2\sqrt{2}, 0\right)$

$x - 1 = 1e \pm 2\sqrt{2}$; $y - 2 = 0$

$\Rightarrow$ $x=1\pm 2\sqrt{2}, y=2$

Required foci : $\left(1\pm 2\sqrt{2},\ 2\right)$

87. If the foci of an ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide, then the value of b^2 is

(1) 1 (2) 5 (3) 7 (4) 6

Sol. Answer (3)

Hyperbola $\frac{x^2}{\left(\frac{12}{5}\right)^2}-\frac{y^2}{\left(\frac{9}{5}\right)^2}=1$

$$\therefore e=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{1+\frac{\frac{81}{25}}{\frac{144}{25}}}=\frac{\sqrt{81+144}}{12}=\frac{15}{12}=\frac{5}{4}$$

$\because$ focus coincide

$\therefore$ $\underset{E}{(ae,\ 0)}=\underset{H}{(a'e',\ 0)}$

$$4\sqrt{1-\frac{b^2}{16}}=\frac{12}{5}\cdot\frac{5}{4}$$

$$\sqrt{1-\frac{b^2}{16}}=\frac{3}{4}$$

$$1-\frac{b^2}{16}=\frac{9}{16}$$

$$1-\frac{9}{16}=\frac{b^2}{16}$$

$$b^2=7$$

88. If $3x^2 - 3y^2 - 18x + 12y + 2 = 0$ represents a hyperbola, then the equation of directrices are

(1) $x=3\pm\sqrt{\frac{6}{13}}$ (2) $x=3\pm\sqrt{\frac{13}{6}}$ (3) $2\pm\sqrt{\frac{13}{6}}$ (4) $2\pm\sqrt{\frac{6}{13}}$

Sol. Answer (2)

Hyperbola $3x^2 - 3y^2 - 18x + 12y + 2 = 0$

$\Rightarrow$ $3(x^2 - 6x) - 3(y^2 - 4y) + 2 = 0$

$3(x - 3)^2 - 3(y - 2)^2 = 27 - 12 - 2 = 13$

$$\Rightarrow \frac{(x-3)^2}{\frac{13}{3}}-\frac{(y-2)^2}{\frac{13}{3}}=1$$

$$\therefore e=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{1+1}=\sqrt{2}$$

Directrices $x-3=\pm\frac{\sqrt{13}}{\sqrt{2}\sqrt{3}}=\pm\sqrt{\frac{13}{6}}$

$$\Rightarrow x=3\pm\sqrt{\frac{13}{6}}$$

[Tangents and Normals to Hyperbola]

89. The equation of a common tangent to the curves $y^2 = 8x$ and $xy = -1$ is

(1) $y = x + 2$ (2) $y = 2x + 1$ (3) $y=\frac{x}{2}+4$ (4) $y=3x+\frac{2}{3}$

Sol. Answer (1)

Equation of tangent to the parabola $y^2 = 8x$ is

$$y=mx+\frac{2}{m} \quad \ldots \text{(i)}$$

Equation (i) is also a tangent of $xy = -1$

$$\therefore \quad x\left(mx+\frac{2}{m}\right) = -1$$

$$mx^2 + \frac{2}{m}.x + 1 = 0$$

Condition of tangency, Discrimination of the equation = 0

$$\frac{4}{m^2} - 4m = 0$$

$\Rightarrow$ $m^3 = 1$

$m = 1$

Common tangent is $y = x + 2$

90. A normal to the parabola $y^2 = 4px$ with slope m touches the rectangular hyperbola $x^2 - y^2 = p^2$, if

(1) $m^6 - 4m^4 - 3m^2 + 1 = 0$ (2) $m^6 + 4m^4 + 3m^2 + 1 = 0$

(3) $m^6 + 4m^4 - 3m^2 + 1 = 0$ (4) $m^6 - 4m^4 + 3m^2 - 1 = 0$

Sol. Answer (2)

$y^2 = 4px$

Equation of normal is

$y = mx - 2pm - pm^3$ … (i)

Equation of tangent of $x^2 - y^2 = p^2$ at (p secθ, p tan) is $x\sec\theta - y\tan\theta = p$ … (ii)

Comparing (i) and (ii)

$$\frac{-m}{\sec\theta} = \frac{1}{-\tan\theta} = \frac{-2pm - pm^3}{p} = -2m - m^3$$

$$\Rightarrow \quad \frac{-m}{\sec\theta} = \frac{1}{-\tan\theta} = -2m - m^3$$

$$\Rightarrow \quad \frac{m}{\sec\theta} = \frac{1}{\tan\theta} = 2m + m^3 \qquad \Rightarrow \quad \sec\theta = \frac{m}{2m+m^3}$$

$$\Rightarrow \quad \tan\theta = \frac{1}{2m+m^3}$$

$1 + \tan^2\theta = \sec^2\theta$

$$\Rightarrow \quad 1+\frac{1}{(2m+m^3)^2} = \frac{(m)^2}{(2m+m^3)^2} \Rightarrow \quad (2m+m^3)^2 + 1 = m^2$$

$\Rightarrow$ $m^6 + 4m^2 + 4m^4 + 1 = m^2$

$\Rightarrow$ $m^6 + 4m^4 + 4m^2 - m^2 + 1 = 0$

$\Rightarrow$ $m^6 + 4m^4 + 3m^2 + 1 = 0$

91. The equation of the tangent to the hyperbola $3x^2 - 4y^2 = 12$, which makes equal intercepts on the axes is

(1) $x - y + 1 = 0$ (2) $x + y + 1 = 0$ (3) $x + y - 1 = 0$ (4) All are correct

Sol. Answer (4)

A line which makes equal intercepts having slope = –1

$y = -x + c$

Condition of tangency

$c = \pm\sqrt{a^2m^2 - b^2} = \pm\sqrt{4-3} = \pm 1$

$y = -x \pm 1 = y + x \pm 1 = 0$

92. The equation of the tangent to the hyperbola $3x^2 - 8y^2 = 24$ and perpendicular to the line $3x - 2y = 4$ is

(1) $3x + 2y \pm \sqrt{5} = 0$ (2) $2x + 3y \pm \sqrt{7} = 0$

(3) $2x + 3y \pm 2 = 0$ (4) $2x + 3y \pm \sqrt{5} = 0$

Sol. Answer (4)

A line perpendicular to given line

$2x + 3y = \lambda$

$m = -\dfrac{2}{3}$ $c = \dfrac{\lambda}{3}$

Condition of tangency

$c^2 = a^2m^2 - b^2$ $a^2 = 8$

$b^2 = 3$

$\dfrac{\lambda^2}{9} = 8.\dfrac{4}{9} - 3$

$\lambda^2 = 32 - 27 = 5$

$\lambda = \pm\sqrt{5}$

Tangents are $2x + 3y \pm \sqrt{5} = 0$

93. If the line $y = mx + \sqrt{a^2m^2 - b^2}$, $m = \dfrac{1}{2}$ touches the hyperbola $\dfrac{x^2}{16} - \dfrac{y^2}{3} = 1$ at the point $(4\sec\theta, \sqrt{3}\tan\theta)$, then θ is

(1) $\dfrac{\pi}{2}$ (2) $\dfrac{\pi}{4}$ (3) $\dfrac{2\pi}{3}$ (4) $\dfrac{\pi}{6}$

Sol. Answer (3)

Here $a = 4$, $b = \sqrt{3}$

$\therefore y = \dfrac{1}{2}x + \sqrt{16 \times \dfrac{1}{4} - 3}$

$\Rightarrow y = \frac{1}{2}x + 1$

$\Rightarrow \sqrt{3}\tan\theta = \frac{1}{2}.4\sec\theta + 1. = 2\sec\theta + 1$

$\Rightarrow 3\tan^2\theta = 4\sec^2\theta + 4\sec\theta + 1$

$\Rightarrow 3(\sec^2\theta - 1) - 4\sec^2\theta - 4\sec\theta - 1 = 0$

$\Rightarrow -\sec^2\theta - 4\sec\theta - 4 = 0$

$\Rightarrow (\sec\theta + 2)^2 = 0$

$\Rightarrow \sec\theta = -2 \Rightarrow \cos\theta = -\frac{1}{2} = \cos\frac{2\pi}{3}$

$\Rightarrow \theta = \frac{2\pi}{3}$

94. A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$ is

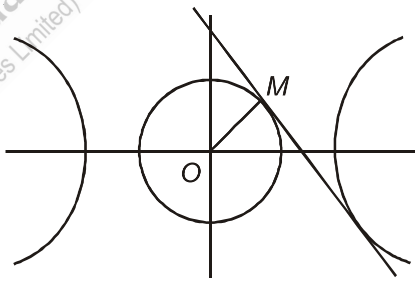
(1) $y\sqrt{7} = \sqrt{2}x + 15$ (2) $y\sqrt{7} = 3\sqrt{2}x + 15$ (3) $y = 3\sqrt{2}x + 15$ (4) $y\sqrt{7} = 3x + 15$

Sol. Answer (2)

Hyperbola : $9x^2 - 16y^2 = 144$

$\Rightarrow \frac{x^2}{16} - \frac{y^2}{9} = 1$

Equation of any tangent to the hyperbola Is



$y = mx + \sqrt{a^2m^2 - b^2}$

$\Rightarrow y = mx + \sqrt{16m^2 - 9}$

$OM = 3.$

$\Rightarrow \left|\frac{\sqrt{16m^2 - 9}}{\sqrt{m^2 + 1}}\right| = 3$

$\Rightarrow 16m^2 - 9 = 9(m^2 + 1)$

$\Rightarrow 7m^2 - 9 = 9$

$\Rightarrow 7m^2 = 18$

$\Rightarrow m^2 = \frac{18}{7}$

$\Rightarrow m \pm \frac{3\sqrt{2}}{\sqrt{7}}$

$\therefore y = mx \pm \sqrt{16m^2 - 9}$

$\Rightarrow y = \pm\dfrac{3\sqrt{2}}{\sqrt{7}}x + \sqrt{16.\dfrac{18}{7} - 9}$

$= \pm\dfrac{3\sqrt{2}}{\sqrt{7}}x + \dfrac{15}{\sqrt{7}}$

$\Rightarrow y\sqrt{7} = \pm 3\sqrt{2}x + 15$

$\Rightarrow y\sqrt{7} = 3\sqrt{2}x + 15$

[Locus and Miscellaneous Problems on Hyperbola]

95. The locus of a point $P(\alpha, \beta)$, such that the line $y = \alpha x + \beta$ is a tangent to $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$, is

(1) An ellipse (2) A circle (3) A parabola (4) A hyperbola

Sol. Answer (4)

Condition of tangent

$c^2 = a^2m^2 - b^2$

$b^2 = a^2\alpha^2 - b^2$

$\therefore$ Locus of $P(\alpha, \beta)$, $a^2x^2 - y^2 = b^2$ is a hyperbola

96. The locus of the middle points of the chords of hyperbola $\dfrac{x^2}{9} - \dfrac{y^2}{4} = 1$, which pass through the fixed point (1, 2) is a hyperbola whose eccentricity is

(1) $\dfrac{3}{2}$ (2) $\dfrac{\sqrt{7}}{2}$ (3) $\dfrac{\sqrt{13}}{2}$ (4) $\dfrac{\sqrt{15}}{2}$

Sol. Answer (3)

Let $P(h, k)$ be the mid point of the chord.

$\therefore$ Equation of chord $S' = T$

$\dfrac{h^2}{9} - \dfrac{k^2}{4} - 1 = \dfrac{x.h}{9} - \dfrac{y.k}{4} - 1$ passes through (1, 2)

$\dfrac{h^2}{9} - \dfrac{k^2}{4} = \dfrac{h}{9} - \dfrac{2k}{4}$

Locus of $P(h, k)$

$\dfrac{x^2}{9} - \dfrac{x}{9} - \dfrac{y^2}{4} + \dfrac{2y}{4} = 0$

$\Rightarrow \dfrac{\left(x - \frac{1}{2}\right)^2}{9} - \dfrac{(y-1)^2}{4} = -\dfrac{2}{9}$

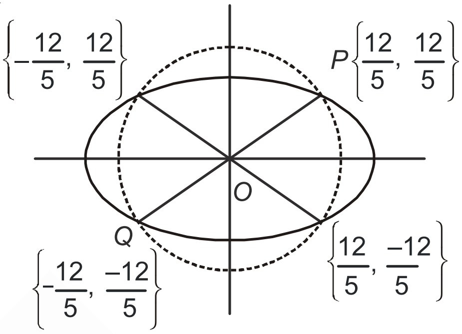
$\Rightarrow \dfrac{(y-1)^2}{\frac{8}{9}} - \dfrac{\left(x - \frac{1}{2}\right)^2}{2} = 1$ $\left[\begin{array}{l} a^2 = \dfrac{8}{9} \\ b^2 = 2 \end{array}\right]$

$\Rightarrow e^2 = 1 + \dfrac{b^2}{a^2} = \dfrac{13}{4}$ $\Rightarrow e = \dfrac{\sqrt{13}}{2}$

97. The radius of the circle passing through the point of intersection of ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2 - y^2 = 0$ is

(1) $\frac{12}{5}$ (2) $\frac{12\sqrt{2}}{5}$ (3) $\frac{6\sqrt{2}}{5}$ (4) $\frac{9\sqrt{2}}{5}$

Sol. Answer (2)



$E : \frac{x^2}{16}+\frac{y^2}{9}=1$

$L : x^2 = y^2$

$\therefore \quad \frac{x^2}{16}+\frac{x^2}{9}=1$

$\Rightarrow \quad \frac{9x^2+16x^2}{144}=1$

$\Rightarrow \quad x^2=\frac{144}{25}$

$\Rightarrow \quad x=\frac{12}{5}, \frac{-12}{5}=y$

$\therefore \quad PQ=\sqrt{\left(\frac{12}{5}+\frac{12}{5}\right)^2+\left(\frac{12}{5}+\frac{12}{5}\right)^2}$

$=\sqrt{\frac{576}{25}\times 2}$

$=\frac{24}{5}\times\sqrt{2}$

Radius of a circle = $\frac{1}{2}\times\frac{24}{5}\times\sqrt{2}=\frac{12\sqrt{2}}{5}$

98. Let $P(a \sec\theta, b \tan\theta)$ and $Q(a \sec\phi, b \tan\phi)$ where $\theta+\phi=\frac{\pi}{2}$, be two points on $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If (h, k) is the point of intersection of the normals at P and Q, then K is equal to

(1) $\frac{a^2+b^2}{a}$ (2) $-\left(\frac{a^2+b^2}{a}\right)$ (3) $\frac{a^2+b^2}{b}$ (4) $-\left(\frac{a^2+b^2}{b}\right)$

Sol. Answer (4)

Equation of normal at $(a \sec\theta, b \tan\theta)$

$ax \cos\theta + by \cot\theta = a^2 + b^2$ … (i)

Equation of normal at ϕ

$ax \cos\phi + by \cot\theta = a^2 + b^2$ $\qquad \phi=\frac{\pi}{2}-\theta$

$ax \sin\phi + by \tan\phi = a^2 + b^2$ … (ii)

(i) $\times$ sinθ (ii) $\times$ cosθ

$by[\cos\theta - \sin\theta] = (a^2 + b^2)[\sin\theta - \cos\theta]$

$\therefore \quad y = \dfrac{-(a^2+b^2)}{b}$

$\therefore \quad k = \dfrac{-(a^2+b^2)}{b}$

99. An ellipse has eccentricity $\dfrac{1}{2}$ and one focus at the point $P\left(\dfrac{1}{2}, 1\right)$. Its one directrix is the common tangent, nearer to the point P, to $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$. The equation of the ellipse is

(1) $\dfrac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}} + \dfrac{(y-1)^2}{\frac{1}{12}} = 1$

(2) $\dfrac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{8}} + \dfrac{(y-1)^2}{\frac{1}{12}} = 1$

(3) $\dfrac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}} + \dfrac{(y-1)^2}{\frac{1}{8}} = 1$

(4) None of these

Sol. Answer (1)

The directrix of the ellipse is x = 1

Clearly the equation of common tangent is $x = 1$

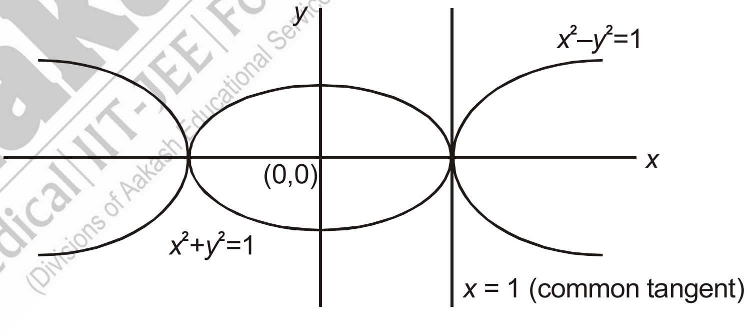
Equation of ellipse

$Ps = ePM$

$\Rightarrow Ps^2 = e^2PM^2$

$\Rightarrow \left(x-\dfrac{1}{2}\right)^2 + (y-1)^2 = \dfrac{1}{4}(x-1)^2$

$\Rightarrow \dfrac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}} + \dfrac{(y-1)^2}{\frac{1}{12}} = 1$



100. If the polars of (x_1, y_1) and (x_2, y_2) w.r.t. the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ are at right angles, then the value of $\dfrac{x_1x_2}{y_1y_2}$ is

(1) $\dfrac{a^4}{b^4}$ (2) $-\dfrac{a^4}{b^4}$ (3) $\dfrac{b^4}{a^4}$ (4) $-\dfrac{b^4}{a^4}$

Sol. Answer (2)

$H : \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$

Points : (x_1, y_1) and (x_2, y_2)

Polar at (x_1, y_1) is $\dfrac{xx_1}{a^2} - \dfrac{yy_1}{b^2} = 1$

$$\Rightarrow \frac{yy_1}{b^2} = \frac{xx_1}{a^2} - 1$$

$$\Rightarrow y = \frac{b^2 xx_1}{a^2 y_1} - \frac{b^2}{y_1}$$

$$\Rightarrow y = \left(\frac{b^2 x_1}{a^2 y_1}\right) x - \frac{b^2}{y_1}$$

$$\therefore m_1 = \frac{b^2 x_1}{a^2 y_1}$$

Similarly $m_2 = \dfrac{b^2 x_2}{b^2 y_2}$

Given $m_1 \times m_2 = -1$

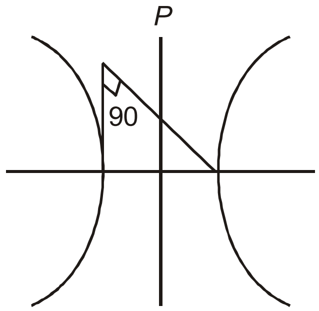
$$\Rightarrow \frac{b^2 x_1}{a^2 y_1} \times \frac{b^2 x_2}{a^2 y_2} = -1$$

$$\Rightarrow \frac{x_1 x_2}{y_1 y_2} = -\frac{a^4}{b^4}$$

101. If the locus of the point of intersection of two perpendicular tangents to a hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ is a circle with centre (0, 0), then the radius of a circle is

(1) 5 (2) 4 (3) 3 (4) 7

Sol. Answer (3)



Hyperbola : $\frac{x^2}{25} - \frac{y^2}{16} = 1$

Clearly, locus of P is a director circle

$\Rightarrow \quad x^2 + y^2 = 25 - 16 = 9$

$\therefore \quad x^2 + y^2 = (3)^2$

radius = 3

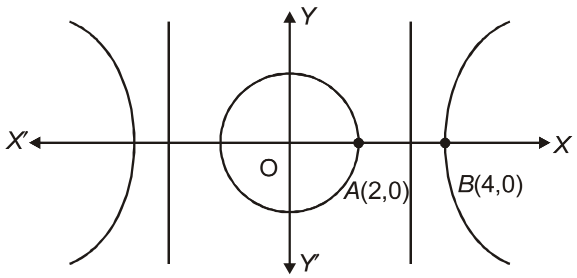
102. Shortest distance between the curves $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{x^2}{4} + \frac{y^2}{4} = 1$ is

(1) 1 (2) 2 (3) 3 (4) 4

Sol. Answer (2)

Shortest distance between the hyperbola and the circle is the distance between their common normal

Hence common normal is x-axis. So shortest distance = AB = 4 – 2 = 2



103. The centre of a rectangular hyperbola lies on the line $y = 3x$. If one of the asymptotes is $x + y + 2 = 0$, then the other asymptotes is

(1) $x + y + 1 = 0$ (2) $x - y + 1 = 0$ (3) $x - y + 2 = 0$ (4) $x + y + 3 = 0$

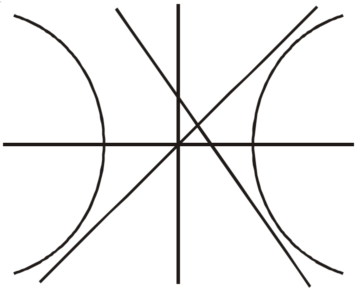
Sol. Answer (2)

The asymptotes of a rectangular hyperbola are perpendicular to each other. Given asymptotes $x + y + 2 = 0$.

Let its other asymptotes be $x - y + \lambda = 0$.

As we know that, asymptotes must pass through the centre of a hyperbola.

Thus $3x - y = 0$, $x + y + 2 = 0$ and $x - y + \lambda = 0$ are con-current



$$\therefore \begin{vmatrix} 3 & -1 & 0 \\ 1 & 1 & 2 \\ 1 & -1 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 3(\lambda + 2) + (\lambda - 2) = 0$$

$$\Rightarrow 3\lambda + 6 + \lambda - 2 = 0$$

$$\Rightarrow 4\lambda + 4 = 0$$

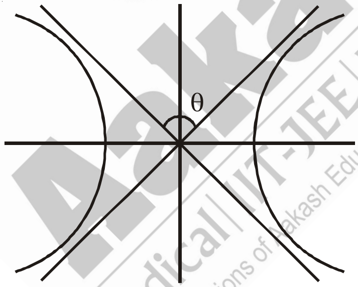
$$\Rightarrow \lambda = -1$$

$\therefore$ Equation of the other asymptotes is $x - y + 1 = 0$

104. If angle between the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

(1) $\tan^{-1}\left(\frac{3}{4}\right)$ (2) $2\tan^{-1}\left(\frac{3}{4}\right)$ (3) $\tan^{-1}\left(\frac{4}{3}\right)$ (4) $2\tan^{-1}\left(\frac{4}{3}\right)$

Sol. Answer (2)



$$H : \frac{x^2}{16} - \frac{y^2}{9} = 1$$

$$A : \frac{x^2}{16} - \frac{y^2}{9} = 0$$

$$\Rightarrow \frac{y^2}{9} = \frac{x^2}{16}$$

$$\Rightarrow y = \pm\frac{3}{4}x$$

$$\therefore m_1 = \frac{3}{4}, m_2 = -\frac{3}{4}$$

Let angle between two asymptotes be θ.

$$\therefore \tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\frac{3}{4} + \frac{3}{4}}{1 - \frac{3}{4}.\frac{3}{4}} = \frac{2.\frac{3}{4}}{1 - \left(\frac{3}{4}\right)^2}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{2.\frac{3}{4}}{1 - \left(\frac{3}{4}\right)^2}\right) = 2\tan^{-1}\left(\frac{3}{4}\right)$$

105. Area of the triangle formed by any arbitrary tangents of the hyperbola $xy = 4$, with the co-ordinate axes is

(1) 2 (2) 4 (3) 6 (4) 8

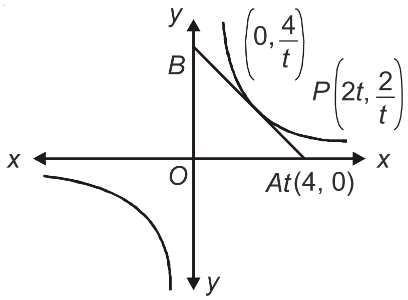
Sol. Answer (4)

Tangent as P

$xy_1 + x_1y = 8$

$\Rightarrow x.\dfrac{2}{t} + y.2t = 8$

$\Rightarrow \dfrac{x}{t} + yt = 4$



$\text{ar}(\Delta\, OAB) = \dfrac{1}{2} \times 4t \times \dfrac{4}{t}$

$= 8$ sq units.

106. If the normal to the rectangular hyperbola $xy = 4$ at the point $\left(2t_1, \dfrac{2}{t_1}\right)$ meets the curve again at $\left(2t_2, \dfrac{2}{t_2}\right)$, then

(1) $t_1{}^3t_2 = 1$ (2) $t_1{}^3t_2 = -1$ (3) $t_2{}^3t_1 = 1$ (4) $t_1t_2{}^3 = -1$

Sol. Answer (2)

$H : xy = 4$

$\Rightarrow y = \dfrac{4}{x}$

$\Rightarrow \dfrac{dy}{dx} = -\dfrac{4}{x^2}$

$\Rightarrow m = \left.\dfrac{dy}{dx}\right|_{\left(2t_1, \frac{2}{t_2}\right)} = \dfrac{-4}{4t_1^2} = \dfrac{-1}{t_1^2}$

Normal at $\left(2t_1, \dfrac{2}{t_1}\right)$ is $y - \dfrac{2}{t_1} = t_1^2(x - 2t_1)$

Which is passing through $\left(2t_2, \dfrac{2}{t_2}\right)$

$\dfrac{2}{t_2} - \dfrac{2}{t_1} = t_1^2(2t_2 - 2t_1)$

$\Rightarrow 2\left(\dfrac{t_1 - t_2}{t_1t_2}\right) = 2t_1^2(t_2 - t_1)$

$\Rightarrow \dfrac{1}{t_1t_2} = -t_1^2$

$\Rightarrow t_1^3\ t_2 = -1$

107. If the circle $x^2 + y^2 = 4$ intersect the hyperbola $xy = 4$ in four point (x_i, y_i), $i = 1, 2, 3, 4$ then

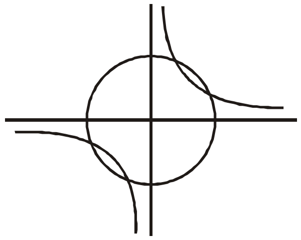
(1) $x_1x_2x_3x_4 = 4$ (2) $x_1x_2x_3x_4 = 16$ (3) $x_1x_2x_3x_4 = 8$ (4) $x_1x_2x_3x_4 = 32$

Sol. Answer (2)

$x^2 + y^2 = 4$, $xy = 4$

$\Rightarrow \quad x^2 + \dfrac{16}{x^2} = 4$

$\Rightarrow \quad x^2 - 4x^2 + 16 = 0$



Let its roots be x_1, x_2 x_3, x_4.

$\therefore \quad x_1 + x_2 + x_3 + x_4 = 0$

$x_1x_2x_3x_4 = 16$

[Miscellaneous Problems on Parabola, Ellipse and Hyperbola]

108. A hyperbola, having the transverse axis of length 2 sinθ, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is **[IIT-JEE 2007]**

(1) $x^2 \operatorname{cosec}^2\theta - y^2 \sec^2\theta = 1$ (2) $x^2 \sec^2\theta - y^2 \operatorname{cosec}^2\theta = 1$

(3) $x^2 \sin^2\theta - y^2 \cos^2\theta = 1$ (4) $x^2 \cos^2\theta - y^2 \sin^2\theta = 1$

Sol. Answer (1)

$3x^2 + 4y^2 = 12$

$\Rightarrow \dfrac{x^2}{4} + \dfrac{y^2}{3} = 1$

Foci of this ellipse are (–1, 0) and (1, 0)

Now, $2ae = 2$, $2a = 2\sin\theta$

a being the length of semi-transverse axis, 'b' the length of semi-conjugate axis can be calculated using $b^2 = a^2(e^2–1)$

109. Let a and b non-zero real numbers. Then, the equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents **[IIT-JEE 2008]**

(1) Four straight lines, when $c = 0$ and a, b are of the same sign

(2) Two straight lines and a circle, when $a = b$, and c is of sign opposite to that of a

(3) Two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a

(4) A circle and an ellipse, when a and b are of the same sign and c is of sign opposite to that of a

Sol. Answer (2)

Let a and b be non-zero real numbers.

The equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ implies either

$x^2 - 5xy + 6y^2 = 0$

$\Rightarrow \quad (x - 2y)(x - 3y) = 0$

$\Rightarrow \quad x = 2y$ and $x = 3y$

$\Rightarrow \quad x^2 - 5xy + 6y^2 = 0$ represents two straight lines passing through origin.

or $\quad ax^2 + by^2 + c = 0$

When $c = 0$ and a and b are of same signs then

$ax^2 + by^2 + c = 0 \Rightarrow x = 0$ and $y = 0$

which is a point specified as the origin.

When $a = b$ and c is of sign opposite of that of a, $ax^2 + by^2 + c = 0$ represents a circle.

110. Consider a branch of the hyperbola

$$x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$$

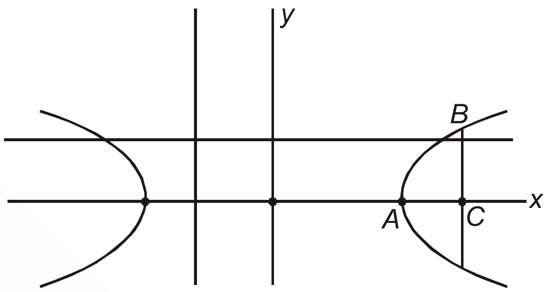
with vertex at the point A. Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A, then the area of the triangle ABC is **[IIT-JEE 2008]**

(1) $1-\sqrt{\frac{2}{3}}$ (2) $\sqrt{\frac{3}{2}}-1$ (3) $1+\sqrt{\frac{2}{3}}$ (4) $\sqrt{\frac{3}{2}}+1$

Sol. Answer (2)

$$\frac{(x-\sqrt{2})^2}{4} - \frac{(y+\sqrt{2})^2}{2} = 1$$

for A(x, y)



$$e = \sqrt{1+\frac{2}{4}} = \sqrt{\frac{3}{2}}$$

$$x - \sqrt{2} = 2 \Rightarrow x = 2 + \sqrt{2}$$

For C(x, y)

$$x - \sqrt{2} = ae = \sqrt{6}$$

$$x = \sqrt{6} + \sqrt{2}$$

AC = $\sqrt{6} + \sqrt{2} - 2 - \sqrt{2} = \sqrt{6} - 2$

$$BC = \frac{b^2}{a} = \frac{2}{2} = 1$$

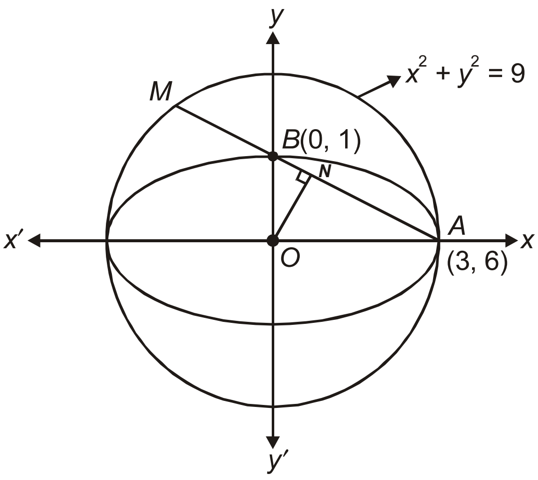
Area = $\frac{1}{2} \times BC \times (\sqrt{6}-2) = \frac{1}{2} \times 1 \times (\sqrt{6}-2) = \sqrt{\frac{3}{2}} - 1$

111. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices at A, M and the origin O is **[IIT-JEE 2009]**

(1) $\frac{31}{10}$ (2) $\frac{29}{10}$ (3) $\frac{21}{10}$ (4) $\frac{27}{10}$

Sol. Answer (4)

The given ellipse is



$$x^2 + 9y^2 = 9 \Rightarrow \frac{x^2}{9} + \frac{y^2}{1} = 1$$

Now, equation of auxiliary circle is $x^2 + y^2 = 9$

AM is chord of circle, $x^2 + y^2 = 9$

The equation of AM is $\frac{x}{3} + \frac{y}{1} = 1 \Rightarrow x + 3y = 3$

Now, perpendicular distance $ON = \left|\frac{3}{\sqrt{10}}\right|$

$$NA = \sqrt{9 - \frac{9}{10}} = \sqrt{\frac{81}{10}} = \frac{9}{\sqrt{10}}$$

$AM = 2\times\frac{9}{\sqrt{10}}$

Required area of $\Delta AOM = \frac{1}{2}\times 2\times\frac{9}{\sqrt{10}}\times\frac{3}{\sqrt{10}} = \frac{27}{10}$

112. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x-axis at Q. If M is the mid point of the line segment PQ, then the locus of M intersects the latus rectums of the given ellipse at the points

[IIT-JEE 2009]

(1) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$ (2) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$ (3) $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$ (4) $\left(\pm 2\sqrt{3}, \pm\frac{4\sqrt{3}}{7}\right)$

Sol. Answer (3)

The given equation of ellipse is $\frac{x^2}{16}+\frac{y^2}{4}=1$

Any point on ellipse (4 cosθ, 2 sinθ) may be taken as

The equation of normal at (4 cosθ, 2 sinθ) is given by

$4\sec\theta\, x - 2y\,\mathrm{cosec}\theta = 16 - 4$

$\Rightarrow$ $2\sec\theta x - y\,\mathrm{cosec}\theta = 6$ …(i)

$Q = (3\cos\theta, 0)$

Now $M = (h, k)$

$h = \frac{3\cos\theta + 4\cos\theta}{2}$ $\Rightarrow$ $\cos\theta = \frac{2h}{7}$

$k = \sin\theta$ $\Rightarrow$ $\sin\theta = 1$

$\frac{4h^2}{49} + k^2 = 1$

Locus of M is

$4x^2 + 49y^2 = 49$

Equation of latus rectum is $x = \pm 2\sqrt{3}$

at $x = \pm 2\sqrt{3}$, $y^2 = 49 - 48 = 1$

$y = \pm\frac{1}{7}$

113. Let $P(6, 3)$ be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If the normal at the point P intersects the x-axis at (9, 0), then the eccentricity of the hyperbola is **[IIT-JEE 2011]**

(1) $\sqrt{\frac{5}{2}}$ (2) $\sqrt{\frac{3}{2}}$ (3) $\sqrt{2}$ (4) $\sqrt{3}$

Sol. Answer (2)

The equation of the normal at $P(6, 3)$ to the given hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

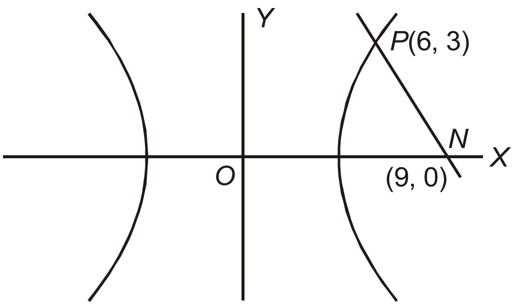
$\frac{a^2 x}{6}+\frac{b^2 y}{3} = a^2 + b^2$

which meets x-axis at (9, 0), hence

$$\frac{9a^2}{6} = a^2 + b^2$$

$$\Rightarrow \frac{3}{2} = 1 + \frac{b^2}{a^2} = e^2$$

$$\Rightarrow e = \frac{\sqrt{3}}{2}$$



114. Let (x, y) be any point on the parabola $y^2 = 4x$. Let P be the point that divides the line segment from (0, 0) to (x, y) in the ratio 1 : 3. Then the locus of P is **[IIT-JEE 2011]**

(1) $x^2 = y$ (2) $y^2 = 2x$

(3) $y^2 = x$ (4) $x^2 = 2y$

Sol. Answer (3)

Let $P(\alpha, \beta)$ be the point intersecting the line-segment joining $O(0, 0)$ and $Q(t^2, 2t)$ in the ratio 1 : 3.

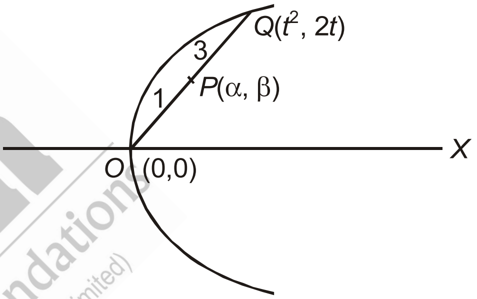
Then $\alpha = \dfrac{t^2 + 0}{4}$, $\beta = \dfrac{2t + 0}{4}$

$$\Rightarrow \alpha = \frac{t^2}{4}, \quad \beta = \frac{t}{2}$$

$$\Rightarrow 4\alpha = (2\beta)^2$$

$$\Rightarrow \beta^2 = \alpha$$

Locus of (α, β) is $y^2 = x$.



115. The ellipse $E_1 : \dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse E_2 passing through the point (0, 4) circumscribes the rectangle R. The eccentricity of the ellipse E_2 is **[IIT-JEE 2012]**

(1) $\dfrac{\sqrt{2}}{2}$ (2) $\dfrac{\sqrt{3}}{2}$ (3) $\dfrac{1}{2}$ (4) $\dfrac{3}{4}$

Sol. Answer (3)

Let the equation of E_2 be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Clearly the point P has coordinates (3, 2)

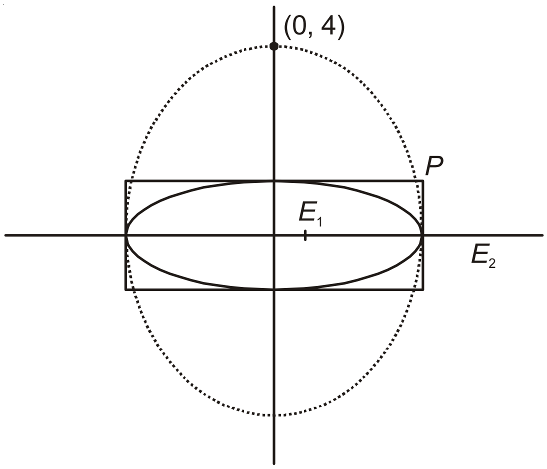
So, (3, 2) and (0, 4) satisfy E_2

$$\frac{16}{b^2} = 1 \Rightarrow b^2 = 16$$

and $\dfrac{9}{a^2} + \dfrac{4}{b^2} = 1$

$$\frac{9}{a^2} = \frac{3}{4} \Rightarrow a^2 = 12$$

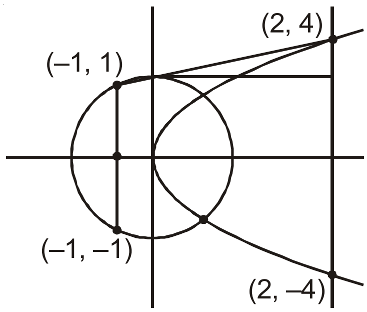
$$e = \sqrt{1 - \frac{12}{16}} = \frac{1}{2}$$



116. The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the poitns R, S. Then the area of the quadrilateral $PQRS$ is **[JEE(Advanced)-2014]**

(1) 3 (2) 6 (3) 9 (4) 15

Sol. Answer (4)



$x^2 + y^2 = 2$

Equation of tangent to circle $x^2 + y^2 = 2$ is $y = mx \pm \sqrt{2}\sqrt{1+m^2}$

Equation of tangent to $y^2 = 8x$ is $y = mx + \dfrac{2}{m}$

$$\sqrt{2}(\sqrt{1+m^2}) = \frac{2}{m}$$

$$2(1+m^2) = \frac{4}{m^2}$$

$$m^2(1 + m^2) = 2$$

$$m^4 + m^2 - 2 = 0$$

$$(m^2 + 2)(m^2 - 1) = 0$$

$$m = \pm 1$$

$$y = x + 2$$

$$y = -x - 2$$

Area $= \left(3\times 1+\dfrac{1}{2}\times 3\times 3\right)\times 2 = 6 + 9 = 15$ sq. unit

SECTION - B

Objective Type Questions (More than one options are correct)

1. Equation $y^2 - 2x - 2y + 5 = 0$ represents

(1) A pair of straight line (2) A circle with centre (1, 1)

(3) A parabola with vertex (2, 1) (4) A parabola with directrix $x = \dfrac{3}{2}$

Sol. Answer (3, 4)

$y^2 - 2y = 2x - 5$

$(y - 1)^2 = 2(x - 2)$ represents a parabola with vertex (2, 1) and directrix $x - 2 = -\dfrac{1}{2}$

$$x = \frac{3}{2}$$

2. The locus of the mid-point of the focal radii of a variable point moving on the parabola, $y^2 = 4ax$ is a parabola whose

(1) Focus has the coordinates $(a, 0)$

(2) Directrix is $x = 0$

(3) Vertex is $\left(\dfrac{a}{2},\ 0\right)$

(4) Latus rectum is half the latus rectum of the original parabola

Sol. Answer (1, 2, 3, 4)

$S(a, 0)$ is a focus

Moving point $Q(x, y)$

Let $P(h, k)$ be mid-point of SQ

$$h = \frac{a+x}{2}, k = \frac{y}{2}$$

$x = 2h - a, y = 2k$

$Q(x, y)$ will satisfy the $y^2 = 4ax$

$\therefore\quad 4k^2 = 4a(2h - a)$

$\therefore\quad$ Locus of $P(h, k)$

$$y^2 = 2a\left(x - \frac{a}{2}\right)$$

Shifting origin at $\left(\frac{a}{2}, 0\right)$

$\therefore\quad y^2 = 2ax$

Length of latus rectum = $2a$

Vertex $\left(\frac{a}{2}, 0\right)$

Equation of directrix $x = -\frac{a}{2}$

$\Rightarrow\quad x - \frac{a}{2} = -\frac{a}{2}$

$\Rightarrow\quad x = 0$

Focus of parabola

Focus with respect to original axis $(a, 0)$

3. If tangents PA and PB are drawn from $P(-1, 2)$ to $y^2 = 4x$ then

(1) Equation of AB is $y = x - 1$

(2) Length of AB is 8

(3) Length of AB is 4

(4) Equation of AB is $y = x + 1$

Sol. Answer (1, 2)

Equation of chord of contact $T = 0$

$\therefore\quad y.2 = 2(x - 1)$

$\Rightarrow\quad y = x - 1 \quad \ldots \text{(i)}$

$\Rightarrow\quad y^2 = 4x \quad \ldots \text{(ii)}$

Solve equation (i) and (ii) $A(3+2\sqrt{2}, 2+2\sqrt{2})$

$B(3-2\sqrt{2}, 2-2\sqrt{2})$

$AB = \sqrt{(4\sqrt{2})^2 + (4\sqrt{2})^2} = \sqrt{64} = 8$

4. Two parabolas have the same focus. If their directrices are the x-axis and the y-axis respectively, then the slope of their common chord is

(1) 1 (2) –1 (3) $\frac{3}{4}$ (4) $\frac{4}{3}$

Sol. Answer (1, 2)

Let focus be (a, b) and directrix $x = 0$

$\therefore$ Vertex of parabola $\left(\frac{a}{2}, b\right)$

$\therefore$ Equation of parabola

$$(y-b)^2 = 2a\left(x-\frac{a}{2}\right) \quad \ldots \text{(i)}$$

Similarly equation of other parabola

$$(x-a)^2 = 2b\left(y-\frac{b}{2}\right) \quad \ldots \text{(ii)}$$

Equation of common chord (i) – (ii)

$y^2 = x^2$

$y = \pm x$

$\therefore$ Slope of common chord are 1 and –1

5. Equation of a common tangent to the circle $x^2 + y^2 = 18$ and $y^2 = 24x$ is

(1) $x + y + 6 = 0$ (2) $x - y + 6 = 0$ (3) $x + y - 6 = 0$ (4) $x - y - 6 = 0$

Sol. Answer (1, 2)

Equation of tangent to $y^2 = 24x$

be $y = mx + \frac{6}{m}$ ... (i)

Line (i) is a tangent to the circle

$\therefore$ Distance from centre = radius

$\Rightarrow \left|\frac{\frac{6}{m}}{\sqrt{m^2+1}}\right| = \sqrt{18}$

$\Rightarrow \left(\frac{6}{m}\right)^2 = 18(m^2+1)$

$\Rightarrow 2 = m^2(m^2+1)$

$\Rightarrow m^4 + m^2 - 2 = 0$

$\Rightarrow (m^2 + 2)(m^2 - 1) = 0$

$\Rightarrow m = \pm 1$

$\therefore x - y + 6 = 0$

$x + y + 6 = 0$

6. The normals to parabola $y^2 = 4ax$ from the point $(5a, -2a)$ are

(1) $y = x - 3a$ (2) $y + 2x = 12a$

(3) $y = -3x + 33a$ (4) $y = x + 3a$

Sol. Answer (1, 2)

Any normal to parabola $y^2 = 4ax$ may be

$y - 2at = -t\{x - at^2\}$

Or $y + tx - 2at - at^3 = 0$ ...(i)

$(5a, -2a)$ lies on (i), then

$t^3 - 3t + 2 = 0$

$\Rightarrow$ $t = -1, 2, -1$

$\therefore$ Equation of normals are

$y = x - 3a$ and $y + 2x = 12a$

7. Equation $x^2 - 2x - 2y + 5 = 0$ represents

(1) Parabola with vertex (1, 1) (2) Parabola with vertex (1, 2)

(3) Parabola with directrix, $y = 5/2$ (4) Parabola with directrix, $y = -1/2$

Sol. Answer (2, 3)

Given, $x^2 - 2x - 2y + 5 = 0$

$\Rightarrow$ $(x - 1)^2 = 2\{y - 2\}$ ...(i)

Equation (i) is a parabola whose vertex is (1, 2)

Its directrix is $y - 2 = a = \dfrac{1}{2}$ or $y = \dfrac{5}{2}$

8. The coordinates of a focus of the ellipse $4x^2 + 9y^2 = 1$ are

(1) $\left(\dfrac{\sqrt{5}}{6}, 0\right)$ (2) $\left(-\dfrac{\sqrt{5}}{6}, 0\right)$

(3) $\left(\dfrac{\sqrt{5}}{3}, 0\right)$ (4) $\left(-\dfrac{\sqrt{5}}{3}, 0\right)$

Sol. Answer (1, 2)

$\dfrac{x^2}{\frac{1}{4}} = \dfrac{Y^2}{\frac{1}{9}} = 1$ $a^2 = \dfrac{1}{4},\ b^2 = \dfrac{1}{9}$

$\Rightarrow$ $e^2 = 1 - \dfrac{b^2}{a^2} = 1 - \dfrac{4}{9} = \dfrac{5}{9}$

$\Rightarrow$ $e = \dfrac{\sqrt{5}}{3}$

Foci $(\pm\, ae, 0) = \left(\pm\dfrac{\sqrt{5}}{6}, 0\right)$

9. On the ellipse $4x^2 + 9y^2 = 1$, the points at which the tangents are parallel to $8x = 9y$ are

(1) $\left(\frac{2}{5}, \frac{1}{5}\right)$ (2) $\left(-\frac{2}{5}, \frac{1}{5}\right)$ (3) $\left(-\frac{2}{5}, -\frac{1}{5}\right)$ (4) $\left(\frac{2}{5}, -\frac{1}{5}\right)$

Sol. Answer (2, 4)

Equation of line parallel to $8x = 9y$ is $y = \frac{8}{9}x + c$

Condition of tangency

$$c^2 = a^2m^2 + b^2$$

$$c^2 = \frac{1}{4}.\frac{64}{81} + \frac{1}{9} \quad \Rightarrow \quad c = \pm\frac{5}{9}$$

$\therefore \quad y = \frac{8}{9}x \pm \frac{5}{9}$

$8x - 9y \pm 5 = 0$ … (i)

Let point of contact be $P(h, k)$

$\therefore$ Equation of tangent at $P(h, k)$

$4.x.h + 9.y.k - 1 = 0$ … (ii)

Line (i) and (ii) are coincident line

$\therefore \quad \frac{4h}{8} = \frac{9k}{-9} = \frac{-1}{\pm 5}$

$$h = \mp\frac{2}{5}, k = \pm\frac{1}{5}$$

Point of contact $\left(\frac{-2}{5}, \frac{1}{5}\right)$ and $\left(\frac{2}{5}, -\frac{1}{5}\right)$

10. The focal distances of the point $(4\sqrt{3}, 5)$ on the ellipse $25x^2 + 16y^2 = 1600$ may be

(1) 7 (2) 6 (3) 13 (4) 11

Sol. Answer (1, 3)

$$\frac{x^2}{64} + \frac{y^2}{100} = 1$$

$$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{64}{100} = \frac{36}{100}$$

$$e = \frac{3}{5}$$

Focal distance = $a \pm ey$ = $10 \pm \frac{3}{5}(5) = 13, 7$

11. Let the ellipse be $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the circle $x^2 + y^2 = 16$. Then

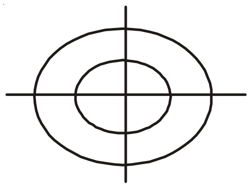
(1) Number of common tangents is zero

(2) Number of distinct common normals is two

(3) Area of the bounded region is 16π

(4) Angle between the two curves is 60°

Sol. Answer (1, 2)



$E:\frac{x^2}{9}+\frac{y^2}{4}=1;$

$C: x^2 + y^2 = 16$

From the figure, it is clear that, no common tangent can be drawn but 2 distinct normals X and Y axes can be drawn.

12. Let P be a variable point on the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ with foci at F_1 & F_2

(1) Area of ΔPF_1F_2 is $12\sin\theta$

(2) Area of ΔPF_1F_2 is maximum when $\theta=\frac{\pi}{2}$

(3) Co-ordinates of P are (0, 4)

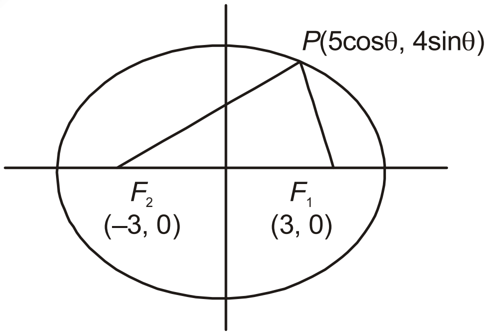
(4) Centre of the ellipse is (1, 2)

Sol. Answer (1, 2, 3)

$E:\frac{x^2}{25}+\frac{y^2}{16}=1$

$e=\sqrt{1-\frac{16}{25}}=\frac{3}{5}$

$ae=5\times\frac{3}{5}=3$



$$ar(\Delta PF_1F_2)=\left|\frac{1}{2}\begin{vmatrix}5\cos\theta & 4\sin\theta & 1\\ 3 & 0 & 1\\ -3 & 0 & 1\end{vmatrix}\right|$$

$$=\left|-\frac{1}{2}\times 6\times 4\sin\theta\right|$$

$$= 12\sin\theta$$

Max. ar$(\Delta PF_1F_2) = 12$

Co-ordinates of $P\equiv\left(5\cdot\cos\frac{\pi}{2}, 4\sin\frac{\pi}{2}\right)$

$\equiv (0, 4)$

13. Tangents are drawn to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ at the ends of a latus rectum then

(1) Co-ordinates of one end of a latus rectum are $\left(2,\frac{5}{3}\right)$ (2) Equation of tangent is $\frac{2}{9}x+\frac{y}{3}=1$

(3) Area of a quadrilateral so formed is 27 sq. units (4) Centre of the ellipse is (2, 3)

Sol. Answer (1, 2, 3)

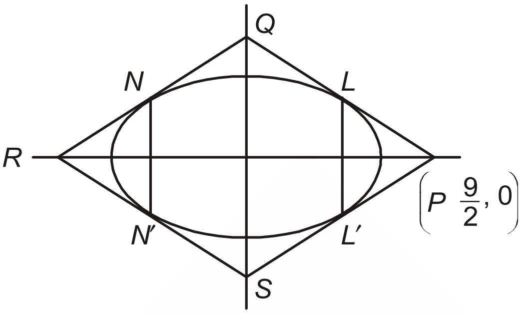
$E:\frac{x^2}{9}+\frac{y^2}{5}=1$

$\therefore \quad e-\sqrt{1-\frac{5}{9}}=\frac{2}{3}$

$ae=3\times\frac{2}{3}=2$

$\frac{b^2}{a}=\frac{5}{3}$

$L\equiv\left(ae,\frac{b^2}{a}\right)\equiv\left(2,\frac{5}{3}\right)$



Tangent at L is $\frac{xx_1}{9}+\frac{yy_1}{5}=1$

$\Rightarrow \quad \frac{2x}{9}+\frac{y}{3}=1$

$\therefore \quad P:\left(\frac{9}{2},0\right),\ Q:(0,3)$

Area ($\square$ $PQRS$) = 4 × ar(ΔPOQ) = $4\times\frac{1}{2}\times3\times\frac{9}{2}=27$ sq. units.

14. The equation of common tangent of the curve $x^2+4y^2=8$ and $y^2=4x$ are

(1) $x-2y+4=0$ (2) $x+2y+4=0$ (3) $2x-y+4=0$ (4) $2x+y+4=0$

Sol. Answer (1, 2)

$E:\frac{x^2}{8}+\frac{y^2}{2}=1$

$P: y^2=4x$

Equations of any tangent to the parabola is

$$y=mx+\frac{a}{m}=mx+\frac{1}{m} \quad \text{...(i)}$$

$E:\frac{x^2}{8}+\frac{y^2}{2}=1.\ ;\ T: y=mx+\frac{1}{m}$

$c^2=a^2m^2+b^2$

$\Rightarrow \quad \frac{1}{m^2}=8m^2+2$

$\Rightarrow \; 8m^4 + 2m^2 - 1 = 0$

$\Rightarrow \; 8m^4 + 4m^2 - 2m^2 - 1 = 0$

$\Rightarrow \; 4m^2(2m^2 + 1) - 1\,(2m^2 + 1) = 0$

$\Rightarrow \; 4m^2 = 1$

$\Rightarrow \; m = \pm\dfrac{1}{2}$

$T : y = \pm\dfrac{1}{2}x \pm 2$

$\Rightarrow \; 2y = \pm\, x \pm 4$

$\Rightarrow \; x - 2y + 4 = 0,\; x + 2y + 4 = 0$

15. Let the equation of the ellipse be $2x^2 + y^2 = 4$ and the point is (2, 1)

(1) Equation of tangent is $4x + y = 4$

(2) Equation of normal is $x - 4y + 2 = 0$

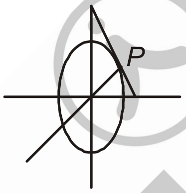
(3) Equation of polar is $4x + y = 4$

(4) Equation of diameter is $x - 2y = 0$

Sol. Answer (1, 2, 3, 4)

$E : 2x^2 + y^2 = 4$

$\Rightarrow \; \dfrac{x^2}{2} + \dfrac{y^2}{4} = 1$



Tangent at $P : \dfrac{xx_1}{2} + \dfrac{yy_1}{4} = 1$

$\dfrac{2x}{2} + \dfrac{y}{4} = 1$

$\Rightarrow \; 4x + y = 4$

Normal at $P : x - 4y = \lambda$

It passes through (2, 1), then $\lambda = 2 - 4 = -2$

$\therefore \; x - 4y + 2 = 0$

Polar at $P : 4x + y = 4$

Diameter $y = \dfrac{1}{2}x$

$\Rightarrow \; x - 2y = 0$

16. Chord of contact of tangents drawn from the point $M(h, k)$ to the ellipse $x^2 + 4y^2 = 4$ intersects at P & Q, subtends a right angle of the centre 'O'

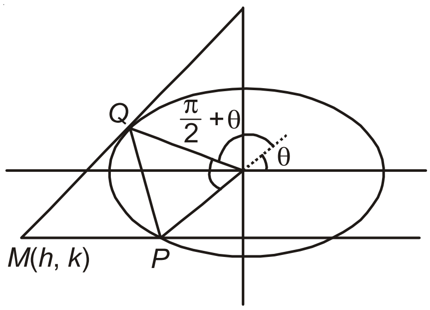
(1) Locus of M is $x^2 + 16y^2 = 20$

(2) $\dfrac{1}{OP^2} + \dfrac{1}{OQ^2} = \dfrac{5}{4}$

(3) Equation of chord of contact is $hx + 4ky = 4$

(4) Equation of normal is $3x + 2y + 5 = 0$

Sol. Answer (1, 2, 3)



$E : x^2 + 4y^2 = 4 \Rightarrow \frac{x^2}{4} + \frac{y^2}{1} = 1$

PQ : chord of contact

$hx + 4ky = 4$

$\Rightarrow \quad \frac{hx + 4ky}{4} = 1$

Now, $x^2 + 4y^2 = 4\left(\frac{hx + 4ky}{4}\right)^2$

$\Rightarrow \quad 4(x^2 + 4y^2) = (hx + 4ky)^2 = h^2x^2 + 16k^2y^2 + 8hkxy$

$\Rightarrow \quad (4 - h^2)x^2 + (16 - 16k^2)y^2 + (\ ) = 0$

Chord of contact subtends a right angle at the centre, so $4 - h^2 + 16 - 16k^2 = 0$

Locus of M is $x^2 + 16y^2 = 20$

$\Rightarrow \quad h^2 + 16k^2 = 20$

Let OP makes an angle θ with the x-axis then OQ makes an angle $\frac{\pi}{2} + \theta$

$\therefore \quad P : (OP\cos\theta,\ OP\sin\theta)$

$Q : (-OQ\sin\theta,\ OQ\cos\theta)$

Since P & Q lies on the ellipse, so we have

$OP^2\cos^2\theta + 4.OP^2\sin^2\theta = 4$

$\Rightarrow \quad \frac{4}{OP^2} = \cos^2\theta + 4\sin^2\theta$

Also $OQ^2\sin^2\theta + 4.OQ^2\cos^2\theta = 4$

$\Rightarrow \quad \frac{4}{OQ^2} = \sin^2\theta + 4\cos^2\theta$

Now, $\frac{4}{OP^2} + \frac{4}{OQ^2} = (\cos^2\theta + 4\sin^2\theta) + (\sin^2\theta + 4\cos^2\theta) = 5(\cos^2\theta + \sin^2\theta) = 5$

$\Rightarrow \quad \frac{1}{OP^2} + \frac{1}{OQ^2} = \frac{5}{4}$

17. Let the ellipse be $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and P is one end of a minor axis of the ellipse. Then

(1) Equation of the incident ray is $4x - 3y - 12 = 0$

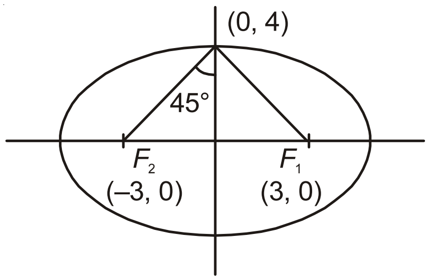
(2) Equation of the reflection ray is $4x + 3y + 12 = 0$

(3) Angle between the focal radii normal at P is 45°

(4) Distance between two foci is 6

Sol. Answer (1, 2, 3, 4)

$$E:\frac{x^2}{25}+\frac{y^2}{16}=1$$



$$\therefore \quad e=\sqrt{1-\frac{16}{25}}=\frac{3}{5}$$

$F_1 = (ae, 0)$

$= (3, 0)$

$F_2 = (–3, 0)$

$\therefore \quad F_1F_2 = 3 + 3 = 6$

Equation of the incident ray is

$$y=0=\frac{4}{3}(x-3)$$

$\Rightarrow \quad 3y = 4x – 12$

$\Rightarrow \quad 4x – 3y = 12$

Equation of the reflection ray is

$$y-0=-\frac{4}{3}(x+3)$$

$\Rightarrow \quad 3y = – 4x – 12$

$\Rightarrow \quad 4x + 3y + 12 = 0$

Angle between the focal radii and normal is 45°

18. If a quadrilateral formed by four tangents to the ellipse $3x^2 + 4y^2 = 12$ is a square, then

(1) The vertices of the square lie on $y = \pm x$

(2) The vertices of the square lie on $x^2 + y^2 = 7^2$

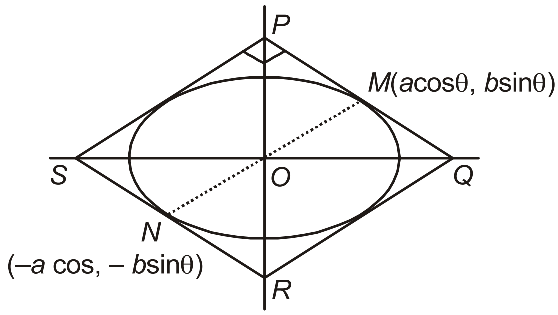
(3) The area of all such squares is constant

(4) Only two such squares are possible

Sol. Answer (2, 3)

$E : 3x^2 + 4y^2 = 12$

$$\Rightarrow \quad \frac{x^2}{4}+\frac{y^2}{3}=1$$



Clearly, the vertices of the squares will lie on the director circle of the ellipse

i.e., $x^2 + y^2 = 4 + 3 = 7$

Area of a square *PQRS*

$= 2 (a^2 + b^2)$

$= 2(4 + 3)$

$= 14$ sq. units

Only one such square is possible.

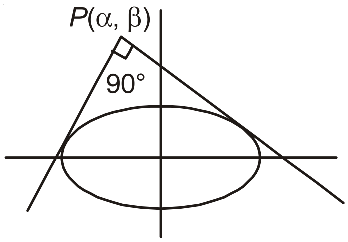
19. Let (α, β) be a point from which two perpendicular tangent can be drawn to the ellipse $4x^2 + 5y^2 = 20$. If $F = 4\alpha + 3\beta$, then

(1) Domain of F is $[-3, 3]$
(2) Range of F is $[-15, 15]$
(3) Equation of the director circle is $x^2 + y^2 = 9$
(4) Maximum value of F is 15

Sol. Answer (1, 2, 3, 4)

$E : 4x^2 + 5y^2 = 20$

$\Rightarrow \dfrac{x^2}{5} + \dfrac{y^2}{4} = 1$



Clearly locus of $P(\alpha, \beta)$ is a director circle

$\therefore \quad x^2 + y^2 = 5 + 4 = 9$

Which is passes through $P(\alpha, \beta)$

$\therefore \quad \alpha^2 + \beta^2 = 9$

Now, $F = 4\alpha + 3\beta = 4\alpha + 3\sqrt{9-\alpha^2}$

$\Rightarrow \quad D_F = [-3, 3]$

$R_F = [-15, 15]$

Maximum value of F is 15

20. Equation of a tangent passing through (2, 8) to the hyperbola $5x^2 - y^2 = 5$ is

(1) $3x - y + 2 = 0$
(2) $23x - 3y - 22 = 0$
(3) $3x - 23y + 178 = 0$
(4) $3x + y + 14 = 0$

Sol. Answer (1, 2)

Equation of a line through (2, 8)

$y - 8 = m(x - 2)$

$y = m.x + 8 - 2m$ ... (i)

Line (i) is a tangent to the hyperbola $\dfrac{x^2}{1} - \dfrac{y^2}{5} = 1$

$\therefore \quad$ Condition of tangency $c^2 = a^2m^2 - b^2$

$(8 - 2m)^2 = m^2 - 5$

$3m^2 - 32m + 69 = 0$

Gives

$m = 3, \; m = \dfrac{23}{3}$

$\therefore \quad$ Tangents are $3x - y + 2 = 0$ and $23x - 3y - 22 = 0$

21. The circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points (x_1, y_1) (x_2, y_2) (x_3, y_3) and (x_4, y_4) then

(1) $\Sigma x_i = 0$
(2) $\Sigma y_i = 0$
(3) $x_1x_2x_3x_4 = C^4$
(4) $y_1y_2y_3y_4 = C^4$

Sol. Answer (1, 2, 3, 4)

$y = \dfrac{c^2}{x}$ put in $x^2 + y^2 = a^2$

$x^2 + \dfrac{c^4}{x^2} = a^2$

$x^4 - a^2x^2 + c^4 = 0$

Let roots are x_1, x_2, x_3, and x_4

$x_1 x_2 x_3 x_4 = c^4$

Similarly $y_1 y_2 y_3 y_4 = c^4$

$x_1 + x_2 + x_3 + x_4 = 0$

$\Sigma x_i = 0$

Similarly $\Sigma y_i = 0$

22. The equation $\dfrac{x^2}{12-k} + \dfrac{y^2}{8-k} = 1$ represents

(1) No locus if $k > 12$

(2) A hyperbola, if $8 < k < 12$

(3) An ellipse if $k < 8$

(4) An ellipse if $k > 12$ or $k < 8$

Sol. Answer (1, 2, 3)

$$\frac{x^2}{12-k} + \frac{y^2}{8-k} = 1$$

Condition of ellipse : $12 - k > 0$ and $8 - k > 0$

$K < 12$ and $k < 8$

$\therefore \quad k < 8$

Condition of hyperbola

$12 - k > 0$ and $8 - k < 0$

$k < 12$ and $k > 8$

$8 < k < 12$

or

$12 - k < 0$ and $8 - k > 0$

$k > 12$ and $k < 8$

Does not exist

$\therefore \quad$ No locus of $k > 12$

23. Let AB be a double ordinate of hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$. If O be the centre of the hyperbola and OAB is an equilateral triangle, then which of the following is/are true, if $A(\alpha, \beta)$?

(1) $e > \dfrac{2}{\sqrt{3}}$

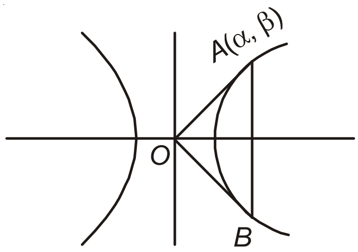
(2) $e > \dfrac{4}{\sqrt{3}}$

(3) $\alpha^2 = 3\beta^2$

(4) $\alpha^2 > 3\beta^2$

Sol. Answer (1, 3)

$$\tan 30° = \frac{\beta}{\alpha} = \frac{1}{\sqrt{3}}$$



$3\beta^2 = \alpha^2$

$$\Rightarrow \quad \frac{\alpha^2}{a^2} - \frac{\beta^2}{b^2} = 1$$

$\Rightarrow \dfrac{3\beta^2}{a^2} - \dfrac{\beta^2}{b^2} = 1 \qquad \Rightarrow \dfrac{3}{a^2} - \dfrac{1}{b^2} = \dfrac{1}{\beta^2}$

$\Rightarrow \dfrac{3}{a^2} - \dfrac{1}{b^2} > 0$

$\Rightarrow \dfrac{3}{a^2} > \dfrac{1}{b^2}$

$\Rightarrow e > \dfrac{2}{\sqrt{3}}$

24. The angle between a pair of tangent drawn from *a* point *P* to the parabola $y^2 = 4ax$ is 45°. If locus of point *P* is hyperbola then its foci are

(1) $(a, 0)$ (2) $(-7a, 0)$ (3) $(4a, 0)$ (4) $(-4a, 0)$

Sol. Answer (1, 2)

$y = mx + \dfrac{a}{m}$

$k = mh + \dfrac{a}{m}$

$m^2h - km + a = 0$

$\Rightarrow \tan 45 = \dfrac{m_1 - m_2}{1 + m_1 m_2}$

$\Rightarrow 1 = \dfrac{\sqrt{\left(\dfrac{k}{h}\right)^2 - 4\dfrac{a}{h}}}{1 + \dfrac{a}{h}} = \dfrac{\sqrt{k^2 - 4ah}}{h + a}$

$\Rightarrow k^2 - 4ah = h^2 + a^2 + 2ah$

$x^2 + 6ax - y^2 + a^2 = 0$

$(x + 3a)^2 - y^2 = 8a^2$

Its centre is $(-3a, 0)$

Distance of focus from centre is $= 2\sqrt{2}a \cdot \sqrt{2} = 4a$

Hence focii are $(a, 0)$, $(-7a, 0)$

25. Which of the following is/are true for locus represented by $x = \dfrac{1}{2}a\left(t + \dfrac{1}{t}\right)$, $y = \dfrac{1}{2}a\left(t - \dfrac{1}{t}\right)$

(1) Locus is ellipse (2) Locus is hyperbola

(3) Directrix are $x = \pm\dfrac{a}{\sqrt{2}}$ (4) Directrix are $x = \pm a\sqrt{2}$

Sol. Answer (2, 3)

$x^2 - y^2 = \dfrac{1}{4}a^2\left(t^2 + \dfrac{1}{t^2} + 2 - t^2 - \dfrac{1}{t^2} + 2\right)$

$x^2 - y^2 = a^2$

Its directrices are $x = \pm\dfrac{a}{\sqrt{2}}$

26. If line joining point (0, 3) and (5, –2) is tangent to curve $y = \frac{1}{x+c}$, then value of c is/are

(1) –5 (2) –1 (3) 5 (4) 1

Sol. Answer (1, 2)

Equation of line is

$y - 3 = \frac{-2-3}{5-0}(x - 0)$

$x + y = 3$

Since it is tangent

$(3 - x)(x + c) = 1$

$3x + 3c - x^2 - xc = 1$

$x^2 + (c - 3)x + 1 - 3c = 0$

$(c - 3)^2 - 4(1 - 3c) = 0$

$c^2 + 9 - 6c - 4 + 12c = 0$

$c^2 + 6c + 5 = 0$

$(c + 5)(c + 1) = 0$

$c = -5, -1$

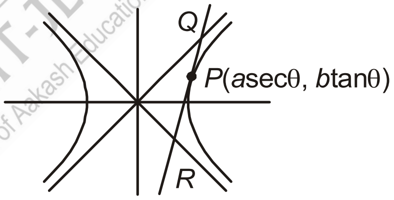
27. Tangents at any point P is drawn to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ intersects asymptotes at Q and R, if O is the centre of hyperbola then

(1) Area of triangle OQR is ab (2) Area of triangle OQR is $2ab$

(3) P is mid-point of QR (4) P trisect QR

Sol. Answer (1, 3)

Tangent at P is $\frac{x - \sec\theta}{a} - \frac{y\tan\theta}{b} = 1$



Equation of asymptotes are $y = \pm\frac{b}{a}x$

$Q(a(\sec\theta + \tan\theta), b(\sec\theta + \tan\theta))$

$R(a(\sec\theta - \tan\theta), b(\sec\theta - \tan\theta))$

Area of OQR is = ab, P is mid-point of QR

28. The locus of a point whose chord of contact touches the circle described on the straight line joining the foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ as diameter is/has

(1) Hyperbola (2) Ellipse (3) $e = \frac{\sqrt{a^4 - b^4}}{a^2}$ (4) $e = \frac{\sqrt{a^4 + b^4}}{a^2}$

Sol. Answer (2, 3)

Given circle is

$(x - ae)(x + ae) + y^2 = 0$

$x^2 + y^2 = a^2e^2 = a^2 + b^2$

Chord of contact is

$\frac{xh}{a^2} - \frac{yk}{b^2} = 1$

$$\frac{-1}{\sqrt{\frac{h^2}{a^4}+\frac{k^2}{b^4}}}=\sqrt{a^2+b^2}$$

Locus is $\frac{x^2}{a^4}+\frac{y^2}{b^4}=\frac{1}{a^2+b^2}$

Its eccentricity

$$\frac{b^4}{a^2+b^2}=\frac{a^4}{a^2+b^2}(1-e^2)$$

$$\Rightarrow \quad e^2=1-\frac{b^4}{a^4}=\frac{a^4-b^4}{a^4}$$

$$\Rightarrow \quad e=\frac{\sqrt{a^4-b^4}}{a^2}$$

29. The normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ meets the axes in M and N, and lines MP and NP are drawn at right angle to the axes then locus of P is a hyperbola with eccentricity e', if eccentricity of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is e then

(1) e' is eccentricity of conjugate of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

(2) $\frac{1}{e^2}+\frac{1}{e'^2}=1$

(3) $e^2 + e'^2 = 3$

(4) $e^2 + e'^2 = 4$

Sol. Answer (1, 2)

$$\frac{ax}{\sec\theta}+\frac{by}{\tan\theta}=a^2+b^2$$

$$M\left(\frac{(a^2+b^2)}{a}\sec\theta, 0\right)$$

$$N\left(0, \frac{(a^2+b^2)}{b}\tan\theta\right)$$

Locus is $a^2x^2 - b^2y^2 = (a^2 + b^2)^2$

$\Rightarrow \quad a^2 = b^2(e'^2 - 1)$

Which is same as conjugate hyperbola

30. If the axis of a varying central hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be fixed in magnitude and position, then locus of point of contact of a tangent drawn to it from a fixed point on x-axis is

(1) Parabola (2) Ellipse (3) Hyperbola (4) Straight line

Sol. Answer (3, 4)

Points of contact is $\left(\frac{-a^2m}{\sqrt{a^2m^2-b^2}}, \frac{-b^2}{\sqrt{a^2m^2-b^2}}\right)$

If tangents are drawn from $(\alpha, 0)$

Then $0 = m\alpha + \sqrt{a^2m^2-b^2}$

$\Rightarrow \quad m = \frac{b^2}{a^2-\alpha^2}$

$h = -\frac{a}{\alpha}$

$k = \frac{-b(a^2-\alpha^2)}{\alpha^2}$

Hence locus may be straight line or hyperbola

31. If e_1 and e_2 are the eccentricity of hyperbola $xy = c^2$ and $x^2 - y^2 = c^2$, then

(1) $\frac{1}{e_1^2}+\frac{1}{e_2^2}=1$ (2) $e_1^2 + e_2^2 = 1$ (3) $e_1^2 - e_2^2 = 0$ (4) $e_1 \cdot e_2 = 2$

Sol. Answer (1, 3, 4)

Eccentricity of rectangular hyperbola is constant and it is equal to $\sqrt{2}$

32. If equation of hyperbola is $xy + 3x - 2y - 10$, then

(1) Eccentricity $= \sqrt{2}$ (2) Centre = (2, –3)

(3) Lengths of latus rectum = $4\sqrt{2}$ (4) Asymptotes are $x = 2$ and $y = -3$

Sol. Answer (1, 2, 3, 4)

Given equation $xy + 3x - 2y - 10 = 0$

$x(y + 3) - 2y - 6 + 6 - 10 = 0$

$x(y + 3) - 2(y + 3) - 4 = 0$

$(x - 2)(y + 3) = 4$

Length of latus rectum = $\frac{2b^2}{a} = 2a$

And $c^2 = \frac{a^2}{2} = 4 \Rightarrow a = 2\sqrt{2}$

$\therefore$ Length of latus rectum = $4\sqrt{2}$

33. If (5, 12) and (24, 7) are the focii of a conic passing through the origin, then eccentricity of conic is

(1) $\frac{\sqrt{386}}{12}$ (2) $\frac{\sqrt{386}}{13}$ (3) $\frac{\sqrt{386}}{38}$ (4) $\frac{\sqrt{386}}{25}$

Sol. Answer (1, 3)

Conic may be ellipse or hyperbola

Case –I: If conic is ellipse, then

$SP + S'P = 2a$ and $2ae = SS'$

Where S and S' are the focii and e is the eccentricity

Let P is origin

Now, $SP + S'P = \sqrt{5^2+12^2} + \sqrt{24^2+7^2} = 13 + 25 = 38$

$\therefore \quad 2ae = SS' = \sqrt{386}$

$\Rightarrow \quad e = \frac{\sqrt{386}}{38}$

Case-II: If conic is hyperbola, then

$|SP - S'P| = 2a$ and $2ae = SS'$

$|SP - S'P| = |13 - 25| = 12$

$\therefore \quad |SP - S'P| = |13 - 25| = 12$

Now, $e = \frac{SS'}{2a} = \frac{\sqrt{386}}{12}$

34. If equation of hyperbola is $2x^2 + 5xy + 2y^2 - 11x - 7y - 4 = 0$ then

(1) Conjugate hyperbola is $2x^2 + 5xy + 2y^2 - 11x - 7y + 4 = 0$

(2) Conjugate hyperbola is $2x^2 + 5xy + 2y^2 - 11x - 7y + 14 = 0$

(3) Asymptotes is $2x^2 + 5xy + 2y^2 - 11x - 7y + 5 = 0$

(4) Asymptotes is $2x^2 + 5xy + 2y^2 - 11x - 7y - 5 = 0$

Sol. Answer (2, 3)

Given equation of hyperbola is $2x^2 + 5xy + 2y^2 - 11x - 7y - 4 = 0$

Then equation of asymptotes

$2x^2 + 5xy + 2y^2 - 11x - 7y + \lambda = 0$

It will represent two straight line, then $\Delta = 0$

$\therefore \quad a = 2, b = 2, c = \lambda, f = \frac{-7}{2}, g = \frac{-11}{2}$ and $h = \frac{5}{2}$

$\Rightarrow \quad 2\times 2\times \lambda + 2\left(\frac{-7}{2}\right)\left(\frac{-11}{2}\right)\frac{5}{2} - 2\left(\frac{-7}{2}\right)^2 - 2\left(\frac{-11}{2}\right)^2 - \lambda\left(\frac{5}{2}\right)^2 = 0$

$\Rightarrow \lambda = 5$

Then equation of asymptotes is $2x^2 + 5xy + 2y^2 - 11x - 7y + 5 = 0$

For conjugate hyperbola

$\boxed{H + C.H = 2ASM}$

$\therefore$ Conjugate hyperbola

$2(2x^2 + 5xy + 2y^2 - 11x - 7y + 5) - (2x^2 + 5xy + 2y^2 - 11x - 7y - 4) = 0$

$\Rightarrow$ $2x^2 + 5xy + 2y^2 - 11x - 7y + 14 = 0$

35. For the equation of rectangular hyperbola $xy = 18$

(1) Length of transverse axis = length of conjugate axis = 12

(2) Vertices are $(3\sqrt{2}, 3\sqrt{2})$ or $(-3\sqrt{2}, -3\sqrt{2})$

(3) Focii are (6, + 6), (–6, –6)

(4) Equation of tangent with slope 1 cannot be possible

Sol. Answer (1, 2, 3, 4)

(1) Equation of rectangular hyperbola

$$xy = \frac{a^2}{2}$$

By comparing with given hyperbola

$$\frac{a^2}{2} = 18 \Rightarrow a = 6$$

$\therefore$ Length of transverse axis = $2a$ = 12

And length of transverse axis = length of conjugate axis

(2) Vertices are $\left(\frac{+a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ or $\left(\frac{-a}{\sqrt{2}}, \frac{-a}{\sqrt{2}}\right)$

$\Rightarrow$ $(+3\sqrt{2}, 3\sqrt{2})$ or $(-3\sqrt{2}, -3\sqrt{2})$

(3) Focii are (a, a) or $(-a, -a)$

$\Rightarrow$ (6, 6) or (–6, –6)

(4) If $m < 0$, then tangent can possible to the hyperbola

36. If the normals at (x_i, y_i), i = 1, 2, 3, 4 on the rectangular hyperbola $xy = c^2$ meet at the point (α, β), then

(1) $\Sigma x_i = \alpha$ (2) $\Sigma y_i = \beta$ (3) $\Sigma x_i^2 = \alpha^2$ (4) $\Sigma y_i^2 = \beta^2$

Sol. Answer (1, 2, 3, 4)

Let $(x_i, y_i) = \left(ct_i, \frac{c}{t_i}\right)$, i = 1, 2, 3, 4 are the points on the rectangular hyperbola $xy = c^2$,

Equation of normal to the hyperbola $xy = c^2$ at $\left(ct, \frac{c}{t}\right)$ is

$ct^4 - t^3x + ty - c = 0$

It passes through (α, β), then

$ct^4 - t^3\alpha + t\beta - c = 0$

Its biquadratic equation in t. Let the roots of equation are t_1, t_2, t_3, t_4 then

$$\Sigma t_1 = \frac{\alpha}{c}, \Sigma t_1t_2 = 0, \Sigma t_1t_2t_3 = \frac{-\beta}{c}, t_1t_2t_3t_4 = -1$$

(1) $\Sigma x_i = \Sigma ct_i = c\Sigma t_i = c\left(\frac{\alpha}{c}\right) = \alpha$

(2) $\Sigma y_i = \Sigma \frac{c}{t_i} = c\Sigma \frac{1}{t_i} = \frac{c\Sigma t_1t_2t_3}{t_1t_2t_3t_4} = \frac{c\left(\frac{-B}{C}\right)}{-1} = B$

(3) $\Sigma x_i^2 = (\Sigma x_i)^2 - 2\Sigma x_i x_2 = \alpha^2 - 0 = \alpha^2$

(4) $\Sigma y_i^2 = (\Sigma y_i)^2 - 2\Sigma y_1 y_2 = \beta^2 - 0 = \beta^2$

37. The feet of the normals to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ from (h, k) lie on

(1) $a^2y(x - h) + b^2x(y - k) = 0$

(2) $b^2x(x - h) + a^2y(y - k) = 0$

(3) $(a^2 + b^2)xy - a^2hy - b^2xk = 0$

(4) None of these

Sol. Answer (1, 3)

Let $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$ and $S(x_4, y_4)$ be co-normal points so that normals drawn from them meet in $T(h, k)$. The equation of normal at $p(x_1, y_1)$

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$$

Or $a^2xy_1 + b^2yx_1 = (a^2 + b^2)x_1y_1$

Or $(a^2 + b^2)x_1y_1 - a^2xy_1 - b^2yx_1 = 0$

The point $T(h, k)$ lies on it

$(a^2 + b^2)x_1y_1 - a^2hy_1 - b^2x_1k = 0$

Similarly, for points Q, R and S are

$(a^2 + b^2)x_2y_2 - a^2hy_2 - b^2x_2k = 0$

$(a^2 + b^2)x_3y_3 - a^2hy_3 - b^2x_3k = 0$

$(a^2 + b^2)x_4y_4 - a^2hy_4 - b^2x_4k = 0$

Hence P, Q, R, S lie on the curve

$(a^2 + b^2)xy - a^2hy - b^2xk = 0$

Or $a^2y(x - h) + b^2x(y - k) = 0$

38. The equation of the asymptotes of a hyperbola are $4x - 3y + 8 = 0$ and $3x + 4y - 7 = 0$, then

(1) Eccentricity is $\sqrt{2}$

(2) Centre is $\left(\frac{-11}{25}, \frac{52}{25}\right)$

(3) Centre is $\left(\frac{11}{25}, \frac{-52}{25}\right)$

(4) Equation of axes are $x - 7y + 15 = 0$ and $7x + y + 1 = 0$

Sol. Answer (1, 2, 4)

(1) Asymptotes are perpendicular, then hyperbola is rectangular hyperbola

$\therefore \quad e = \sqrt{2}$

(2) Centre of rectangular hyperbola is the point of intersection of asymptotes

$\therefore \quad 4x - 3y + 8 = 0$ and $3x + 4y - 7 = 0$

(4) Axes are along the bisector of angles between the asymptotes

Then equation of axes are

$$\frac{4x-3y+8}{\sqrt{4^2+(-3)^2}} = \pm\frac{3x+4y-7}{\sqrt{3^2+4^2}}$$

$\therefore \quad x - 7y + 15 = 0$ and $7x + y + 1 = 0$

39. If the tangent at the point (a secα, b tanα) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ meets the transverse axis at T. Then the distance of T form a focus of the hyperbola is

(1) $a(e - \cos\alpha)$ (2) $b(e + \cos\alpha)$

(3) $a(e + \cos\alpha)$ (4) $\sqrt{a^2e^2+b^2\cot^2\alpha}$

Sol. Answer (1, 3)

Equation of tangent at ($a$secα, $b$tanα) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $\frac{x}{a}\sec\alpha-\frac{y}{b}\tan\alpha=1$ which meets the transverse axis $y = 0$ at the point T ($a$cosα, 0) whose distance from the focus $ae = 0$ is $ae - a\cos\alpha$ and from the focus ($-ae$, 0) is ($ae + a\cos\alpha$)

Note that $ae > a\cos\alpha$

Since $e > 1$

40. If $ax + by + c = 0$ is the normal to the hyperbola $xy + 1 = 0$, then

(1) $a > 0, b > 0$ (2) $a < 0, b > 0$

(3) $a > 0, b < 0$ (4) $a < 0, b < 0$

Sol. Answer (1, 4)

Slope of normal $= -\frac{a}{b}$

And $xy = -1 \Rightarrow y = -\frac{1}{x}$

$\Rightarrow \quad \frac{dy}{dx} = +\frac{1}{x^2}$

Slope of normal $= -x^2 = -\frac{a}{b}$ (given)

$\frac{a}{b} > 0$

It can be possible if a and b are same sign

41. If equation of hyperbola is $4(2y - x - 3)^2 - 9(2x + y - 1)^2 = 80$, then

(1) Eccentricity is $\frac{\sqrt{13}}{3}$

(2) Centre of hyperbola is $\left(-\frac{1}{5}, -\frac{7}{5}\right)$

(3) Transverse axis is $2x + y - 1 = 0$

(4) Conjugate axis is $x - 2y + 3 = 0$

Sol. Answer (1, 2, 3, 4)

$$4.5\left(\frac{2y-x-3}{\sqrt{5}}\right)^2 - 9.5\left(\frac{2x+y-1}{\sqrt{5}}\right)^2 = 80$$

$\Rightarrow$ $20x^2 - 9.5y^2 = 80$

$\Rightarrow$ $\frac{x^2}{4} - \frac{y^2}{\left(\frac{16}{9}\right)} = 1$

(1) Eccentricity = $\sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+\frac{16}{9.(4)}} = \frac{\sqrt{13}}{3}$

(2) Centre: $2y - x - 3 = 0$

And $2x + y - 1 = 0$

(3) Transverse axis : $Y = 0$

(4) Conjugate axis : $X = 0$

42. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$, be the end points of the latus rectum of the ellipse $x^2 + 4y^2 = 4$. The equations of parabolas with latus rectum PQ are **[IIT-JEE 2008]**

(1) $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$

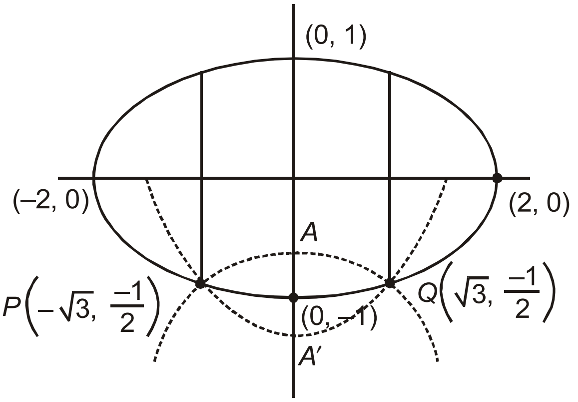
(2) $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$

(3) $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$

(4) $x^2 - 2\sqrt{3}y = 3 - \sqrt{3}$

Sol. Answer (2, 3)

The equation $x^2 + 4y^2 = 4$ represents an ellipse with 2 and 1 as semi-major and semi-minor axes and eccentricity $\frac{\sqrt{3}}{2}$. Thus the ends of latera recta are $\left(\sqrt{3}, \frac{1}{2}\right)$, $\left(\sqrt{3}, -\frac{1}{2}\right)$, $\left(-\sqrt{3}, \frac{1}{2}\right)$ and $\left(-\sqrt{3}, -\frac{1}{2}\right)$.



According to the question, $P\left(-\sqrt{3}, -\frac{1}{2}\right)$ and $Q\left(\sqrt{3}, -\frac{1}{2}\right)$.

$PQ = 2\sqrt{3}$

Thus the coordinates of the vertex of the parabolas are $A\left(0, \frac{-1+\sqrt{3}}{2}\right)$ and $A'\left(0, \frac{-1-\sqrt{3}}{2}\right)$ and corresponding equations are

$$(x-0)^2 = 4\cdot\frac{\sqrt{3}}{2}\left(y+\frac{1-\sqrt{3}}{2}\right)$$

and $(x-0)^2 = 4\cdot\frac{\sqrt{3}}{2}\left(y-\frac{-1-\sqrt{3}}{2}\right)$

i.e., $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$

and $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$

43. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then **[IIT-JEE 2009]**

(1) Equation of ellipse is $x^2 + 2y^2 = 2$
(2) The foci of ellipse are $(\pm 1, 0)$
(3) Equation of ellipse is $x^2 + 2y^2 = 4$
(4) The foci of ellipse are $(\pm\sqrt{2}, 0)$

Sol. Answer (1, 2)

We have

$2x^2 - 2y^2 = 1$

$\Rightarrow \quad x^2 - y^2 = \frac{1}{2}$ ...(i)

Eccentricity $e = \sqrt{1+\frac{1}{1}} = \sqrt{2}$

Let the equation of the ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

given that $\sqrt{1-\frac{b^2}{a^2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow \quad \frac{b^2}{a^2} = \frac{1}{2} \Rightarrow a^2 = 2b^2$

Hence equation of the ellipse is

$$\frac{x^2}{2b^2}+\frac{y^2}{b^2}=1$$

$\Rightarrow \quad x^2 + 2y^2 = 2b^2$ ...(ii)

Solving (i) and (ii)

$$x^2 = \frac{1+2b^2}{3},\ y^2 = \frac{4b^2-1}{6}$$

Differentiating (i)

$$2x - 2y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x}{y} \quad \text{...(iii)}$$

By (ii)

$$2x + 4y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{2y} \quad \text{...(iv)}$$

As the curves are orthogonal. Hence

$$\left(\frac{x}{y}\right)\left(-\frac{x}{2y}\right) = -1$$

$$x^2 = 2y^2$$

$$\Rightarrow \frac{1+2b^2}{3} = 2.\frac{4b^2-1}{6}$$

$$\Rightarrow 1 + 2b^2 = 4b^2 - 1$$

$$2b^2 = 2$$

$$b^2 = 1$$

Hence equation of the ellipse is

$$x^2 + 2y^2 = 2$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{1} = 1$$

$$\text{Foci} = \left(\pm\sqrt{2}\times\frac{1}{\sqrt{2}}10\right) = (\pm 1, 0)$$

44. The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N, respectively. The locus of the centroid of the triangle PTN is a parabola whose **[IIT-JEE 2009]**

(1) Vertex is $\left(\frac{2a}{3}, 0\right)$

(2) Directrix is $x = 0$

(3) Latus rectum is $\frac{2a}{3}$

(4) Focus is $(a, 0)$

Sol. Answer (1, 4)

Let $P \equiv (at^2, 2at)$

Equation of tangent at P is

$yt = x + at^2$

at, $y = 0$ we get $x = -at^2$

Hence $T \equiv (-at^2, 0)$

Similarly equation of normal at P is

$y + xt = 2at + at^3$

at, $y = 0$, we get $x = 2a + at^2$

Hence $N \equiv (2a + at^2, 0)$

Let centroid of the triangle PTN is (h, k)

$\Rightarrow \quad h = \dfrac{at^2 - at^2 + 2a + at^2}{3}$

$\Rightarrow \quad h = \dfrac{2a + at^2}{3}$ ...(i)

$k = \dfrac{2at + 0 + 0}{3} = \dfrac{2at}{3}$ ...(ii)

By (i) and (ii)

$$h = \frac{2a + a\left(\frac{3k}{2a}\right)^2}{3}$$

$\Rightarrow \quad 3h = 2a + a.\dfrac{9k^2}{4a^2}$

$3h = 2a + \dfrac{9k^2}{4a}$

$\Rightarrow \quad \dfrac{9k^2}{4a} = 3h - 2a$

$k^2 = \dfrac{4a}{9}(3h - 2a)$

$k^2 = \dfrac{4a}{3}\left(h - \dfrac{2a}{3}\right)$

Hence locus of (h, k) is

$y^2 = \dfrac{4a}{3}\left(x - \dfrac{2a}{3}\right)$

which represents a parabola whose vertex is $\left(\dfrac{2a}{3}, 0\right)$

Directrix : $x - \dfrac{2a}{3} = -\dfrac{a}{3} \Rightarrow x = \dfrac{a}{3}$

Latus rectum : $\dfrac{4a}{3}$

Focus : $x - \dfrac{2a}{3} = \dfrac{a}{3}$ and $y = 0$

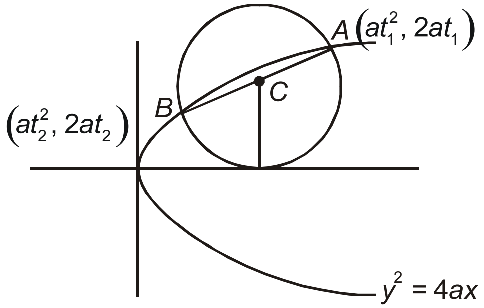
$\Rightarrow \quad x = a, y = 0$

$\Rightarrow \quad$ Focus : $(a, 0)$

45. Let A and B be two distinct points on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be **[IIT-JEE 2010]**

(1) $-\dfrac{1}{r}$ (2) $\dfrac{1}{r}$ (3) $\dfrac{2}{r}$ (4) $-\dfrac{2}{r}$

Sol. Answer (3, 4)



Mid - point of A & B

$a(t_1 + t_2) = r$

$t_1 + t_2 = \frac{r}{a}$

Slope of $AB = \frac{2a\left(t_1 - t_2\right)}{\left(t_1^2 - t_2^2\right)} = \frac{2}{t_1 + t_2} = \frac{2a}{r}$

In $y^2 = 4x$

$a = 1$

$\therefore$ slope = $\frac{2}{r}$

If circle in below x-axis, then

Slope = $-\frac{2}{r}$

46. Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be reciprocal to that of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then **[IIT-JEE 2011]**

(1) The equation of the hyperbola is $\frac{x^2}{3} - \frac{y^2}{2} = 1$

(2) A focus of the hyperbola is (2, 0)

(3) The eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$

(4) The equation of the hyperbola is $x^2 - 3y^2 = 3$

Sol. Answer (2, 4)

The equation of the given ellipse is $\frac{x^2}{4} + \frac{y^2}{1} = 1$

Whose eccentricity $e = \frac{\sqrt{3}}{2}$

Foci of ellipse are $\left(\sqrt{3}, 0\right)$ and $\left(-\sqrt{3}, 0\right)$

Eccentricity of hyperbola = $\frac{2}{\sqrt{3}}$

Since given hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ passes through $\left(\sqrt{3},0\right)$,

Hence $\frac{3}{a^2}-\frac{0}{b^2}=1$

$\Rightarrow$ $a^2 = 3$

$b^2 = a^2(e^2 - 1)$

$\Rightarrow$ $b^2=3\left(\frac{4}{3}-1\right)=1$

Thus, the equation of the hyperbola is

$$\frac{x^2}{3}-\frac{y^2}{1}=1$$

$\Rightarrow$ $x^2 - 3y^2 = 3$

Foci of the hyperbola are $(\pm 2, 0)$.

47. Let L be a normal to the parabola $y^2 = 4x$. If L passes through the point (9, 6), then L is given by **[IIT-JEE 2011]**

(1) $y - x + 3 = 0$ (2) $y + 3x - 33 = 0$ (3) $y + x - 15 = 0$ (4) $y - 2x + 12 = 0$

Sol. Answer (1, 2, 4)

The equation of the normal to the given parabola $y^2 = 4x$ in slope form is

$y = mx - 2m - m^3$

which will pass through (9, 6) if

$6 = 9m - 2m - m^3$

$\Rightarrow$ $m^3 - 7m + 6 = 0$

$\Rightarrow$ $m = 1, 2, -3$

Consequently the equation of the normal L is

$y = x - 3 \quad \Rightarrow \quad y - x + 3 = 0$

or $y = 2x - 12 \quad \Rightarrow \quad y - 2x + 12 = 0$

or $y = -3x + 33 \quad \Rightarrow \quad y + 3x - 33 = 0$

48. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$, parallel to the straight line $2x - y = 1$. The point of contact of the tangents on the hyperbola are **[IIT-JEE 2012]**

(1) $\left(\frac{9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (2) $\left(-\frac{9}{2\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$ (3) $(3\sqrt{3},-2\sqrt{2})$ (4) $(-3\sqrt{3},2\sqrt{2})$

Sol. Answer (1, 2)

As, points of contact $=\left(\frac{\pm a^2m}{\sqrt{a^2m^2-b^2}},\frac{\pm b^2}{\sqrt{a^2m^2-b^2}}\right)$

Here, $a^2 = 9$, $b^2 = 4$, $m = 2$

so, required points are $\left(\frac{\pm 9}{2\sqrt{2}},\frac{\pm 1}{\sqrt{2}}\right)$

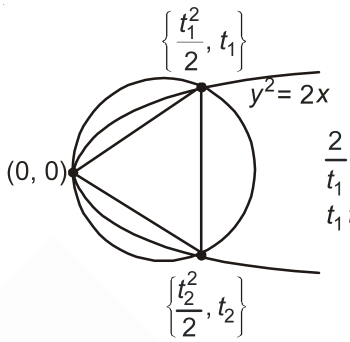
49. Let P and Q be distinct points on the parabola $y^2 = 2x$ such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle ΔOPQ is $3\sqrt{2}$, then which of the following is(are) the coordinates of P? **[JEE(Advanced)-2015]**

(1) $(4, 2\sqrt{2})$ (2) $(9, 3\sqrt{2})$ (3) $\left(\frac{1}{4}, \frac{1}{\sqrt{2}}\right)$ (4) $(1, \sqrt{2})$

Sol. Answer (1, 4)

$\Delta = 3\sqrt{2}$

$$\frac{1}{2}\begin{vmatrix} 0 & 0 \\ \frac{t_1^2}{2} & t_1 \\ \frac{t_2^2}{2} & t_2 \\ 0 & 0 \end{vmatrix} = \pm 3\sqrt{2}$$



$\frac{2}{t_1} \times \frac{2}{t_2} = 1$

$t_1 t_2 = -4$

$$\frac{1}{2}\left(\frac{t_1^2 t_2}{2} - \frac{t_2^2 t_1}{2}\right) = \pm 3\sqrt{2}$$

$$\frac{1}{4} t_1 t_2 (t_1 - t_2) = \pm 3\sqrt{2}$$

$$t_1 t_2 (t_1 - t_2) = \pm 12\sqrt{2}$$

$$t_1 - t_2 = \pm 3\sqrt{2}$$

$$\Rightarrow \quad t_1 - t_2 = 3\sqrt{2} \qquad \text{(as } t_1 > t_2\text{)}$$

$$t_1 + t_2 = \sqrt{18 - 16} = \pm\sqrt{2}$$

$$2t_1 = 3\sqrt{2} + \sqrt{2}$$

$$t_1 = 2\sqrt{2}$$

$$2t_1 = 2\sqrt{2}$$

$$t_1 = \sqrt{2}$$

$P(4, 2\sqrt{2})$ and $P(1, \sqrt{2})$

50. Let E_1 and E_2 be two ellipses whose centers are at the origin. The major axes of E_1 and E_2 lie along the x-axis and the y-axis, respectively. Let S be the circle $x^2 + (y - 1)^2 = 2$. The straight line $x + y = 3$ touches the curves S, E_1 and E_2 at P, Q and R respectively. Suppose that $PQ = PR = \frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2, respectively, then the correct expression(s) is (are) **[JEE(Advanced)-2015]**

(1) $e_1^2 + e_2^2 = \frac{43}{40}$ (2) $e_1 e_2 = \frac{\sqrt{7}}{2\sqrt{10}}$ (3) $\left|e_1^2 - e_2^2\right| = \frac{5}{8}$ (4) $e_1 e_2 = \frac{\sqrt{3}}{4}$

Sol. Answer (1, 2)

Let Q be (x_1, y_1)

So equation tangent at Q will be

$xx_1 + yy_1 - (y + y_1) - 1 = 0$

Comparing with $x + y = 3$

$(x_1 y_1) \equiv (1, 2)$

So R & Q will be $\left(1 \pm \frac{2\sqrt{2}}{3}\cos\frac{3\pi}{4}, 2 \pm \frac{2\sqrt{2}}{3}\sin\frac{3\pi}{4}\right)$

$\Rightarrow Q\left(\frac{5}{3}, \frac{4}{3}\right) \& \left(\frac{1}{3}, \frac{8}{3}\right)$

Let Q lies on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

So tangent at P is $\frac{5x}{3a^2}+\frac{4y}{3b^2}=1$

Comparing with $x + y = 3$

$a^2=5,\ b^2=4 \Rightarrow e_1=\frac{1}{\sqrt{5}}$

and R lies on $\frac{x^2}{a_1^2}+\frac{y^2}{b_1^2}=1$

So tangent at $\frac{x}{3a_1^2}+\frac{8y}{3b_1^2}=3$

Comparing with $x + y = 3$

$\Rightarrow a_1^2=1, b_1^2=8 \quad \Rightarrow e_2=\sqrt{\frac{7}{8}}$

$\Rightarrow e_1^2+e_2^2=\frac{43}{40} \ \& \ e_1e_2=\frac{\sqrt{7}}{2\sqrt{10}}$

51. The circle $C_1 : x^2 + y^2 = 3$, with centre at O, intersects the parabola $x^2 = 2y$ at the point P in the first quadrant. Let the tangent to the circle C_1 at P touches other two circles C_2 and C_3 at R_2 and R_3, respectively. Suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centers Q_2 and Q_3, respectively. If Q_2 and Q_3 lie on the y-axis, then

[JEE(Advanced)-2016]

(1) $Q_2Q_3 = 12$

(2) $R_2R_3 = 4\sqrt{6}$

(3) Area of the triangle OR_2R_3 is $6\sqrt{2}$

(4) Area of the triangle PQ_2Q_3 is $4\sqrt{2}$

Sol. Answer (1, 2, 3)

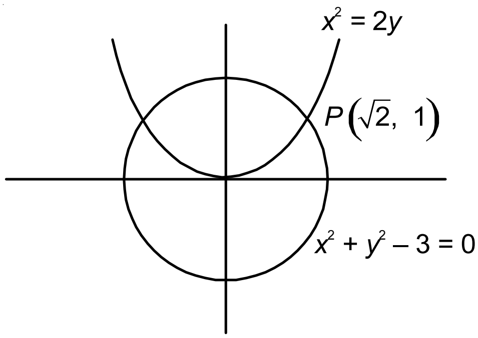
$x^2 + y^2 = 3$

$x^2 = 2y$

$\therefore \quad P\left(\sqrt{2},\ 1\right)$

Tangent at P, $\sqrt{2}x+y-3=0$

Let the centre of circle be $(0, a)$



$\therefore \quad \frac{\sqrt{2}(0)+a-3}{\sqrt{3}}=\pm 2\sqrt{3}$

$a - 3 = \pm 6$

$a = 9, -3$

$\therefore \quad Q_2\ (0, 9)$

$Q_3\ (0, -3)$

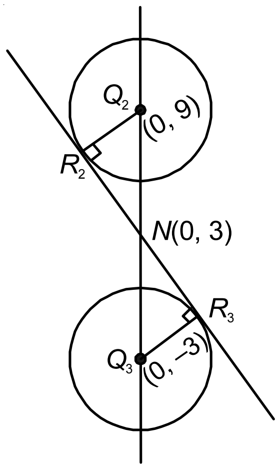
$Q_2Q_3 = 12$

$\therefore \quad NR_3 = \sqrt{6^2 - 12}$

$= \sqrt{24}$

$= 2\sqrt{6}$

$\therefore \quad R_2R_3 = 4\sqrt{6}$

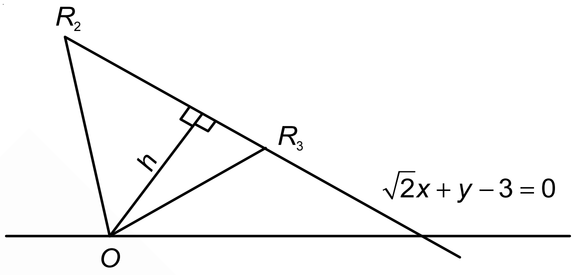


Area of OR_2R_3

$$\text{Area of } \Delta = \frac{1}{2} R_2R_3 \times h$$

$$= \frac{1}{2} 4\sqrt{6} \times \frac{3}{\sqrt{3}}$$

$$= \frac{4\sqrt{3} \times \sqrt{2} \times \sqrt{3}}{2}$$

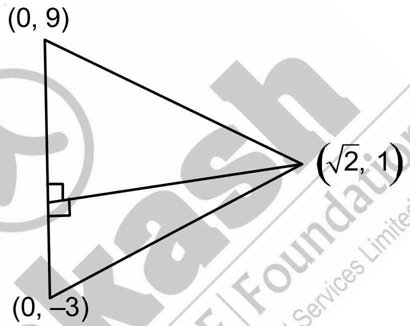


ar $(\Delta OR_2R_3) = 6\sqrt{2}$

ar (ΔPQ_2Q_3)

$$\text{Area of } \Delta PQ_2Q_3 = \frac{1}{2} \times 12 \times \sqrt{2}$$

$$= 6\sqrt{2}$$



52. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where S is the point (1, 0). Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q. The normal to the circle at P intersects a line drawn through Q parallel to RS at point E. Then the locus of E passes through the point(s). **[JEE(Advanced)-2016]**

(1) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ (2) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (3) $\left(\frac{1}{3}, -\frac{1}{\sqrt{3}}\right)$ (4) $\left(\frac{1}{4}, -\frac{1}{2}\right)$

Sol. Answer (1, 3)

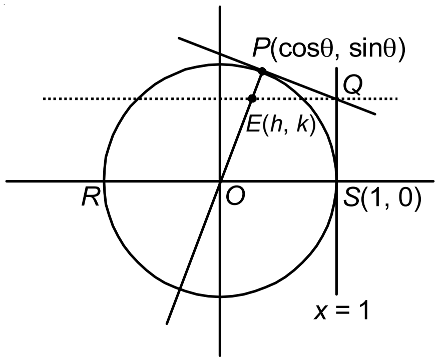
Tangent at P : $x\cos\theta + y\sin\theta = 1$

So, $Q\left(1, \frac{1-\cos\theta}{\sin\theta}\right) \equiv \left(1, \tan\frac{\theta}{2}\right)$

Normal at P : $y = x\tan\theta$

So, $E\left(\frac{\tan\frac{\theta}{2}}{\tan\theta}, \tan\frac{\theta}{2}\right)$

$\Rightarrow \quad h = \frac{\tan\frac{\theta}{2}}{\tan\theta}, \; k = \tan\frac{\theta}{2}$



On eliminating θ

$k^2 = 1 - 2h$

$\Rightarrow$ Locus of E is $y^2 = 1 - 2x$

53. Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then

[JEE(Advanced)-2016]

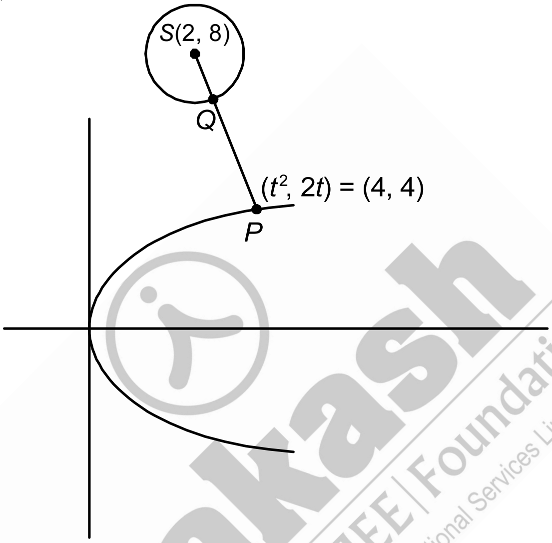
(1) $SP = 2\sqrt{5}$

(2) $SQ : QP = (\sqrt{5}+1) : 2$

(3) The x-intercept of the normal to the parabola at P is 6

(4) The slope of the tangent to the circle at Q is $\frac{1}{2}$

Sol. Answer (1, 3, 4)



$(\text{Distance})^2 = SP^2 = (t^2 - 2)^2 + (2t - 8)^2$

$$\frac{d(SP)^2}{dt} = 2(t^2 - 2).2t + 4(2t - 8)$$

$= 4[t^3 - 8]$, for minimum $t^3 = 8$, $t = 2$

$$\frac{d^2(SP)^2}{dt^2} = 12t^2 > 0 \text{ at } t = 2$$

Point $P(4, 4)$

$SP^2 = 4 + 16 = 20$

$SP = 2\sqrt{5}$

$$\frac{SQ}{QP} = \frac{2}{2\sqrt{5}-2} = \frac{1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} = \frac{\sqrt{5}+1}{4}$$

Equation of normal at P.

$y = -tx + 2at + at^3$ for $y^2 = 4ax$

$y = -2x + 4 + 8$

x intercept, $y = 0$ $x = 6$

Slope of $SP = \frac{4}{2} = -2$

Slope of tangent at $Q = \frac{1}{2}$

54. If $2x - y + 1 = 0$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$, then which of the following cannot be sides of a right angled triangle? **[JEE(Advanced)-2017]**

(1) a, 4, 2 (2) $2a$, 8, 1 (3) $2a$, 4, 1 (4) a, 4, 1

Sol. Answer (1, 2, 4)

$y = 2x + 1$ be tangent to $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$

$\therefore$ $y = 2x + 1$ compare with $y = mx \pm \sqrt{a^2m^2 - 16}$.

$\Rightarrow$ $m = 2$ and $a^2m^2 - 16 = 1$

$\Rightarrow$ $4a^2 = 17$

$\therefore$ $2a$, 4, 1 are the sides of right angled triangle.

55. If a chord, which is not a tangent, of the parabola $y^2 = 16x$ has the equation $2x + y = p$, and midpoint (h, k), then which of the following is(are) possible value(s) of p, h and k? **[JEE(Advanced)-2017]**

(1) $p = -2$, $h = 2$, $k = -4$ (2) $p = 5$, $h = 4$, $k = -3$

(3) $p = -1$, $h = 1$, $k = -3$ (4) $p = 2$, $h = 3$, $k = -4$

Sol. Answer (4)

Equation of chord is $2x + y = p$

For point of intersection with parabola,

$$(p - 2x)^2 = 16x$$

$\Rightarrow$ $4x^2 - (4p + 16)x + p^2 = 0$

$\Rightarrow$ $p = 128p + 256$

Hence, $p = -2$

Equation of chord for midpoint (h, k)

$$yk - 8(x + h) = k^2 - 16h$$

$\Rightarrow$ $yk - 8x = k^2 - 8h$

$-8x + ky = k^2 - 8h$

On comparing,

$2x + y = p$

$-\frac{8}{2} = \frac{k}{1} = \frac{k^2 - 8h}{p}$

$k^2 - 8h = -4p$

$16 - 8h = -4p$

$\Rightarrow$ $-p + 2h = 4$

$\therefore$ $k = -4$, $p = 2$, $h = 3$

56. Consider two straight lines, each of which is tangent to both the circle $x^2 + y^2 = \frac{1}{2}$ and the parabola $y^2 = 4x$. Let these lines intersect at the point Q. Consider the ellipse whose center is at the origin $O(0, 0)$ and whose semi-major axis is OQ. If the length of the minor axis of this ellipse is $\sqrt{2}$, then which of the following statement(s) is (are) TRUE? **[JEE(Advanced)-2018]**

(1) For the ellipse, the eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is 1

(2) For the ellipse, the eccentricity is $\frac{1}{2}$ and the length of the latus rectum is $\frac{1}{2}$

(3) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and x = 1 is $\frac{1}{4\sqrt{2}}(\pi - 2)$

(4) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and x = 1 is $\frac{1}{16}(\pi - 2)$

Sol. Answer (1, 3)

$y = mx + \frac{1}{m}$ is also tangent to $x^2 + y^2 = \frac{1}{2}$

$$\Rightarrow \frac{\left|\frac{1}{m}\right|}{\sqrt{1+m^2}} = \frac{1}{\sqrt{2}} \Rightarrow m = \pm 1$$

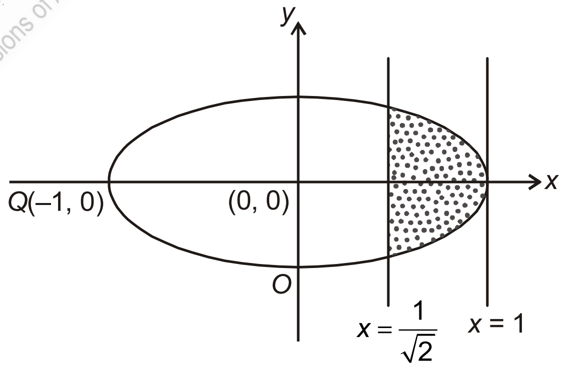
Common tangents are $y = x + 1$ and $y = -x - 1 \Rightarrow Q(-1, 0)$

Equation of ellipse is $\frac{x^2}{1^2} + \frac{y^2}{\left(\frac{1}{\sqrt{2}}\right)^2} = 1$

$b^2 = a^2(1 - e^2)$

$$\Rightarrow \frac{1}{2} = 1(1 - e^2) \Rightarrow e = \frac{1}{\sqrt{2}}$$

Length of latus rectum $= \frac{2b^2}{a} = \frac{2\times\frac{1}{2}}{1} = 1$



$$y = \pm\frac{1}{\sqrt{2}}\sqrt{1-x^2}$$

$$A = 2\times\frac{1}{\sqrt{2}}\int_{\frac{1}{\sqrt{2}}}^{1}\sqrt{1-x^2}\,dx$$

$$= \sqrt{2}\left[\frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_{\frac{1}{\sqrt{2}}}^{1}$$

$$= \frac{\pi - 2}{4\sqrt{2}} \text{ sq. units}$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Let $C : y = x^2 - 3$, $D : y = kx^2$ be two parabolas and $L_1 : x = a$, $L_2 : x = 1$ ($a \neq 0$) be two straight lines.

1. If C and D intersect at a point A on the line L_1, then the equation of the tangent line L at A to the parabola D is

(1) $2(a^3 - 3)x - ay + (a^3 - 3a) = 0$
(2) $2(a^2 - 3)x - ay - a^3 + 3a = 0$
(3) $(a^3 - 3)x - 2ay - 2a^3 + 6a = 0$
(4) None of these

Sol. Answer (2)

Put $x = a$ on both parabolas $y = a^2 - 3$ and $y = ka^2$

$$a^2 - 3 = ka^2$$

$$k = \frac{a^2 - 3}{a^2}$$

Point of intersection $A(a, a^2 - 3)$

$\therefore$ Equation of tangent L to the curve

$y = k\,.\,x^2$ at A

$$\frac{1}{2}(y + a^2 - 3) = k.x.a$$

$$\frac{1}{2}(y + a^2 - 3) = \left(\frac{a^2 - 3}{a^2}\right).xa$$

$a(y + a^2 - 3) = 2(a^2 - 3)x$

$2(a^2 - 3)x - ay - a^3 + 3a = 0$

2. If the line L meets the parabola C at a point B on the line L_2, other than A, then a may be equal to

(1) –3
(2) –2
(3) 2
(4) None of these

Sol. Answer (2)

On L_2, $x = 1$

$C : y_2 = x^2 - 3 = 1 - 3 = -2$

$B(1, -2)$

$B(1, -2)$ lie on L

$2(a^2 - 3) + 2a - a^3 + 3a = 0$

$$a^3 - 2a^2 - 5a + 6 = 0$$

$$(a - 1)(a + 2)(a - 3) = 0$$

$a = 1, -2, 3$

3. If $a > 0$, the angle subtended by the chord AB at the vertex of the parabola C is

(1) $\tan^{-1}\left(\frac{5}{7}\right)$ (2) $\tan^{-1}\left(\frac{1}{2}\right)$ (3) $\tan^{-1}(2)$ (4) $\tan^{-1}\left(\frac{1}{8}\right)$

Sol. Answer (2)

$a > 0$, a may be 1 or 3

$a = 3$, [$a \neq 1$, if $a = 1$, A and B coincidence]

$A(3, 6)$, $B(1, -2)$

Vertex of $e(0, -3)$

Slope of $AC = \frac{6+3}{3} = 3 = m_1$ say

Slope of $BC = \frac{1}{1} = 1 = m_2$ say

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{3-1}{1+3}\right| = \frac{2}{4}$$

$$\theta = \tan^{-1}\left(\frac{1}{2}\right)$$

Comprehension-II

Let $P_1 : y^2 = 4ax$ and $P_2 : y^2 = -4ax$ be two parabolas and $L : y = x$ be a straight line.

1. If $a = 4$, then the equation of the ellipse having the line segment joining the foci of the parabolas P_1 and P_2 as the major axis and eccentricity equal to $\frac{1}{2}$ is

(1) $\frac{x^2}{4} + \frac{y^2}{3} = 1$ (2) $\frac{x^2}{3} + \frac{y^2}{4} = 1$ (3) $\frac{x^2}{16} + \frac{y^2}{12} = 1$ (4) $\frac{x^2}{12} + \frac{y^2}{16} = 1$

Sol. Answer (3)

Foci $P_1(4, 0)$ and $P_2(4, 0)$

$2a = 8 \Rightarrow a = 4$

$$b^2 = a^2(1 - e^2) = 16\left(1 - \frac{1}{4}\right) = 12$$

Equation of ellipse $\frac{x^2}{16} + \frac{y^2}{12} = 1$

2. Equation of the tangent at the point on the parabola P_1 where the line L meets the parabola is

(1) $x - 2y + 4a = 0$ (2) $x + 2y - 4a = 0$ (3) $x + 2y - 8a = 0$ (4) $x - 2y + 8a = 0$

Sol. Answer (1)

Solve equations $y = x$ and $y^2 = 4ax$

Points of intersections are $O(0, 0)$ and $A(4a, 4a)$

Equation of tangent at $O(0, 0)$ is $x = 0$

and equation of tangent at $A(4a, 4a)$

$y.4a = 2a(x + 4a)$

$x - 2y + 4a = 0$

3. The co-ordinates of the other extremity of a focal chord of the parabola P_2, one of whose extremity is the point of intersection of L and P_2 is

(1) $(-a, 2a)$ (2) $\left(-\frac{a}{4}, a\right)$ (3) $\left(-\frac{a}{4}, -a\right)$ (4) $(-a, -2a)$

Sol. Answer (2)

Solve equation $y = x$ and $y^2 = -4ax$

$O(0, 0)$, $B(-4a, -4a)$

Let B, $B(-at_1^2, -2at_1)$

Focal chord again intersect parabola at t_2 where $t_2 = -\frac{1}{t_1}$

$\therefore \quad t_1 = 2$ and $t_2 = -\frac{1}{2}$

$B'(-at_2^2, -2at_2)$

$\Rightarrow \quad B'\left(-\frac{a}{4}, a\right)$

Comprehension-III

The equation of the curve represented by

$c \equiv 9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$, then

1. The locus of curve 'c' given in the above statement is

(1) Circle (2) Pair of straight line

(3) Parabola (4) Ellipse

2. The equation of axis of curve 'c' is

(1) $x = 4y$ (2) $3x = y$

(3) $3x + 4y = 0$ (4) $3x = 4y$

3. The equation of directrix of curve 'c' is

(1) $16x + 9y = 53$ (2) $16x + 12y + 53 = 0$

(3) $16x + 12y = 53$ (4) $16x + y = 53$

4. The length of latus rectum of curve 'c' is

(1) 1 (2) 2

(3) 4 (4) 8

Solution of Comprehension-III

1. Answer (3)
2. Answer (4)
3. Answer (2)
4. Answer (1)

Comparing given equation with

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, we get

$a = 9$, $b = 16$, $c = -60$, $h = -12$, $g = -10$, $f = -\frac{15}{2}$

$\Rightarrow$ $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$$= -8640 - 1800 - 0.9\left\{\frac{225}{4}\right\} - 16\{1000\} + 8640$$

$\neq 0$

And $h^2 - ab = 144 - 144 = 0$

So, 'c' represents a parabola

Now, $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$

$\Rightarrow$ $\{3x - 4y\}^2 = 5\{4x + 3y + 12\}$

$$\Rightarrow \left\{\frac{3x-4y}{\sqrt{3^2+4^2}}\right\}^2 = \frac{4x+3y+12}{\sqrt{4^2+3^2}}$$

Let $\frac{3x-4y}{\sqrt{3^2+4^2}} = Y$ and $\frac{4x+3y+12}{\sqrt{4^2+3^2}} = X$

$\therefore$ Equation reduces to, $Y^2 = X$

Comparing $Y^2 = X$ with $y^2 = 4ax$

We get, $4a = 1 \Rightarrow a = \frac{1}{4}$

So, axis of parabola is $y = 0$

$\Rightarrow$ $3x - 4y = 0$

Directrix of parabola is $X = -\frac{1}{4}$

$$\Rightarrow \frac{4x+3y+12}{\sqrt{3^2+4^2}} = -\frac{1}{4}$$

$$\Rightarrow 4x + 3y + \frac{53}{4} = 0 \Rightarrow 16x + 12y + 53 = 0$$

Hence, length of latus reactum = $4a = 1$

Comprehension-IV

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S. **[IIT-JEE 2007]**

1. The ratio of the areas of the triangles PQS and PQR is

(1) $1 : \sqrt{2}$ (2) $1 : 2$ (3) $1 : 4$ (4) $1 : 8$

2. The radius of the circumcircle of the triangle *PRS* is

(1) 5 (2) $3\sqrt{3}$ (3) $3\sqrt{2}$ (4) $2\sqrt{3}$

3. The radius of the incircle of the triangle *PQR* is

(1) 4 (2) 3 (3) $\frac{8}{3}$ (4) 2

Solution of Comprehension-IV

1. Answer (3)
2. Answer (2)
3. Answer (4)

Point of intersection of the circle and parabola are $P(1, 2\sqrt{2})$ and $Q(1, -2, \sqrt{2})$.

Tangents at P and Q to the circle meet the x-axis in $R(9, 0)$

Tangents at *P* and *Q* to the parabola meet the *x*-axis in $S(-1, 0)$.

$\frac{\text{area }(PQS)}{\text{area }(PQR)} =$ Ratio of altitudes of the triangle, where *PQ* serves as the common base $= \frac{1}{4} = 1:4$

Radius of circumcircle of triangle *PRS* can be obtained using

$$R = \frac{abc}{4\Delta} = 3\sqrt{3}$$

Radius of incircle of the triangle *PQR* can be obtained using

$$r = \frac{\Delta}{S} = 2$$

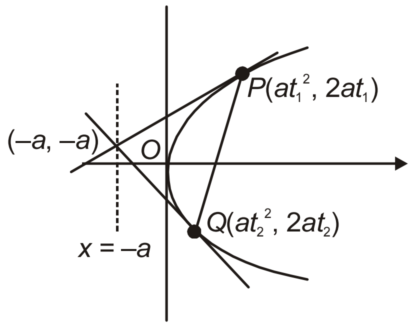
Comprehension-V

Let *PQ* be a focal chord of the parabola $y^2 = 4ax$. The tangents to the parabola at *P* and *Q* meet at a point lying on the line $y = 2x + a$, $a > 0$. **JEE(Advanced) 2013]**

1. Length of chord *PQ* is

(1) $7a$ (3) $5a$ (3) $2a$ (4) $3a$

Sol. Answer (2)



$$a(t_1 + t_2) = a$$

$$t_1 + t_2 = 1$$

$$PQ = a(t_1 - t_2)^2$$

$$= a\left[(t_1 + t_2)^2 - 4t_1t_2\right]$$

$$= a[1 + 4] = 5a \qquad \text{use (1)}$$

2. If chord *PQ* subtends an angle θ at the vertex of $y^2 = 4ax$, then tan θ =

(1) $\frac{2}{3}\sqrt{7}$ (2) $\frac{-2}{3}\sqrt{7}$ (3) $\frac{2}{3}\sqrt{5}$ (4) $\frac{-2}{3}\sqrt{5}$

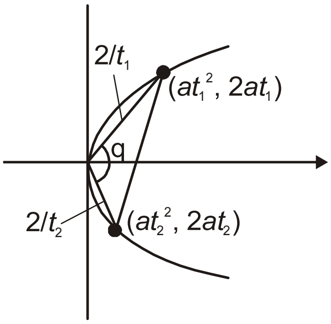
Sol. Answer (4)

$t_1 + t_2 = 1$

$t_1 t_2 = -1$

$t_1 = \dfrac{1 \pm \sqrt{5}}{2}$, $t > 0$

$\therefore \quad t_1 = \dfrac{1+\sqrt{5}}{2}$



$$\tan\theta = \dfrac{\dfrac{2}{t_1} - \dfrac{2}{t_2}}{1 + \dfrac{4}{t_1 t_2}}$$

$$= \dfrac{2(t_2 - t_1)}{t_1 t_2 + 4} = \dfrac{2(1-(1+\sqrt{5}))}{3} = -\dfrac{2\sqrt{5}}{3}$$

Comprehension-VI

Let a, r, s, t be nonzero real numbers. Let $P(at^2, 2at)$, Q, $R(ar^2, 2ar)$ and $S(as^2, 2as)$ be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point $(2a, 0)$. **[JEE(Advanced)-2014]**

1. The value of r is

(1) $-\dfrac{1}{t}$ (2) $\dfrac{t^2+1}{t}$ (3) $\dfrac{1}{t}$ (4) $\dfrac{t^2-1}{t}$

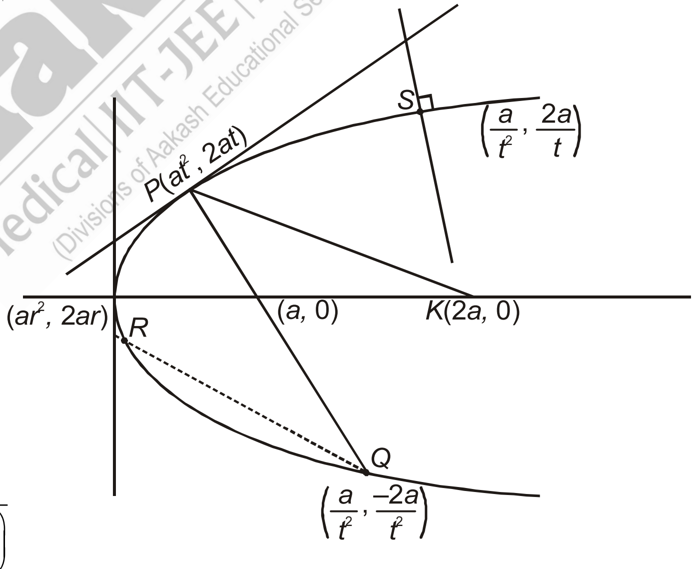
Sol. Answer (4)

Let co-ordinates of Q be $(at_2^2, 2at_2)$

Since PQ is focal chord

$t \cdot t_2 = -1$

$$\boxed{t_2 = -\dfrac{1}{t}}$$



$\therefore$ Co-ordinates of $Q\left(\dfrac{a}{t^2}, -\dfrac{2a}{t}\right)$

Slope of $PK = \dfrac{2at - 0}{at^2 - 2a} = \dfrac{2t}{t^2 - 2}$

Slope of $RQ = \dfrac{2ar + \dfrac{2a}{t}}{ar^2 - \dfrac{a}{t^2}} = \dfrac{2\left(r + \dfrac{1}{t}\right)}{r^2 - \dfrac{1}{t^2}} = \dfrac{2}{\left(r - \dfrac{1}{t}\right)}$

Slope of PK = Slope of RQ

$$\dfrac{2t}{t^2 - 2} = \dfrac{2}{r - \dfrac{1}{t}}$$

$$r - \dfrac{1}{t} = \dfrac{t^2 - 2}{t} \quad \Rightarrow \quad r = \dfrac{t^2 - 1}{t}$$

2. If $st = 1$, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is

(1) $\dfrac{(t^2+1)^2}{2t^3}$ (2) $\dfrac{a(t^2+1)^2}{2t^3}$ (3) $\dfrac{a(t^2+1)^2}{t^3}$ (4) $\dfrac{a(t^2+2)^2}{t^3}$

Sol. Answer (2)

Since $st = 1$

$$s = \frac{1}{t}$$

Co-ordinates of $S = \left(\dfrac{a}{t^2}, \dfrac{2a}{t}\right)$

Equation of tangent at P

$$y = \frac{1}{t}(x + at^2)$$

$$\boxed{x = yt - at^2} \quad ....(i)$$

Slope of normal at $S \Rightarrow -s$

$$= -\frac{1}{t}$$

Equation of normal

$$y - \frac{2a}{t} = \frac{-1}{t}\left(x - \frac{a}{t^2}\right)$$

$$-ty + 2a = x - \frac{a}{t^2}$$

$$\Rightarrow \boxed{x = -ty + 2a + \frac{a}{t^2}} \quad ....(ii)$$

From (i) and (ii),

$$y.t - at^2 = -ty + 2a + \frac{a}{t^2}$$

$$2ty = at^2 + \frac{a}{t^2} + 2a$$

$$2ty = a\left(t + \frac{1}{t}\right)^2 \Rightarrow y = \frac{a\left(t+\frac{1}{t}\right)^2}{2t}$$

$$\boxed{y = \frac{a(t^2+1)^2}{2t^3}}$$

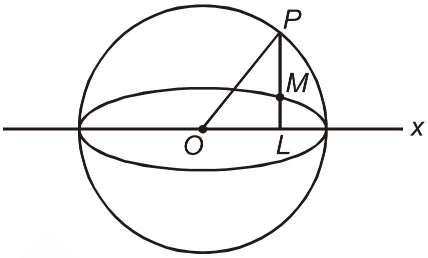
Comprehension-VII

Let $C : x^2 + y^2 = 9$, $E : \dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ and $L : y = 2x$ be three curves. P be a point on C and PL be the perpendicular to the major axis of ellipse E. PL cuts the ellipse at point M.

1. $\dfrac{ML}{PL}$ is equal to

(1) $\dfrac{1}{3}$ (2) $\dfrac{2}{3}$ (3) $\dfrac{1}{2}$ (4) 1

Sol. Answer (2)



$M(3\cos\theta, 2\sin\theta)$

$P(3\cos\theta, 3\sin\theta)$

$L(3\cos\theta, 0)$

$ML = 2\sin\theta$

$PL = 3\sin\theta$

$$\frac{ML}{PL} = \frac{2}{3}$$

2. If equation of normal to C at point P be $L : y = 2x$ then the equation of the tangent at M to the ellipse E is

(1) $x+3y\pm 3\sqrt{5}=0$ (2) $4x+3y\pm\sqrt{5}=0$ (3) $x + y \pm 3 = 0$ (4) None of these

Sol. Answer (1)

OP is a normal

$\therefore$ $P(3\cos\theta, 3\sin\theta)$ lie on $y = 2x$

$3\sin\theta = 2 \,.\, 3\cos\theta$

$$\frac{\sin\theta}{2} = \frac{\cos\theta}{1} = \frac{\sqrt{\sin^2\theta+\cos^2\theta}}{\pm\sqrt{4+1}} = \frac{1}{\pm\sqrt{5}}$$

$\therefore$ $\sin\theta = \pm\dfrac{2}{\sqrt{5}}$ and $\cos\theta = \pm\dfrac{1}{\sqrt{5}}$

Equation of tangent at M

$$\frac{x}{3}.\cos\theta + \frac{y}{2}.\sin\theta = 1$$

$$\frac{x}{3}.\frac{1}{\pm\sqrt{5}} + \frac{y}{2}.\frac{2}{\pm\sqrt{5}} = 1$$

$x + 3y \pm 3\sqrt{5} = 0$

3. If R is the point of intersection of the line L with the line $x = 1$, then

(1) R lies inside both C and E (2) R lies outside both C and E

(3) R lies on both C and E (4) R lies inside C but outside E

Sol. Answer (4)

Put $x = 1$ in $y = 2x$

$\therefore$ $R(1, 2)$

$C \equiv 1 + 4 - 9 < 0$

$E \equiv \frac{1}{9} + \frac{4}{4} - 1 > 0$

$\therefore$ R lies inside C and outside E.

Comprehension-VIII

An ellipse has its centre $C(1, 3)$ focus at $S(6, 3)$ and passing through the point $P(4, 7)$ then

1. The product of the lengths of perpendicular segments from the focii on tangent at point P is

(1) 20 (2) 45

(3) 40 (4) Cannot be determined

Sol. Answer (1)

Equation of ellipse will be assumed as

$$\frac{(x-1)^2}{a^2} + \frac{(y-3)^2}{b^2} = 1$$

$$ae = \sqrt{(6-1)^2 + 0} = 5$$

$P(4,7)$ lies on ellipse, hence $\frac{9}{a^2} + \frac{16}{b^2} = 1$

Hence, $e = \frac{\sqrt{5}}{3}$

$a^2 = 45$

$b^2 = 20$

Since product of length of perpendicular segments from the foci on tangents at P is b^2

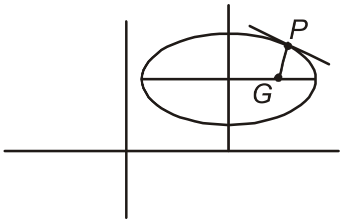
2. The point of the lines joining each focus to the foot of the perpendicular from the other focus upon the tangent at point P is

(1) $\left(\frac{5}{3}, 5\right)$ (2) $\left(\frac{4}{3}, 3\right)$ (3) $\left(\frac{8}{3}, 3\right)$ (4) $\left(\frac{10}{3}, 5\right)$

Sol. Answer (4)

Lines joining each focus to the foot of perpendicular to other focus bisects the normal at P, hence required point is mid-point of normal PG. Equation of normals at $P(4, 7)$ is

$3x - y - 5 = 0$



Point $G\left(\frac{8}{3}, 3\right)$

Required point is $\left(\frac{10}{3}, 5\right)$

3. If the normal at a variable point on the ellipse meets its axes in Q and R then the locus of the mid-point of QR is a conic with an eccentricity(e') then

(1) $e' = \frac{3}{\sqrt{10}}$ (2) $e' = \frac{\sqrt{5}}{3}$ (3) $e' = \frac{3}{\sqrt{5}}$ (4) $e' = \frac{\sqrt{10}}{3}$

Sol. Answer (2)

Locus of mid-point of QR will be ellipse of same eccentricity $e' = \frac{\sqrt{5}}{3}$

Comprehension-IX

Equation of chord of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ can be written as $T = S_1$ if its mid point is (x_1, y_1) which is

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

1. Tangents drawn from any point on circle $x^2 + y^2 = c^2$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then locus of mid point of chord of contact is

(1) $x^2 - y^2 = c^2\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right)$ (2) $x^2 + y^2 = c^2\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)$

(3) $c^2(x^2 - y^2) = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ (4) $c^2(x^2 + y^2) = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

Sol. Answer (2)

Equation of chord of contact is

$$\frac{xc\cos\theta}{a^2} + \frac{yc\sin\theta}{b^2} = 1$$

Which is same as

$$\frac{xh}{a^2} + \frac{yk}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \text{ mid point is } (h, k)$$

$$\frac{\frac{c\cos\theta}{a^2}}{\frac{h}{a^2}} = \frac{\frac{c\sin\theta}{b^2}}{\frac{k}{b^2}} = \frac{1}{\frac{h^2}{a^2} + \frac{k^2}{b^2}}$$

$$c\cos\theta = \frac{h}{\frac{h^2}{a^2} + \frac{k^2}{b^2}} \quad c\sin\theta = \frac{k}{\frac{h^2}{a^2} + \frac{k^2}{b^2}}$$

$$c^2 = \frac{h^2 + k^2}{\frac{h^2}{a^2} + \frac{k^2}{b^2}}$$

$$x^2 + y^2 = c^2\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)$$

2. Locus of mid-point of chord if it subtend right angle at origin

(1) $\frac{x^2}{a^4}+\frac{y^2}{b^4}=\left(\frac{x^2}{a^2}+\frac{y^2}{b^4}\right)^2\left(\frac{1}{a^2}+\frac{1}{b^2}\right)$

(2) $\frac{x^2}{a^4}+\frac{y^2}{b^4}=\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)\left(\frac{1}{a^2}+\frac{1}{b^2}\right)$

(3) $\frac{x^2}{a^4}+\frac{y^2}{b^4}=\frac{x^2}{a^2}-\frac{y^2}{b^2}$

(4) $\frac{x^2}{a^4}-\frac{y^2}{b^4}=\frac{x^2}{a^2}-\frac{y^2}{b^2}$

Sol. Answer (1)

$$\frac{xh}{a^2}+\frac{yk}{b^2}=\frac{h^2}{a^2}+\frac{k^2}{b^2}$$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{\frac{xh}{a^2}+\frac{yk}{b^2}}{\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2}$$

$$\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)=\left(\frac{xh}{a^2}+\frac{yk}{b^2}\right)^2$$

For 90°

$$\frac{1}{a^2}\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2-\frac{h^2}{a^4}+\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2\frac{1}{b^2}-\frac{k^2}{b^4}=0$$

$$\frac{x^2}{a^4}+\frac{y^2}{b^4}=\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)^2\left(\frac{1}{a^2}+\frac{1}{b^2}\right)$$

3. If $\left(\frac{1}{2},\frac{2}{5}\right)$ be the mid point of the chord of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ then its length is

(1) $7\sqrt{41}$ (2) $\frac{7}{5}\sqrt{41}$ (3) $\frac{7}{10}\sqrt{41}$ (4) $\frac{7}{15}\sqrt{41}$

Sol. Answer (2)

The chord whose mid-point is $\left(\frac{1}{2},\frac{2}{5}\right)$

Chord is $4x + 5y = 4$

Solving with ellipse we get points

$\left(4,\frac{-12}{5}\right)$ and $\left(-3,\frac{16}{5}\right)$

Length = $\frac{7}{5}\sqrt{41}$

Comprehension-X

Two tangents can be drawn from any external point on ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, which can be observed by equation $y = mx + \sqrt{a^2m^2 + b^2}$, since this is quadratic.

1. The equation of tangent from the point (2, 2) to the ellipse $4x^2 + 9y^2 = 36$

(1) $y = 2$ (2) $x = 2$ (3) $8x + 5y = 26$ (4) $8x + 5y = 13$

Sol. Answer (1, 3)

$$\frac{x^2}{9}+\frac{y^2}{4}=1$$

$$y = mx + \sqrt{9m^2 + 4}$$

$$2 = 2m + \sqrt{9m^2 + 4}$$

$$4 + 4m^2 - 8m = 9m^2 + 4$$

$$5m^2 + 8m = 0,\ m = 0, -\frac{8}{5}$$

Tangents are $y = 2$

$8x + 5y = 26$

2. The equation of the locus of a point from which two tangents can be drawn to the ellipse making angles θ_1, θ_2 with the major axis such that $\tan^2\theta_1 + \tan^2\theta_2 = 0$

(1) $2x^2y^2 = (x^2 - a^2)(y^2 - b^2)$ (2) $x^2y^2 = (x^2 - a^2)(y^2 - b^2)$

(3) $2x^2y^2 = (x^2 + a^2)(y^2 + b^2)$ (4) $x^2y^2 = (x^2 + a^2)(y^2 + b^2)$

Sol. Answer (1)

$$y = mx + \sqrt{a^2 + m^2 + b^2}$$

$$(y - mx)^2 = a^2m^2 + b^2$$

$$k^2 + m^2h^2 - 2hkm = a^2m^2 + b^2$$

$$m^2(h^2 - a^2) - 2khm + k^2 - b^2 = 0$$

$$m_1^2 + m_2^2 = c$$

$$(m_1 + m_2)^2 - 2m_1m_2 = c$$

$$\left(\frac{2kh}{h^2-a^2}\right)^2 - 2\frac{k^2-b^2}{h^2-a^2} = 0$$

$$2x^2y^2 = (x^2 - a^2)(y^2 - b^2)$$

3. The equation of the locus of a points from which two tangents can be drawn to the ellipse making angles θ_1, θ_2 wtih the major axis such that $\theta_1 + \theta_2 = 2\alpha$(constant)

(1) $x^2 + y^2 - 2xy\cot 2\alpha = a^2 - b^2$ (2) $x^2 - y^2 - 2xy\cot 2\alpha = a^2 - b^2$

(3) $x^2 - y^2 + 2xy\cot 2\alpha = a^2 - b^2$ (4) $x^2 - y^2 + 2xy\cot 2\alpha = a^2 + b^2$

Sol. Answer (2)

$\theta_1 + \theta_2 = 2\alpha$

$\tan(\theta_1 + \theta_2) = \tan 2\alpha$

$$\frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2} = \tan 2\alpha$$

$$\frac{m_1 + m_2}{1 - m_1 m_2} = \tan 2\alpha$$

$$\frac{\dfrac{2kh}{h^2 - a^2}}{1 - \dfrac{k^2 - b^2}{h^2 - a^2}} = \tan 2\alpha$$

$$\frac{2kh}{h^2 - k^2 + b^2 - a^2} = \tan 2\alpha$$

$\cot 2\alpha . 2xy = x^2 - y^2 + b^2 - a^2$

$x^2 - y^2 - 2xy\cot 2\alpha = a^2 - b^2$

Comprehension-XI

Tangents are drawn from the point $P(3, 4)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B. **[IIT-JEE 2010]**

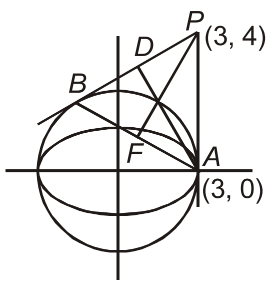
1. The coordinates of A and B are

(1) (3, 0) and (0, 2) (2) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

(3) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and (0, 2) (4) (3, 0) and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

Sol. Answer (4)

Figure is self explanatory

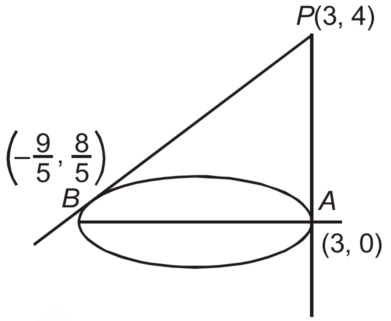


2. The orthocentre of the triangle *PAB* is

(1) $\left(5, \frac{8}{7}\right)$ (2) $\left(\frac{7}{5}, \frac{25}{8}\right)$ (3) $\left(\frac{11}{5}, \frac{8}{5}\right)$ (4) $\left(\frac{8}{25}, \frac{7}{5}\right)$

Sol. Answer (3)

Equation of *AB* is



$$y - 0 = \frac{\frac{8}{5}}{-\frac{9}{5} - 3}(x - 3) = \frac{8}{-24}(x - 3)$$

$\Rightarrow \quad y = -\frac{1}{3}(x - 3)$

$\Rightarrow \quad x + 3y = 3$ ...(i)

Equation of the straight line perpendicular to *AB* through *P* is $3x - y = 5$

Equation of *PA* is $x - 3 = 0$

The equation of straight line perpendicular to *PA* through $B\left(\frac{-9}{5}, \frac{8}{5}\right)$ is $y = \frac{8}{5}$.

Hence the orthocentre is $\left(\frac{11}{5}, \frac{8}{5}\right)$

3. The equation of the locus of the point whose distances from the point *P* and the line *AB* are equal, is

(1) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$ (2) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

(3) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$ (4) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

Sol. Answer (1)

Equation of $AB = y - 0 = -\frac{1}{3}(x - 3)$

$x + 3y - 3 = 0$

$|x + 3y - 3|^2 = 10[(x - 3)^2 + (y - 4)^2]$

(Look at coefficient of x^2 & y^2 in the answers)

Comprehension-XII

Let $F_1(x_1, 0)$ and $F_2(x_2, 0)$, for $x_1 < 0$ and $x_2 > 0$, be the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{8} = 1$. Suppose a parabola having vertex at the origin and focus at F_2 intersects the ellipse at point *M* in the first quadrant and at point *N* in the fourth quadrant. **[JEE(Advanced)-2016]**

1. The orthocentre of the triangle F_1MN is

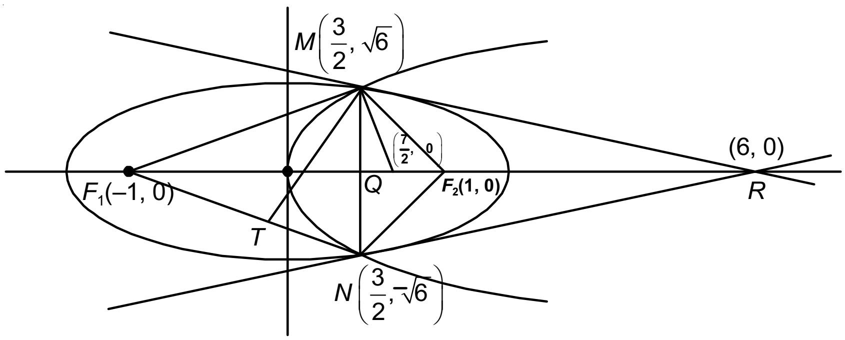
(1) $\left(-\frac{9}{10}, 0\right)$ (3) $\left(\frac{2}{3}, 0\right)$ (3) $\left(\frac{9}{10}, 0\right)$ (4) $\left(\frac{2}{3}, \sqrt{6}\right)$

Sol. Answer (1)

The parabola is $y^2 = 4x$

$\therefore$ Coordinate of $M\left(\dfrac{3}{2}, \sqrt{6}\right)$

$N\left(\dfrac{3}{2}, -\sqrt{6}\right)$



$\therefore$ Tangent to $8x^2 + 9y^2 - 72 = 0$

$\therefore$ $8\times\dfrac{3}{2}+9y\sqrt{6}-72=0$

$\therefore$ $R(6, 0)$

Normal to $y^2 = 4x$ at $\left(\dfrac{3}{2}, \sqrt{6}\right)$

$$y-\sqrt{6}=m\left(x-\frac{3}{2}\right)$$

$$y-\sqrt{6}=\frac{-\sqrt{6}}{2}\left(x-\frac{3}{2}\right)$$

$\therefore$ $Q\left(\dfrac{7}{2}, 0\right)$

$\therefore$ Orthocentre is point of intersection of line MT with x axis.

$\therefore$ Equation of MT $y-\sqrt{6}=\dfrac{5}{2\sqrt{6}}\left(x-\dfrac{3}{2}\right)$

$\therefore$ The required point is $\left(-\dfrac{9}{10}, 0\right)$

2. If the tangents to the ellipse at M and N meet at R and the normal to the parabola at M meets the x-axis at Q, then the ratio of area of the triangle MQR to area of the quadrilateral MF_1NF_2 is

(1) 3 : 4 (2) 4 : 5 (3) 5 : 8 (4) 2 : 3

Sol. Answer (3)

Area of ΔMQR = $\dfrac{1}{2}\left(6-\dfrac{7}{2}\right)\sqrt{6}$

Area of $\square MF_1NF_2$ = $2\times\dfrac{1}{2}(2)\sqrt{6}$

$\therefore$ Ratio = $\dfrac{\frac{5}{4}\sqrt{6}}{2\sqrt{6}}=\dfrac{5}{8}$

Comprehension-XIII

Let $H : x^2 - y^2 = 9$, $P : y^2 = 4(x - 5)$, $L : x = 9$ be three curves.

1. If L is the chord of contact of the hyperbola H, then the equation of the corresponding pair of tangents is

(1) $9x^2 - 8y^2 + 18x + 9 = 0$ (2) $9x^2 - 8y^2 - 18x - 9 = 0$

(3) $9x^2 - 8y^2 - 18x + 9 = 0$ (4) $9x^2 + 8y^2 + 18x + 9 = 0$

Sol. Answer (3)

Let $R(h, k)$ be point of intersection of tangents

$\therefore$ Equation of chord of contact $T = 0$

$x.h - y\,.\,k - 9 = 0$ … (i)

$x - 9 = 0$ … (ii)

These lines are coincident lines

$\frac{h}{1} = -\frac{k}{0} = \frac{-9}{-9}$

$\therefore$ $h = 1, k = 0$

$R(1, 0)$

Equation of pair of tangents

$SS' = T^2$

$(x^2 - y^2 - 9)(1 - 0\ \ 9) = (x.1 - y.0 - 9)^2$

$-8x^2 + 8y^2 + 72 = x^2 + 81 - 18x$

$9x^2 - 8y^2 - 18x + 9 = 0$

2. If R is the point of intersection of the tangent to H at the extremities of the chord L, then equation of the chord of contact of R with respect to the parabola P is

(1) $x = 7$ (2) $x = 9$ (3) $y = 7$ (4) $y = 9$

Sol. Answer (2)

$R(1, 0)$

Equation of chord of contact to the parabola

$y^2 = 4(x - 5)$

$T = 0$

$y.0 = 2(x + 1) - 20$

$0 = 2x - 18$

$x = 9$

3. If the chord of contact of R with respect to the parabola P meets the parabola at T and T', S is the focus of the parabola, then the area of the $\Delta STT'$ is in sq. units

(1) 8 (2) 9 (3) 12 (4) 16

Sol. Answer (3)

Solve equations $x = 9$ and $y^2 = 4(x - 5)$

$y^2 = 16$

$y = \pm 4$

$T(9, 4)$ and $T'(9, 4)$

$S(6, 0)$

Area of triangle = $\left|\frac{1}{2}\begin{vmatrix} 9 & 4 & 1 \\ 9 & -4 & 1 \\ 6 & 0 & 1 \end{vmatrix}\right| = \left|\frac{1}{2}[9.(-4) - 4(9-6) + 1(24)]\right|$

$\left|\frac{1}{2}[-24]\right| = |-12| = 12$ sq. units

Comprehension-XIV

Rectangular hyperbola is the hyperbola whose asymptotes are perpendicular hence its equation is $x^2 - y^2 = a^2$, if axes are rotated by 45° in clockwise direction then its equation becomes $xy = c^2$.

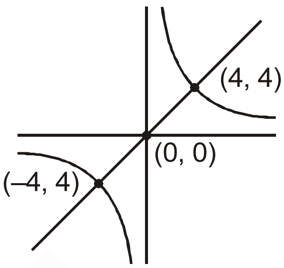
1. Focus of hyperbola $xy = 16$, is

(1) $(4\sqrt{2}, 4\sqrt{2})$ (2) $(4\sqrt{2}, 0)$ (3) $(0, 4\sqrt{2})$ (4) $(4, 0)$

Sol. Answer (1)

a = distance of vertex from centre

$= 4\sqrt{2}$



$e = \sqrt{2}$

$ae = 4\sqrt{2} \cdot \sqrt{2} = 8$

Distance of focii from centre = 8

Hence its coordinates are

$x^2 + x^2 = 8^2$

$2x^2 = 64$

$x = 4\sqrt{2}$

2. Directrix of hyperbola $xy = 16$ are

(1) $x + y = 4\sqrt{2}$ (2) $x - y = 4\sqrt{2}$ (3) $x + y = 4$ (4) $x + y = -4$

Sol. Answer (1)

Distance of direction from centre $= \dfrac{a}{e} = \dfrac{4\sqrt{2}}{\sqrt{2}} = 4$

Equation of directrix is $x + y = \pm\lambda$

$$\frac{\pm\lambda}{\sqrt{2}} = 4$$

$$\lambda = \pm 4\sqrt{2}$$

$$x + y = \pm 4\sqrt{2}$$

3. Length of minor axis of hyperbola $xy = 16$ is

(1) $4\sqrt{2}$ (2) 4 (3) $8\sqrt{2}$ (4) 8

Sol. Answer (3)

Length of minor axis is same as major axis in rectangular hyperbola

$L = 8\sqrt{2}$

Comprehension-XV

If P is a variable point and F_1 and F_2 are two fixed points such that $(PF_1 - PF_2) = 2a$. Then the locus of the point P is a hyperbola with points F_1 and F_2 as the two focii ($F_1F_2 > 2a$). If $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ is a hyperbola, then its conjugate hyperbola is $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$. Let $P(x, y)$ is a variable point such that

$$\left| \sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2} \right| = 3$$

1. If the locus of the point P represents a hyperbola of eccentricity e, then the eccentricity e' of the corresponding conjugate hyperbola is

(1) $\frac{5}{3}$ (2) $\frac{4}{3}$ (3) $\frac{5}{4}$ (4) $\frac{3}{\sqrt{7}}$

Sol. Answer (3)

$2ae = 5$

$2a = 3$

$e = \frac{5}{3}$, $b^2 = \frac{9}{4}\left(\frac{25}{9} - 1\right) = 4$

For eccentricity of conjugate hyperbola

$$\frac{1}{e^2} + \frac{1}{e'^2} = 1$$

$$\frac{1}{\frac{25}{9}} + \frac{1}{e'^2} = 1$$

$$e' = \frac{5}{4}$$

2. Locus of intersection of two perpendicular tangent to the given hyperbola is

(1) $(x-3)^2 + \left(y - \frac{7}{2}\right)^2 = \frac{55}{4}$ (2) $(x-3)^2 + \left(y - \frac{7}{2}\right)^2 = \frac{25}{4}$

(3) $(x-3)^2 + \left(y - \frac{7}{2}\right)^2 = \frac{7}{4}$ (4) None of these

Sol. Answer (4)

Equation of directrix circle will be

$$(x-3)^2 + \left(y - \frac{7}{2}\right)^2 = a^2 - b^2$$

$$= \frac{9}{4} - 4 = \text{negative}$$

Hence, no points.

3. If origin is shifted to point $\left(3, \frac{7}{2}\right)$ and the axes are rotated through an angle θ in anticlockwise sense so that equation of given hyperbola changes to the standard form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Then θ is

(1) $\tan^{-1}\left(\frac{4}{3}\right)$ (2) $\tan^{-1}\left(\frac{3}{4}\right)$ (3) $\tan^{-1}\left(\frac{5}{3}\right)$ (4) $\tan^{-1}\left(\frac{3}{5}\right)$

Sol. Answer (2)

Rotation must be equal to slope of major axis.

$$\text{Slope} = \frac{5-2}{5-1} = \frac{3}{4}$$

$$\tan\theta = \frac{3}{4}$$

Comprehension-XVI

The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B. **[IIT-JEE 2010]**

1. Equation of a common tangent with positive slope to the circle as well as to the hyperbola is

(1) $2x - \sqrt{5}y - 20 = 0$ (2) $2x - \sqrt{5}y + 4 = 0$

(3) $3x - 4y + 8 = 0$ (4) $4x - 3y + 4 = 0$

Sol. Answer (2)

Equation of tangent to hyperbola having slope m is

$$y = mx + \sqrt{9m^2 - 4} \quad ...(1)$$

Equation of tangent to circle is

$$y = m(x-4) + \sqrt{16m^2 + 16} \quad ...(2)$$

(1) and (2) will be identical for $m = \frac{2}{\sqrt{5}}$ satisfy

$\therefore$ Equation of common tangent is $2x - \sqrt{5}y + 4 = 0$

2. Equation of the circle with AB as its diameter is

(1) $x^2 + y^2 - 12x + 24 = 0$ (2) $x^2 + y^2 + 12x + 24 = 0$

(3) $x^2 + y^2 + 24x - 12 = 0$ (4) $x^2 + y^2 - 24x - 12 = 0$

Sol. Answer (1)

The equation of the hyperbola is

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

and that of circle is

$x^2 + y^2 - 8x = 0$

For their points of intersection

$$\frac{x^2}{9} + \frac{x^2 - 8x}{4} = 1$$

$\Rightarrow$ $4x^2 + 9x^2 - 72x = 36$

$\Rightarrow$ $13x^2 - 72x - 36 = 0$

$\Rightarrow$ $13x^2 - 78x + 6x - 36 = 0$

$\Rightarrow$ $13x(x-6) + 6(x-6) = 0$

$\Rightarrow$ $x = 6,\ x = -\frac{13}{6}$

$x = -\frac{13}{6}$ not acceptable

Now, for $x = 6$, $y = \pm 2\sqrt{3}$

Required equation is

$$(x-6)^2 + \left(y+2\sqrt{3}\right)\left(y-2\sqrt{3}\right) = 0$$

$\Rightarrow$ $x^2 - 12x + y^2 + 24 = 0$

$\Rightarrow$ $x^2 + y^2 - 12x + 24 = 0$

Comprehension-XVII

Answer Q.1, Q.2 and Q.3 by appropriately matching the information given in the three columns of the following table. Column 1, 2, and 3 contain conics, equations of tangents to the conics and points of contact, respectively.

Column 1	Column 2	Column 3
(I) $x^2 + y^2 = a^2$	(i) $my = m^2x + a$	(P) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$
(II) $x^2 + a^2y^2 = a^2$	(ii) $y = mx + a\sqrt{m^2+1}$	(Q) $\left(\frac{-ma}{\sqrt{m^2+1}}, \frac{a}{\sqrt{m^2+1}}\right)$
(III) $y^2 = 4ax$	(iii) $y = mx + \sqrt{a^2m^2-1}$	(R) $\left(\frac{-a^2m}{\sqrt{a^2m^2+1}}, \frac{1}{\sqrt{a^2m^2+1}}\right)$
(IV) $x^2 - a^2y^2 = a^2$	(iv) $y = mx + \sqrt{a^2m^2+1}$	(S) $\left(\frac{-a^2m}{\sqrt{a^2m^2-1}}, \frac{-1}{\sqrt{a^2m^2-1}}\right)$

[JEE(Advanced)-2017]

1. For $a = \sqrt{2}$, if a tangent is drawn to a suitable conic (Column 1) at the point of contact (–1, 1), then which of the following options is the only correct combination for obtaining its equation?

   (1) (II) (ii) (Q) (1) (III) (i) (P) (3) (I) (ii) (Q) (4) (I) (i) (P)

2. If a tangent to a suitable conic (column 1) is found to be $y = x + 8$ and its point of contact is (8, 16), then which of the following options is the only correct combination?

   (1) (III) (i) (P) (2) (II) (iv) (R) (3) (III) (ii) (Q) (4) (I) (ii) (Q)

3. The tangent to a suitable conic (Column 1) at $\left(\sqrt{3}, \frac{1}{2}\right)$ is found to be $\sqrt{3}x + 2y = 4$, then which of the following options is the only correct combination?

   (1) (IV) (iii) (S) (2) (II) (iv) (R) (3) (IV) (iv) (S) (4) (II) (iii) (R)

Solution of Comprehension-XVII

Correct combinations are

I, ii, Q

II, iv, R

III, i, P

IV, iii, S

1. Answer (3)

   (–1, 1) for $a=\sqrt{2}$ lies on

   $x^2 + y^2 = a^2$, so the only correct answer is (3).

2. Answer (1)

   As $y = x + 8$ and its point of contact (8, 16) satisfy only $y^2 = 4ax$

3. Answer (2)

   Out of the given (1), (2) are the right combination which we need to check. Out of this (Less check for)

   $x^2 + a^2y^2 = a^2$,

   $$\left(\frac{-a^2m}{\sqrt{a^2m^2+1}}, \frac{1}{\sqrt{a^2m^2+1}}\right) = \left(\frac{+4\times\frac{\sqrt{3}}{2}}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}\right)$$

   Which is given so, (2) is correct.

SECTION - D

Assertion-Reason Type Questions

1. Let $P : y^2 = 4ax$ be a parabola.

   STATEMENT-1 : From any fixed point three normals can be drawn to P.

   and

   STATEMENT-2 : Equation of normal to P is $y = mx - 2am - am^3$, and we known that cubic equation has three roots.

Sol. Answer (4)

Equation of normal $y = mx - 2am - am^3$

Let normal passes through (α, β)

$\therefore\ am^3 + (2a - \alpha)m + \beta = 0$

This equation has three roots then its derivative's equation

$3am^2 + (2a - \alpha) = 0$ has two roots

This is possible only when

$2a - \alpha < 0$

$\alpha < 2a$

Statement-1 is false and statement-2 is true.

2. STATEMENT-1 : From point (4, 0) three different normals can be drawn to the parabola $y^2 = 4x$.

   and

   STATEMENT-2 : From any point, atmost three different normals can be drawn to a hyperbola.

Sol. Answer (2)

From a point $(c, 0)$ three different normals can be drawn to the parabola

$y^2 = 4ax$

If $c > 2a \quad \therefore \quad c > 2$

Four different normals can be drawn to a hyperbola

Statement-1 and statement-2 are correct, but statement-2 is not the correct explanation of statement-1

3. STATEMENT-1 : If normal at ends of double ordinate $x = 4$ of parabola $y^2 = 4x$ meet the curve again at P and P' respectively, then $PP' = 12$ units.

and

STATEMENT-2 : If normal at t_1 of $y^2 = 4ax$ meet parabola again at t_2, then $t_2 = -t_1 - \frac{2}{t_1}$.

Sol. Answer (3)

End points of double ordinate $x = 4$ of parabola

$y^2 = 4x$ are $(4, \pm 4)$

$\Rightarrow t_1 = \pm 2$

$\Rightarrow t_2 = -t_1 - \frac{2}{t_1} = \pm 3$

$\Rightarrow P(9, 6)$ and $P'(9, -6) \Rightarrow PP' = 12$ units

4. STATEMENT-1 : If the parabola $y = (a - b)x^2 + (b - c)x + (c - a)$ touches the x-axes in the interval (0, 1) then the line $ax + by + c = 0$ always passes through a fixed point.

and

STATEMENT-2 : The equation $L_1 + \lambda L_2 = 0$ or $\mu L_1 + vL_2 = 0$ represents a line passing through intersection of lines $L_1 = 0$ and $L_2 = 0$ which is a fixed point, when λ, μ, v are constants.

Sol. Answer (1)

$\because\ (a - b) + (b - c) + (c - a) = 0$

$\therefore\ x = 1$ is a root

Solving equation of parabola with x-axis *i.e.* $y = 0$ we get, $(a - b)x^2 + (b - c)x + (c - a) = 0$ which should have two equal values of x, as x-axis touches parabola

$\therefore\ |x| = \frac{c-a}{a-b} \Rightarrow -2a + b + c = 0$ ...(i)

Given, $ax + by + c = 0$ ...(ii)

From (i) and (ii), $a(x + 2) + b(y - 1) = 0$

Which is a family of line

$\therefore\ x + 2 = 0$ and $y - 1 = 0$

$\Rightarrow (-2, 1)$

$\Rightarrow ax + by + c = 0$ always passes through $(-2, 1)$

5. STATEMENT-1 : The lines from vertex to the two extremities of a focal chord of parabola $y^2 = 4ax$ are at an angle $\pi/2$.

and

STATEMENT-2 : If extremities of focal chord of parabola are $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ then $t_1t_2 = -1$.

Sol. Answer (4)

If $P \equiv (at_1^2, 2at_1)$ and $Q \equiv (at_2^2, 2at_2)$, vertex $A \equiv (0, 0)$

$\therefore$ Slope of $AP \times$ slope of AQ = $\left(\dfrac{2at_1 - 0}{2at_1^2 - 0}\right) \times \left(\dfrac{2at_2 - 0}{at_2^2 - 0}\right)$

$= \dfrac{2}{t_1} \times \dfrac{2}{t_2} = \dfrac{4}{t_1 t_2} = \dfrac{4}{-1} = -4 \neq -1 \qquad \{\because t_1 t_2 = -1\}$

$\therefore$ $\angle PQ$, $Q \neq \pi/2$

6. STATEMENT-1 : The length of latus rectum of the parabola $(x - y + 2)^2 = 8\sqrt{2}\{x + y - 6\}$ is $8\sqrt{2}$.

and

STATEMENT-2 : The length of latus rectum of parabola $(y - a)^2 = 8\sqrt{2}(x - b)$ is $8\sqrt{2}$.

Sol. Answer (3)

$\because$ $x - y + 2 = 0$ and $x + y - 6 = 0$ are perpendicular to each other

$\therefore$ $2\left\{\dfrac{x - y + 2}{\sqrt{2}}\right\}^2 = 8\sqrt{2}\left\{\dfrac{x + y - 6}{\sqrt{2}}\right\}\sqrt{2}$

$\therefore$ $\left\{\dfrac{x - y + 2}{\sqrt{2}}\right\}^2 = 8\left\{\dfrac{x + y - 6}{\sqrt{2}}\right\}$

Let $\dfrac{x - y + 2}{\sqrt{2}} = Y$ and $\dfrac{x + y - 6}{\sqrt{2}} = X$

$\Rightarrow$ $Y^2 = 8X$

$\therefore$ Length of latus rectum is 8

7. STATEMENT-1 : The curve $y = \dfrac{-x^2}{2} + x + 1$ is symmetric with respect to the line $x = 1$ **.[IIT-JEE 2007]**

and

STATEMENT-2 : A parabola is symmetric about its axis.

Sol. Answer (1)

Both are true and statement-2 clearly explains statement-1.

8. STATEMENT-1 : Two tangents are drawn from the point $(\sqrt{24}, 1)$ to $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$ then they must be perpendicular.

and

STATEMENT-2 : The equation of the director circle to $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$ is given by $x^2 + y^2 = 25$.

Sol. Answer (1)

Equation of director circle of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $x^2+y^2=a^2+b^2$

$\therefore$ Equation of director circle of $\frac{x^2}{16}+\frac{y^2}{9}=1$ is given by

$x^2 + y^2 = 16 + 9 = 25$

Point ($\sqrt{24}$, 1) lies on this curve

$\therefore$ Statement-2 is correct explanation of statement-1.

9. STATEMENT-1 : The points on the auxiliary circle corresponding to P and D subtend a right angle at the centre.

and

STATEMENT-2 : If the eccentric angles of the ends, P and D of a pair of conjugate diameters be ϕ and ϕ' differ by a right angle.

Sol. Answer (1)

Let P is ($a\cos\phi$, $b\sin\phi$), the equation to CP is

$$y = x\cdot\frac{b}{a}\tan\phi$$

So, the equation to CD is

$$y = x\cdot\frac{b}{a}\tan\phi'$$

These diameters are conjugate if

$$\frac{b^2}{a^2}\tan\phi\cdot\tan\phi' = -\frac{b^2}{a^2}$$

$\tan\phi = -\cot\phi' = \tan(\phi' \pm 90°)$

$\Rightarrow \phi - \phi' = \pm 90°$

The points on the auxiliary circle corresponding to P and D subtends a right angle at the centre

For if p and be these points, then

$\angle PCA' = \phi$ and $\angle dCA' = \phi'$

Hence, $\angle pCd = \angle dcA' - \angle pcA' = \phi - \phi' = 90°$

10. STATEMENT-1 : If P and D be the ends of conjugate diameters, then the locus of mid-point of PD is a circle.

and

STATEMENT-2 : If P and D be the ends of conjugate diameter, then the locus of intersection of tangents at P and D is an ellipse.

Sol. Answer (4)

(i) If (x, y) be the mid-point of PD, we have

$$h = \frac{a\cos\phi - a\sin\phi}{2}$$

$$k = \frac{b\sin\phi + b\cos\phi}{2}$$

Eliminating ϕ we get

$$\left(\frac{2h}{a}\right)^2+\left(\frac{2k}{b}\right)=2$$

Or, $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=\dfrac{1}{2}$

Which is an ellipse NOT a circle

(ii) The tangents are

$$\frac{x}{a}\cos\phi+\frac{y}{b}\sin\phi=1$$

And $-\dfrac{x}{a}\sin\phi+\dfrac{y}{b}\cos\phi=1$

Both of these equations hold at the intersection of the tangents. If we eliminate ϕ, we shall have the equation of the locus of their intersections

So, $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=2$

Which is again an ellipse

11. STATEMENT-1 : The tangent at the extremity of any diameter is parallel to the chords which it bisects.

and

STATEMENT-2 : Chords of the $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ always touches the concentric and coaxal ellipse $\dfrac{x^2}{\alpha^2}+\dfrac{y^2}{\beta^2}=1$, then the locus of their poles is an ellipse.

Sol. Answer (2)

Let (x', y') be the point on the ellipse, the tangent at which is parallel to the chord QR, whose equation is

$y = mx + c$

The tangent at point (x', y') is

$$\frac{xx'}{a^2}+\frac{yy'}{b^2}=1$$

Since equations (i) and (ii) are parallel, we have

$$m_1=-\frac{b^2x'}{a^2y'}$$

i.e. the point lies on the straight line

$$y=-\frac{b^2}{a^2m}x$$

Any tangent to the second ellipse is

$$y=mx+\sqrt{\alpha^2m^2+\beta^2} \qquad \text{...(A)}$$

Let the tangent at the points where it meets the first ellipse meet in (h, k). Then the equation must be the same as the polar of (h, k) w.r.t. the first ellipse *i.e.*, it is the same as

$$\frac{xh}{a^2}+\frac{yk}{b^2}-1=0 \qquad \text{...(B)}$$

Since (A) and (B) must be identical

$$\frac{m}{\frac{h}{a^2}} = -\frac{1}{\frac{k}{b^2}} = \sqrt{\frac{\alpha^2 m^2 + \beta^2}{-1}}$$

$$m = -\frac{b^2}{a^2}\frac{h}{k} \text{ and } \sqrt{\alpha^2 m^2 + \beta^2} = \frac{b^2}{k}$$

Eliminating m, we have

$$\alpha^2 \frac{b^4}{a^4}\frac{h^2}{k^2} + \beta^2 = \frac{b^4}{k^2}$$

i.e., The point (h, k) lies on the ellipse

$$\frac{\alpha^2}{a^4}x^2 + \frac{\beta^2}{b^4}y^2 = 1$$

12. STATEMENT-1 : Equation of director circle of the hyperbola $xy = c^2$ is a point circle.

    and

    STATEMENT-2 : For rectangular hyperbola, $a = b$ *i.e.* $x^2 - y^2 = 0$.

Sol. Answer (1)

13. STATEMENT-1 : If eccentricity of a hyperbola is 2, then eccentricity of its conjugate hyperbola is $\frac{2}{\sqrt{3}}$.

    and

    STATEMENT-2 : If e and e' are the eccentricity of hyperbola and its conjugate hyperbola, then $\frac{1}{e^2} + \frac{1}{e'^2} = 1$.

Sol. Answer (1)

14. STATEMENT-1 : A hyperbola and its conjugate hyperbola have the same asymptotes.

    and

    STATEMENT-2 : In a second degree curve, equation of asymptotes, if exists differ by constant only.

Sol. Answer (1)

15. STATEMENT-1 : The line $3x + 4y = 5$ intersects the hyperbola $9x^2 - 16y^2 = 144$ only at one point.

    and

    STATEMENT-2 : Given line is parallel to an asymptotes of the hyperbola.

Sol. Answer (1)

16. STATEMENT-1 : If lines $y = m_1x$ and $y = m_2x$ are the conjugate diameter of the hyperbola $xy = c^2$, then $m_1 + m_2 = 0$.

    and

    STATEMENT-2 : Two lines are called conjugate diameter of hyperbola if they bisect the chords parallel to each other.

Sol. Answer (1)

17. STATEMENT-1 : Tangents at any point $P(x_1, y_1)$ on the hyperbola $xy = c^2$ meets the co-ordinate axes at points Q and R, the circumcentre of ΔOQR has co-ordinate (x_1y_1).

    and

    STATEMENT-2 : Equation of tangent at point (x_1y_1) to the curve $xy = c^2$ is $\frac{x}{x_1} + \frac{y}{y_1} = 2$.

Sol. Answer (2)

18. STATEMENT-1 : From any point, at most four tangents can be drawn to a hyperbola.

and

STATEMENT-2 : From the centre of a hyperbola, four tangents can be drawn which are known as asymptotes.

Sol. Answer (4)

Statement-1 is false because at most two tangents can be drawn from any point.

19. STATEMENT-1 : The line $y = \frac{b}{a}x$ will not meet the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $(a > b > 0)$.

and

STATEMENT-2 : The line $y = \frac{b}{a}x$ is an asymptote to the hyperbola.

Sol. Answer (1)

Asymptote to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \quad \therefore \quad y = \pm\frac{b}{a}x$$

Statement-2 is correct explanation of statement-1

20. STATEMENT-1 : Lines $3x - 4y + 7 = 0$ and $4x + 3y + 8 = 0$ are the asymptotes of rectangular hyperbola.

and

STATEMENT-2 : In rectangular hyperbola, asymptotes intersect at right angle.

Sol. Answer (1)

21. STATEMENT-1 : Foci at the hyperbola $f(x, y)$ $7y^2 - 9x^2 + 54x - 28y - 116 = 0$ are (3, 6) and (3, –2).

and

STATEMENT-2 : Foci is the solution of $\frac{df(x,y)}{dx} = 0$ and $\frac{df(x,y)}{dy} = 0$.

Sol. Answer (3)

SECTION - E

Matrix-Match Type Questions

1. Normals are drawn at points P, Q and R lying on the parabola $y^2 = 4x$, which intersect at (3, 0). Then

Column-I		Column-II
(A) Area of ΔPQR	(p)	2
(B) Radius of circumcircle of ΔPQR	(q)	$\frac{5}{2}$
(C) Centroid of ΔPQR	(r)	$\left(\frac{5}{2}, 0\right)$
(D) Circumcentre of ΔPQR	(s)	$\left(\frac{2}{3}, 0\right)$

Sol. Answer : A(p), B(q), C(s), D(r)

Equation of normal at $(at^2, 2at)$

$y + tx = 2at + at^3$ Put $a = 1$

$\therefore$ Normal at $(t^2, 2t)$

$y + tx = 2t + t^3$ passes through (3, 0)

$3t = 2t + t^3$

$t^3 = t$ gives $t = 0, 1, -1$

$P(0, 0)$ $Q(1, 2)$ $R(1, -2)$

Area of $\Delta PQR = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & -2 & 1 \end{vmatrix} = \left|\frac{1}{2}(-4)\right| = 2$ sq. unit

Let $c(a, b)$ be the circumcentre of ΔPQR

$\therefore$ $CP^2 = CQ^2 = CR^2$

$a^2 + b^2 = (a-1)^2 + (b-2)^2 = (a-1)^2 + (b+2)^2$

$\therefore$ $b = 0,\ a = \frac{5}{2}$

Circumcentre $c\left(\frac{5}{2}, 0\right)$

Circum radius $= CP = \sqrt{\frac{25}{4}} = \frac{5}{2}$

Centroid of $\Delta PQR = \left(\frac{2}{3}, 0\right)$

2. Match column I to column II according to the given condition.

Column I	**Column II**
(A) Tangents at the ends of focal chord for a parabola	(p) Meet at directrix
(B) Let P is a point on the parabola and tangent at P meets the directrix at Q. If S is the focus of parabola then SP and SQ	(q) Meet at an angle of 90°
(C) If $A\left(at_1^2, 2at_1\right)$ and $B\left(at_2^2, 2at_2\right)$ are the ends of a focal chord of $y^2 = 4ax$ then	(r) $t_1 t_2 = -1$
(D) If the normal at $\left(at_1^2, 2at_1\right)$ meets the parabola $y^2 = 4ax$ again at $\left(at_2^2, 2at_2\right)$ then	(s) $t_2 = -t_1 - \frac{2}{t_1}$
	(t) $t_1t_2 + t_1^2 + 2 = 0$

Sol. Answer : A(p, q), B(q), C(r), D(s, t)

(A) Tangent at the each of focal chord meet at 90° and meet at directrix

(B) The angle between SP and SQ is 90°

(C) Condition for focal chord $t_1t_2 = -1$

(D) Condition for normal chord

$$t_2 = -t_1 - \frac{2}{t_1} \qquad \Rightarrow t_1t_2 = -t_1^2 - 2, \qquad t_1t_2 + t_1^2 + 2 = 0$$

3. Let (x, y) be such that $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$

Column-I	Column-II
(A) If $a = 1$ and $b = 0$, then (x, y)	(p) Lies on the circle $x^2 + y^2 = 1$
(B) If $a = 1$ and $b = 1$, then (x, y)	(q) Lies on $(x^2 - 1)(y^2 - 1) = 0$
(C) If $a = 1$ and $b = 2$, then (x, y)	(r) Lies on $y = x$
(D) If $a = 2$ and $b = 2$, then (x, y)	(s) Lies on $(4x^2 - 1)(y^2 - 1) = 0$

Sol. Answer : A(p), B(q), C(p), D(s)

$$\cos^{-1}y + \cos^{-1}(bxy) = \frac{\pi}{2} - \sin^{-1}(ax)$$

$$\cos^{-1}\left[y.bxy - \sqrt{1-y^2}.\sqrt{1-b^2x^2y^2}\right] = \cos^{-1}(ax)$$

$$bxy^2 - \sqrt{1-y^2}\sqrt{1-b^2x^2y^2} = ax$$

$$(1-y^2)(1-b^2x^2y^2) = (bxy^2 - ax)^2$$

$$a^2x^2 + b(b-2a)x^2y^2 + y^2 = 1$$

(A) If $a = 1$, $b = 0$

$x^2 + y^2 = 1$

(B) If $a = 1$, $b = 1$

$x^2 - x^2y^2 + y^2 - 1 = 0$

$(x^2 - 1)(y^2 - 1) = 0$

(C) $a = 1$, $b = 2$

$x^2 + y^2 = 1$

(D) $a = 2$, $b = 2$

$(4x^2 - 1)(y^2 - 1) = 0$

4. Match the following

	Column I		Column II
(A)	Product of the perpendicular from foci of any tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is	(p)	$x=\pm\frac{16}{\sqrt{7}}$
(B)	The equation $\frac{x^2}{3-a}+\frac{y^2}{2-a}=1$ represents an ellipse, then the value of a is	(q)	$(-\infty, 2)$
(C)	The common tangent of $x^2 + y^2 = 4$ and $x^2 + 2y^2 = 2$ is	(r)	4
(D)	Polar of the focus to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ is	(s)	Not defined
		(t)	$2x - 3y + 4 = 0$

Sol. Answer A(r), B(q), C(s), D(p)

(A) $E:\frac{x^2}{9}+\frac{y^2}{4}=1$

Product of the perpendicular from the foci of any tangent to an ellipse

= square of the semi-minor axis

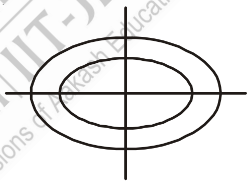
= 4

(B) $3 - a > 0$, $2 - a > 0$

$\Rightarrow$ $a < 3$, $a < 2 \Rightarrow a \in (-\infty, 2)$

(C) $C : x^2 + y^2 = 4$, $E :$ $ax^2 + 2y^2 = 2$

$$\Rightarrow \quad \frac{x^2}{2}=\frac{y^2}{1}=1$$



Not possible

(D) $E:\frac{x^2}{16}+\frac{y^2}{9}=1$

$$\therefore \quad e=\sqrt{1-\frac{9}{16}}=\frac{\sqrt{7}}{4}$$

$$F:(ae,0)=\left(4\cdot\frac{\sqrt{7}}{4},0\right)=(\sqrt{7},0)$$

Polar of the focus $(\sqrt{7},0)$ is

$$\frac{xx_1}{16}+\frac{yy_1}{9}=1$$

$$\Rightarrow \quad \frac{x}{16}\times\sqrt{7}+0=1$$

$$\Rightarrow \quad x=\frac{16}{\sqrt{7}}$$

5. Consider the ellipse $x^2 + 2y^2 - 4x - 4y + 4 = 0$ and match column I to column II according to the given condition.

Column I	Column II
(A) The eccentricity of the ellipse is	(p) $\frac{1}{\sqrt{2}}$
(B) The length of latus rectum	(q) $\sqrt{2}$
(C) The area of parallelogram formed by the tangents at the end of conjugate diameters is always greater than	(r) 4
(D) The sum of squares of two semi-conjugate diameters is greater than	(s) $2\sqrt{2}$
	(t) 2

Sol. Answer : A(p), B(q), C(p, q, r, s, t), D(p, q, s, t)

The given ellipse can be written as

$$\frac{(x-2)^2}{2}+\frac{(y-1)^2}{1}=1$$

$\Rightarrow a^2 = 2,\ b^2 = 1$

(A) Eccentricity = $e=\sqrt{1-\frac{1}{2}}=\frac{1}{\sqrt{2}}$

(B) The length of latus rectum = $\frac{2b^2}{a}=\frac{2\times 1}{\sqrt{2}}=\sqrt{2}$

(C) Area = $4ab = 4\sqrt{2}$

(D) Sum = $a^2 + b^2 = 2 + 1 = 3$

6. Match the following

Column I	Column II
(A) Angle between the pair of tangents drawn from the point (1, 2) to the ellipse $2x^2 + 3y^2 = 6$ is	(p) 30°
(B) If the locus of a point p to the ellipse $3x^2 + 4y^2 = 12$ is $x^2 + y^2 = 7$, then the angle between the tangents drawn at p is	(q) 60°
(C) If θ and ϕ are the eccentric angles of the extremities of two conjugate diameters then $\theta - \phi$ is	(r) 45°
(D) If the angle between the focal radii of an ellipse at a point is 90°, then the angle between a normal and focal radii at that point is	(s) 90°
	(t) 120°

Sol. Answer A(s), B(s), C(s), D(r)

(A) E : $2x^2 + 3y^2 = 6$

$$\Rightarrow \quad \frac{x^2}{3} = \frac{y^2}{2} = 1$$

Any tangent to the ellipse be

$$y = mx + \sqrt{a^2m^2 + b^2}$$

$$\Rightarrow \quad y = mx + \sqrt{3m^2 + 2}$$

Which is passing through (1, 2)

$$\therefore \quad 2 = m + \sqrt{3m^2 + 2}$$

$\Rightarrow$ $(2 - m)^2 = 3m^2 + 2$

$\Rightarrow$ $4 - 4m + m^2 - 3m^2 - 2 = 0$

$\Rightarrow$ $-2m^2 - 4m + 2 = 0$

$\Rightarrow$ $m^2 + 2m - 1 = 0$

m_1, m_2

$\therefore$ $m_1m_2 = -1$

(B) $E : 3x^2 + 4y^2 = 12$

$$\Rightarrow \quad \frac{x^2}{4} + \frac{y^2}{3} = 1$$

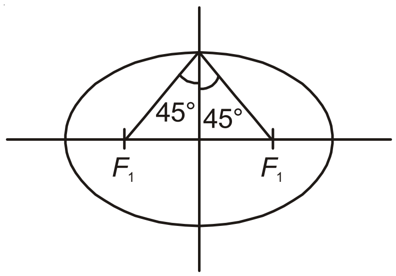
Equation of D.C. is $x^2 + y^2 = 7$

$\Rightarrow$ Angle between them is 90°

(C) $\theta - \phi = 90°$

(D) The tangent and normal at any point of an ellipse bisects the angles between the focal radii of that point

$\therefore$ Angle is 45°



7. Match the following

Column I		**Column II**
(A) Equation of the diameter passes through (0, 0) w.r.t. an ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is	(p)	$\frac{x^2}{9} + \frac{y^2}{4} = 1$

(B) For all real value of m, the straight line $y = mx + \sqrt{9m^2 4}$ is a tangent to the curve is — (q) $y = 3$

(C) Equation of common tangents to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the circle $x^2 + y^2 = 9$ are — (r) $2x + 3y = 6$

(D) From a point $P(3, 2)$ tangents are drawn to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and intersects the ellipse at Q & R. Then the locus of P is — (s) $y = -\frac{16x}{25m}$

(t) $y = -3$

Sol. Answer A(s), B(p), C(q, t), D(r)

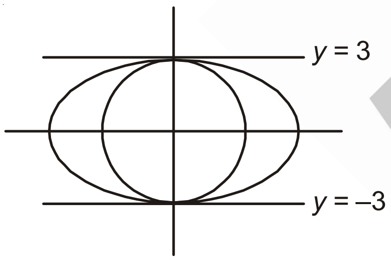
(A) Equation of diameter w.r.t.

An ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is

$$y = -\frac{b^2 x}{a^2 m} = -\frac{16}{25m}x$$

(B) T : $y = mx + \sqrt{9m^2 + 4}$

E : $\frac{x^2}{9} + \frac{y^2}{4} = 1$

(C) 

(D) Locus of P is a chord of contact

$$\therefore \quad \frac{x}{3} + \frac{y}{2} = 1$$

$$\Rightarrow \quad 2x + 3y = 6$$

8. Match the following

Column-I	Column-II
(A) Locus of point of intersection of $x = at^2$, $y = 2at$	(p) $x^2 + y^2 = 2a^2$
(B) Director circle of $x^2 + y^2 = a^2$	(q) $y^2 = 4ax$
(C) Locus of point of intersection of the lines $x \cos\theta = y \cot\theta = a$	(r) $x^2 + y^2 = ax$
(D) The locus of mid point of the chords of the circle $x^2 + y^2 - 2ax = 0$ passing through the origin	(s) $x^2 - y^2 = a^2$

Sol. Answer : A(q), B(p), C(s), D(r)

(A) $x = at^2$ and $y = 2at$ eliminate t

$$\therefore \quad x = a\cdot\left(\frac{y}{2a}\right)^2 \quad \Rightarrow \quad 4ax = y^2$$

(B) Director circle of $x^2 + y^2 = r^2$ is $x^2 + y^2 = 2r^2$.

(C) $\cos\theta = \frac{a}{x}, \ \cot\theta = \frac{a}{y}$

$\sec\theta = \frac{x}{a}, \ \tan\theta = \frac{y}{a}$

w.r.t. $\sec^2\theta - \tan^2\theta = 1$

$\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$

$x^2 - y^2 = a^2$ locus

(D) Let $P(h, k)$ be mid-point of chord AB.

Equal of chord $T = S_1$

$x\cdot h + y\cdot k - a(x+h) = h^2 + k^2 - 2ah$

Chord passes through (0, 0).

$\therefore \quad -a(h) = h^2 + k^2 - 2ab$

$\therefore$ Locus of $P(h, k)$

$x^2 + y^2 = ax$

$S = x^2 + y^2 + x - y - z = 0$

9. Match the following

Column-I		Column-II
(A) Vertex of the parabola $y^2 + 6x - 2y + 13 = 0$	(p)	(5, 3)
(B) Centre of the ellipse $25x^2 + 9y^2 - 250x - 54y + 481 = 0$	(q)	(1, –1)
(C) Focus of the parabola $(y + 1)^2 = 12(x + 2)$	(r)	(–2, 1)
(D) End of latus rectum of the hyperbola $\frac{x^2}{9} - \frac{y^2}{1} = 1$	(s)	$\left(\sqrt{10}, \frac{1}{3}\right)$

Sol. Answer : A(r), B(p), C(q), D(s)

(A) $y^2 - 2y = -6x - 13$

$(y - 1)^2 - 1 = -6x - 13$

$(y - 1)^2 = -6(x + 2)$

Vertex of parabola (–2, 1)

(B) $25(x^2 - 10x) + 9(y^2 - 6y) + 481 = 0$

$25(x^2 - 10x + 25 - 25) + 9(y^2 - 6y + 9 - 9) + 481 = 0$

$25(x - 5)^2 - 625 + 9(y - 3)^2 - 81 + 481 = 0$

$25(x - 5)^2 + 9(y - 3)^2 = 225$

$$\frac{(x-5)^2}{9} + \frac{(y-3)^2}{25} = 1$$

Centre of ellipse (5, 3)

(C) $(y + 1)^2 = 12(x + 2)$

Shifting origin at (–2, –1)

$y^2 = 12x4a = 12$; $a = 3$

Focus of parabola (3, 0) with respect to original axis (1, –1)

(D) $a^2 = 9$, $b^2 = 1$

$$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{1}{9} = \frac{10}{9}$$

$$e = \frac{\sqrt{10}}{3}$$

Focus $(\pm ae, 0) = (\pm\sqrt{10}, 0)$

Semi latus rectum = $\frac{b^2}{a} = \frac{1}{3}$

$\therefore$ End points of latus rectum $\left(\pm\sqrt{10}, \pm\frac{1}{\sqrt{3}}\right)$.

10. If the circle $x^2 + y^2 = 1$ cuts the rectangular hyperbola $xy = 1$ in four distinct points (x_i, y_i), $i = 1, 2, 3, 4$; then

Column-I	**Column-II**
(A) $x_1 + x_2 + x_3 + x_4$ is	(p) 0
(B) $y_1 + y_2 + y_3 + y_4$ is	(q) 5
(C) $x_1 x_2 x_3 x_4$ is	(r) –1
(D) $y_1 y_2 y_3 y_4$ is	(s) 1

Sol. Answer A(p), B(p), C(s), D(s)

R.H : $xy = 1$

$e : x^2 + y^2 = 1$

$\Rightarrow x^2 + \frac{1}{x^2} = 1$

$\Rightarrow x^4 - x^2 + 1 = 0$

x_1, x_2, x_3, x_4

$\therefore x_1 + x_2 + x_3 + x_4 = 0$

$x_1 x_2 x_3 x_4 = 1$

Similarly,

$y_1 + y_2 + y_3 + y_4 = 0$

$y_1 y_2 y_3 y_4 = 1$

11. If the normal of (x_i, y_i), $i = 1, 2, 3, 4$ to the rectangular hyperbola $xy = 2$ meet at the point (3, 4) then

Column-I	**Column-II**
(A) $x_1 + x_2 + x_3 + x_4$ is	(p) 4
(B) $y_1 + y_2 + y_3 + y_4$ is	(q) –4
(C) $x_1 x_2 x_3 x_4$ is	(r) (3)
(D) $y_1 y_2 y_3 y_4$ is	(s) 8

Sol. Answer A(r), B(p), C(q), D(q)

Given R.H. : $xy = 2$ ...(i)

Any point on (i) is $P\left(\sqrt{2}\,t, \frac{\sqrt{2}}{t}\right)$

$\Rightarrow \quad y = \frac{2}{x}$

$\Rightarrow \quad \frac{dy}{dx} = -\frac{2}{x^2}$

$\therefore \quad m = \frac{\left(\frac{dy}{dx}\right)}{P} = -\frac{2}{2t^2} = -\frac{1}{t^2}$

Equation of normal at P is

$$y - \frac{\sqrt{2}}{t} = t^2\,(x - \sqrt{2}\,t)$$

$\Rightarrow \quad ty = t^3x + \sqrt{2} - \sqrt{2}\,t^4$

Which is passing through (3, 4)

$\therefore \quad 4t = 3t^3 + \sqrt{2} - \sqrt{2}\,t^4$

$\Rightarrow \quad \sqrt{2}\,t^4 - 3t^3 + 4t - \sqrt{2} = 0$

Let its roots be t_1, t_2, t_3, t_4.

$\therefore \quad t_1 + t_2 + t_3 + t_4 = \frac{3}{\sqrt{2}}$ $\qquad (x - \sqrt{2}\,t)$

$\Rightarrow \quad \frac{x_1}{\sqrt{2}} + \frac{x_2}{\sqrt{2}} + \frac{x_3}{\sqrt{2}} + \frac{x_4}{\sqrt{2}} = \frac{3}{2}$ $\qquad t_1 = \frac{x_1}{\sqrt{2}}$

$\Rightarrow \quad x_1 + x_2 + x_3 + x_4 = 3$ $\qquad y_1 = \frac{\sqrt{2}}{t_1}$

Also, $\Sigma t_1t_2 = 0$, $\Sigma t_1t_2t_3 = -\frac{4}{\sqrt{2}}$, $\Sigma t_1t_2t_3t_4 = -1$

Now, $y_1 + y_2 + y_3 + y_4$

$$= \sqrt{2}\left(\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}+\frac{1}{t_4}\right)$$

$$= \sqrt{2}\,\frac{\Sigma t_1 t_2 t_3}{t_1 t_2 t_3 t_4}$$

$$= \sqrt{2}\times\frac{-\frac{4}{\sqrt{2}}}{-1}$$

$$= 4$$

$$\therefore\quad x_1x_2x_3x_4 = \sqrt{2}\,t_1\times\sqrt{2}\,t_2\times\sqrt{2}\,t_3\times\sqrt{2}\,t_4$$

$$= 4t_1t_2t_3t_4$$

$$= -4$$

$$\therefore\quad y_1\,y_2\,y_3\,y_4 = \frac{4}{t_1t_2t_3t_4} = -4$$

12. If the length of the transverse and conjugate axes are equal of a hyperbola, then it is called a rectangular hyperbola. Let the rectangular hyperbola be $xy = c^2$. Then

Column-I		**Column-II**
(A) Equation of directrices are	(p)	$2\sqrt{2}c$
(B) The length of latus rectum is	(q)	$\frac{x}{t}+ty2=c$
(C) Equation of tangent in parametric form is	(r)	$y+x=\pm\sqrt{2}c$
(D) Equation of normal in parametric form is	(s)	$xt^3 - yt = c(t^4 - 1)$

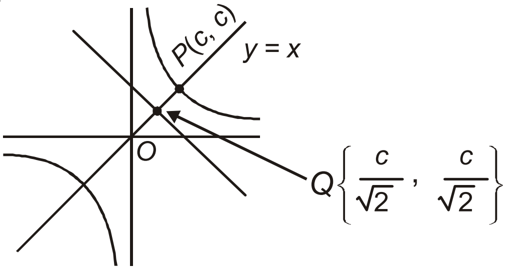
Sol. Answer A(r), B(p), C(q), D(s)

Given R.H. is $xy = c^2$

Axis $x - y = 0$

$D :\ x + y \pm \lambda = 0$

Which is passes through Q



$$\therefore\quad \lambda = \pm\left(\frac{c}{\sqrt{2}}+\frac{c}{\sqrt{2}}\right) = \pm c\sqrt{2}$$

Equation of directrices are

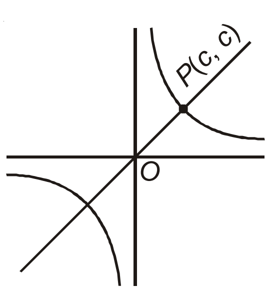
$$x + y = \pm c\sqrt{2}$$

Length of latus rectum

Here $a = OP = \sqrt{c^2+c^2} = 2\sqrt{c}$

$$\text{L.R.} = \frac{2b^2}{a} = \frac{2a^2}{a} = 2a = 2\sqrt{2}\,c$$

Any point on the rectangular hyperbola is $M\left(ct, \dfrac{c}{t}\right)$



Equation of tangent at M is

$$xy_1 + x_1y = c^2$$

$$\Rightarrow \quad x \cdot \frac{c}{t} + ct \cdot y = c^2$$

$$\Rightarrow \quad \frac{x}{t} + yt = c$$

Equation of normal at M is

$$y - \frac{c}{t} = t^2(x - ct)$$

$$\Rightarrow \quad ty - c = t^3(x - ct)$$

$$= t^3x - ct^4$$

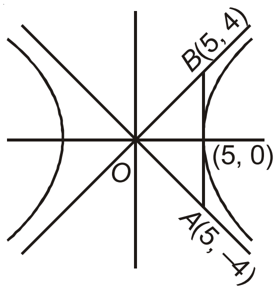
$$\Rightarrow \quad t^3x - yt - ct^4 + c = 0$$

13. Match the following

Column-I	**Column-II**
(A) The area of the Δ that a tangent at a point on the hyperbola $\dfrac{x^2}{25} - \dfrac{y^2}{16} = 1$ makes with its asymptotes is	(p) 96
(B) If λ be the length of the latus rectum of the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$ then 9λ is	(q) $\pm \dfrac{5}{2}$
(C) Equation of the transverse axis of the hyperbola $(x - 3)^2 + (y + 1)^2 = (4x + 3y)^2$ is	(r) 20
(D) If the distance between two parallel tangents to the hyperbola $\dfrac{x^2}{9} - \dfrac{y^2}{49} = 1$ is 2, then their slope is	(s) $3x - 4y = 13$

Sol. Answer A(r), B(p), C(s), D(q)

(A) $H : \dfrac{x^2}{25} - \dfrac{y^2}{16} = 1$



$$\Rightarrow y = \pm \frac{4}{5}x$$

$$\text{ar}(\Delta OAB) = \frac{1}{2} \times 5 \times 8$$

$$= 20 \text{ sq. units}$$

(B) Given hyperbola be

$$16x^2 - 9y^2 + 32x + 36y - 164 = 0$$

$\Rightarrow$ $16(x^2 + 2x) - 9(y^2 - 4y) = 164$

$\Rightarrow$ $16(x^2 + 2x + 1) - 9(y^2 - 4y + 4) = 164 + 16 - 36 = 144$

$\Rightarrow$ $16(x + 1)^2 - 9(y - 2)^2 = 144$

$\Rightarrow$ $\dfrac{(x+1)^2}{9} - \dfrac{(y-2)^2}{16} = 1$

$\therefore$ $\lambda = \text{L.R.} = \dfrac{2b^2}{a} = \dfrac{2\times 16}{3} = \dfrac{32}{3}$

$\Rightarrow$ $9\lambda = 9 \times \dfrac{32}{3} = 96$

(C) Given hyperbola is $(x - 3)^2 + (y + 1)^2 = (4x + 3y)^2$

$\Rightarrow$ $(x - 3)^2 + (y + 1)^2 = 25 \times \left(\dfrac{4x+3y}{5}\right)^2$

$\Rightarrow$ $D : 4x + 3y = 0$ and focus $(3, -1)$

Slope of the transverse axis = $\dfrac{3}{4}$

Equation of T.A. is $y + 1 = \dfrac{3}{4}(x - 3)$

$\Rightarrow$ $4y + 4 = 3x - 9$

$\Rightarrow$ $3x - 4y = 13$

(D) Given hyperbola is $\dfrac{x^2}{9} - \dfrac{y^2}{49} = 1$

Equation of any tangent is

$y = mx \pm\sqrt{a^2m^2 - b^2}$

$= mx \pm \sqrt{9m^2 - 49}$

Distance between two parallel lines = $\left|\dfrac{c_1 - c_2}{\sqrt{a^2+b^2}}\right| = \left|\dfrac{2\sqrt{9m^2-49}}{\sqrt{1+m^2}}\right| = 2$

$\Rightarrow$ $9m^2 - 49 = 1 + m^2$

$\Rightarrow$ $8m^2 = 50$

$\Rightarrow$ $m_2 = \dfrac{50}{8} = \dfrac{25}{4}$

$\Rightarrow$ $m = \pm \dfrac{5}{2}$

14. Match the statements in Column I with the properties in Column II. **[IIT-JEE 2007]**

Column I	**Column II**
(A) Two intersecting circles	(p) have a common tangent
(B) Two mutually external circles	(q) have a common normal
(C) Two circles, one strictly inside the other	(r) do not have a common tangent
(D) Two branches of a hyperbola	(s) do not have a common normal

Sol. Answer : A(p, q) ; B(p, q) ; C(q, r) ; D(q, r)

15. Match the conics in Column I with the statements/expressions in Column II. **[IIT-JEE 2009]**

Column I	**Column II**
(A) Circle	(p) The locus of the point (h, k) for which the line $hx + ky = 1$ touches the circle $x^2 + y^2 = 4$
(B) Parabola	(q) Points z in the complex plane satisfying $\|z + 2\| - \|z - 2\| = \pm 3$
(C) Ellipse	(r) Points of the conic have parametric representation $x = \sqrt{3}\left(\frac{1-t^2}{1+t^2}\right), y = \frac{2t}{1+t^2}$
(D) Hyperbola	(s) The eccentricity of the conic lies in the interva $1 \le x < \infty$
	(t) Points z in the complex plane satisfying $\text{Re}(z + 1)^2 = \|z\|^2 + 1$

Sol. Answer A(p); B(s, t); C(r); D(q, s)

(p) The line $hx + ky = 1$ touches $x^2 + y^2 = 4$

Hence $\frac{1}{\sqrt{h^2+k^2}} = 2$

$\Rightarrow \quad h^2 + k^2 = \frac{1}{4}$

$\Rightarrow$ Locus of (h, k) is $x^2 + y^2 = \frac{1}{4}$, which is a circle.

(q) We have

$|z + 2| - |z - 2| = \pm 3$

Which shows that the difference of distances of the moving point from (–2, 0) and (2, 0) is 3 (less than 4),

Hence locus of the point (z) is a hyperbola.

(r) $\frac{x}{\sqrt{3}} = \frac{1-t^2}{1+t^2}$

$y = \frac{2t}{1+t^2}$

Squaring and adding we get

$$\frac{x^2}{3} + \frac{y^2}{1} = \frac{(1-t^2)^2 + 4t^2}{(1+t^2)^2} = 1$$

which represents an ellipse.

(s) When eccentricity e lies in

$1 \le e < \infty$

For $e = 1$, conic is parabola

For $1 < e < \infty$, conic is hyperbola

(t) We have

$\text{Re}(z + 1)^2 = |z|^2 + 1$

$\Rightarrow (x + 1)^2 = x^2 + y^2 + 1$

$\Rightarrow x^2 + 2x + 1 = x^2 + y^2 + 1$

$\Rightarrow y^2 = 2x$

which represents a parabola.

16. A line $L : y = mx + 3$ meets y –axis at $E(0, 3)$ and the arc of the parabola $y^2 = 16x$, $0 \le y \le 6$ at the point $F(x_0, y_0)$. The tangent to the parabola at $F(x_0, y_0)$ intersects the y-axis at $G(0, y_1)$. The slope m of the line L is chosen such that the area of the triangle EFG has a local maximum. Match Column I with Column II and select the correct answer using the code given below the Column: **[JEE(Advanced) 2013]**

Column I	Column II
(A) m is equal to	(p) $\frac{1}{2}$
(B) Maximum area of ΔEFG is	(q) 4
(C) y_0 is equal to	(r) 2
(D) y_1 is equal to	(s) 1

Codes :

	A	B	C	D
(1)	s	p	q	r
(2)	r	s	p	q
(3)	p	r	q	s
(4)	p	r	s	q

Sol. Answer (1)

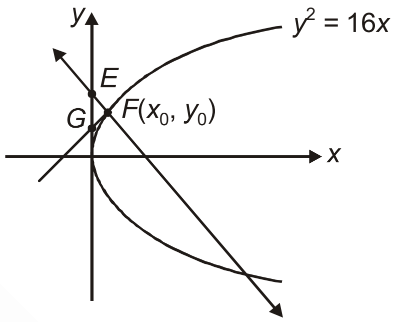
Parabola is $y^2 = 16x$

and line is $y = mx + 3$

Solving,

$(mx + 3)^2 = 16x$

$\Rightarrow$ $m^2x^2 + (6m - 16)x + 9 = 0$...(i)



Also, tangent at F is

$yy_0 = 8(x + x_0)$

So, $y_1 = \dfrac{8x_0}{y_0}$ ...(ii)

Now, area of ΔEFG,

$$\Delta = \frac{1}{2}(3 - y_1) \cdot x_0$$

$$= \frac{1}{2}\left(3 - \frac{8x_0}{y_0}\right)x_0$$

$$= \frac{1}{2}\left(3x_0 - \frac{8x_0^2}{y_0}\right)$$

$$= \frac{1}{2}\left(3x_0 - \frac{8x_0^2}{4\sqrt{x_0}}\right)$$

For Δ is to be maximum,

$$\frac{d\Delta}{dx_0} = 0$$

$\Rightarrow$ $3 - 2 \times \dfrac{3}{2}\sqrt{x_0} = 0$

$\Rightarrow$ $x_0 = 1$

So, $y_0 = 4$

$$y_1 = \frac{8x_0}{y_0} = 2$$

From $y_0 = mx_0 + 3 \Rightarrow m = 1$

17. Let $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$, be a hyperbola in the xy-plane whose conjugate axis LM subtends an angle of 60º at one of its vertices N. Let the area of the triangle LMN be $4\sqrt{3}$. **[JEE(Advanced) 2018]**

Column I	**Column II**
(A) The length of the conjugate axis of H is	(p) 8
(B) The eccentricity of H is	(q) $\frac{4}{\sqrt{3}}$
(C) The distance between the foci of H is	(r) $\frac{2}{\sqrt{3}}$
(D) The length of the latus rectum of H is	(s) 4

The correct option is

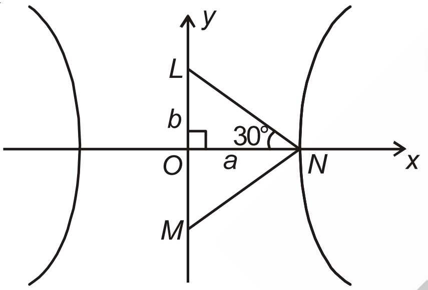
(1) A(s); B(q); C(p); D(r)

(2) A(s); B(r); C(p); D(q)

(3) A(s); B(p); C(r); D(q)

(4) A(r); B(s); C(q); D(p)

Sol. Answer (2)



$$\tan 30^\circ = \frac{b}{a}$$

$$\Rightarrow \quad b = \frac{a}{\sqrt{3}} \qquad \text{...(i)}$$

Now area of $\Delta OLN = \frac{1}{2}ab$

$$\Rightarrow \quad \frac{1}{2}ab = 2\sqrt{3}$$

$$\Rightarrow \quad ab = 4\sqrt{3} \qquad \text{...(ii)}$$

From (i) and (ii) we have $a = 2\sqrt{3}$, $b = 2$

Now $e = \sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+\frac{4}{12}} = \frac{2}{\sqrt{3}}$

Distance between foci = $2ae = 2\times 2\sqrt{3}\cdot\frac{2}{\sqrt{3}} = 8$

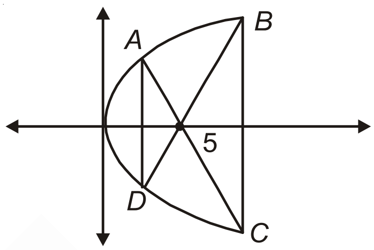
The length of latus rectum = $\frac{2b^2}{a} = \frac{2\times 4}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$

SECTION - F

Integer Answer Type Questions

1. A trapezium is inscribed in parabola $y^2 = 4x$ such that its diagonal pass through the point (1, 0) and each has length $\frac{25}{4}$. If area of trapezium be P then $\frac{4P}{25}$ equals ____.

Sol. Answer (3)



Focus of parabola, $y^2 = 4x$ is (1, 0)

So, diagonals are focal chord

$AS = 1 + t^2 = c$ (say)

$$\because \quad \frac{1}{c}+\frac{1}{\frac{25}{4}-c}=1\left\{\because \frac{1}{AS}+\frac{1}{CS}=\frac{1}{a}\right\}$$

$$\Rightarrow \quad 4c^2-25c+25=0 \Rightarrow c=\frac{5}{4}, 5$$

For $c=\frac{5}{4}, 1+t^2=\frac{5}{4} \Rightarrow t^2=\frac{1}{4} \Rightarrow t=\pm\frac{1}{2}$

For $c = 5$, $1 + t^2 = 5 \Rightarrow t = \pm 2$

$$\therefore \quad A \equiv \left(\frac{1}{4},1\right), B \equiv (4, 4), C \equiv (4, -4) \text{ and } D \equiv \left(\frac{1}{4},-1\right)$$

$AD = 2$ and $BC = 8$, distance between AD and $BC = \frac{15}{4}$

$$\therefore \quad \text{Area of trapezium, } P=\frac{1}{2}(2+8)\cdot\frac{15}{4}=\frac{75}{4} \Rightarrow \frac{4P}{25}=3$$

2. Three normals drawn from any point to parabola $y^2 = 4ax$ cut the line $x = 2a$ in points whose ordinates are in AP. If slopes of normals be m_1, m_2 and m_3 then $\left(\frac{m_1}{m_2}\right)\left(\frac{m_3}{m_2}\right)$ equals ____.

Sol. Answer (1)

Equation of parabola $\rightarrow y^2 = 4\,ax$.

If feet of normals passing through a point be $(am_1^2, -2am_1), (am_2^2, -2am_2)$ and $(am_3^{l} - 2am_3)$

$\therefore$ Equation of normal at $(am_1^2, -2am_1)$ is

$y = m_1x - 2am_1 - am_1^3$

Solving with line $x = 2a$, we get point of intersection as $(2a, m_1^3)$ similarly solving with other normals the ordinates of points of intersection with $x = 2a$ are $-am_2^2$ and $-am_3^2$.

$\therefore$ According to the question

$-\{am_1^3 + am_3^3\} = -2am_2^2$ ...(i)

But $m_1 + m_3 = -m_2$ ...(ii)

$\therefore$ $2m_2^3 = (m_1 + m_3)\{m_1^2 + m_3^2 - m_1m_3\} = 2m_2^2$

$\therefore$ $2m_2^3 = (m_1^2 - m_3^2 + m_1m_3\}$ ...(iii) {using (ii)}

Also $m_2^2 = m_1^2 + m_3^2 + 2m_1m_3$ ...(iv)

(iv) – (iii) $\Rightarrow 3m_2^2 = 3m_1, m_3 \Rightarrow \left\{\frac{m_1}{m_2}\right\}\left\{\frac{m_3}{m_2}\right\} = 1$

3. A chord is drawn from a point $P(1, t)$ to parabola $y^2 = 4x$ which cuts the parabola at A and B. If $PA.PB = 3|t|$, then maximum value of 't' equals ____

Sol. Answer (4)

Let equation of line passing through $P(1, t)$ be

$$\frac{x-1}{\cos\theta} = \frac{y-t}{\sin\theta} = r$$

$x = r\cos\theta = 1$

$y = r\sin\theta + t$

Line meets parabola at A and B

$\Rightarrow$ $\{r\sin\theta + t\}^2 = 4\{r\cos\theta + 1\}$

$r^2\sin^2\theta + 2r\{t\sin\theta - 2\cos\theta\} + t^2 - 4 = 0$

$\therefore$ $PA \cdot PB = \dfrac{t^2-4}{\sin^2\theta} = 3\,|\,t\,|$

$\Rightarrow$ $\left|\dfrac{t^2-4}{3t}\right| = \sin^2\theta \le 1 \Rightarrow \dfrac{|\,t^2-4\,|}{3\,|\,t\,|} \le 1$

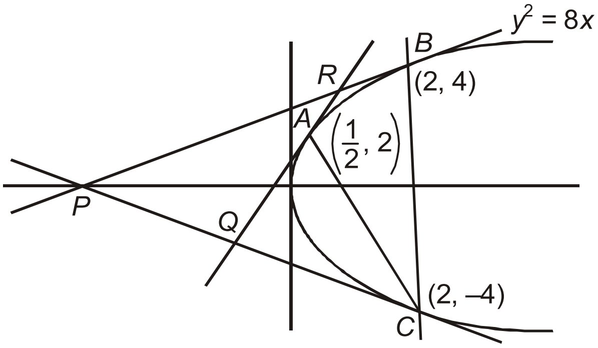
$\Rightarrow$ $t^2 - 3|t| - 4 \le 0$

$\Rightarrow$ $(|t| + 1)\,(|t| - 4) \le 0$

$\Rightarrow$ $|t| \le 4 \Rightarrow$ maximum value of it is 4

4. Consider the parabola $y^2 = 8x$. Let Δ_1 be the area of the triangle formed by the end points of its latus rectum and the point $P\left(\frac{1}{2}, 2\right)$ on the parabola, and Δ_2 be the area of the triangle formed by drawing tangents at P and at the end points of the latus rectum. Then $\frac{\Delta_1}{\Delta_2}$ is **[IIT-JEE 2011]**

Sol. Answer (2)



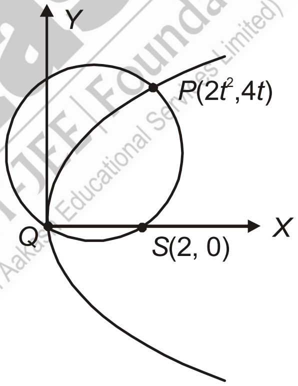
Clearly, ar(ΔABC) = 2(arΔPQR)

$\Delta_1 = 2\Delta_2$

$\Rightarrow \frac{\Delta_1}{\Delta_2} = 2$

5. Let S be the focus of the parabola $y^2 = 8x$ and let PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is **[IIT-JEE 2012]**

Sol. Answer (4)



Clearly $P(2t^2, 4t)$ satisfies the equation of circle,

$x^2 + y^2 - 2x - 4y = 0$

$\Rightarrow 4t^4 + 16t^2 - 4t^2 - 16t = 0$

$\Rightarrow t(t^3 + 3t - 4) = 0$

$\Rightarrow t(t - 1)(t^2 + t + 4) = 0$

$\Rightarrow t = 0, 1$

i.e., P be (2, 4)

So, area of $\Delta PQS = \frac{1}{2} \times 2 \times 4 = 4$ units

6. If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x - 3)^2 + (y + 2)^2 = r^2$, then the value of r^2 is **[JEE(Advanced)-2015]**

Sol. Answer (2)

End points of latus rectum $A(1, 2)$, $B(1, -2)$

Equation of normal at A $\left[\frac{dy}{dx} = 1\right]$

$y - 2 = -1(x - 1)$

$x + y - 3 = 0$ ... (i)

r = Distance of line (i) from (3, –2)

$$r = \left|\frac{3 - 2 - 3}{\sqrt{2}}\right| = \sqrt{2}$$

$r^2 = 2$

7. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is

[JEE(Advanced)-2015]

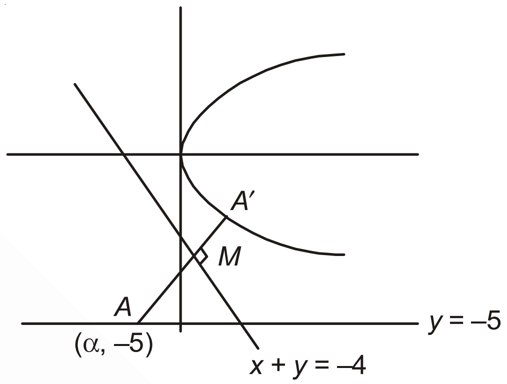
Sol. Answer (4)

Equation of AA' is

$y + 5 = 1(x - \alpha)$

$\Rightarrow$ $y = x - \alpha - 5$

$y - x = -\alpha - 5$ … (i)



Also

$x + y = -4$

$y = \dfrac{-\alpha - 9}{2}$, $2x = \alpha + 1$

$\Rightarrow$ $x = \dfrac{\alpha + 1}{2}$

$M \equiv \left(\dfrac{\alpha+1}{2}, \dfrac{-\alpha-9}{2}\right)$

$A' = (1, -\alpha - 4)$

A' lies on parabola

Now,

$\alpha^2 + 16 + 8\alpha = 4$

$\Rightarrow$ $\alpha^2 + 8\alpha + 12 = 0$

$\Rightarrow$ $\alpha = -2, -6$

AB = distance = 4 units

8. If the locus of the feet of perpendicular from the foci on any tangent to an ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{2} = 1$ is $x^2 + y^2 = k$, then the value of k is ______.

Sol. Answer (4)

The general equation of tangent of slop m is

$y = mx \pm \sqrt{a^2m^2 + b^2}$ …(i)

Equation of perpendicular is

$y - 0 = -\dfrac{1}{m}(x - ae)$ (let focus is $(ae, 0)$)

$ym + x - ae = 0$

$my + x = ae$ …(ii)

Locus of point of intersection of (i) and (ii) is obtained by eliminating m between (i) and (ii)

That is given by

$$(y - xm)^2 + (my + x)^2 = a^2m^2 + b^2 + a^2e^2$$

$$\Rightarrow (1 + m^2)(x^2 + y^2) = a^2m^2 + (a^2 - a^2e^2) + a^2e^2$$

$$\Rightarrow x^2 + y^2 = a^2$$

Here $a^2 = 4$ $\Rightarrow k = 4$

9. The number of points on the ellipse $\frac{x^2}{50}+\frac{y^2}{20}=1$ from which pair of perpendicular tangents are drawn to ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, is ______ .

Sol. Answer (4)

For ellipse, $\frac{x^2}{16}+\frac{y^2}{9}=1$, equation of director circle is $x^2 + y^2 = 25$

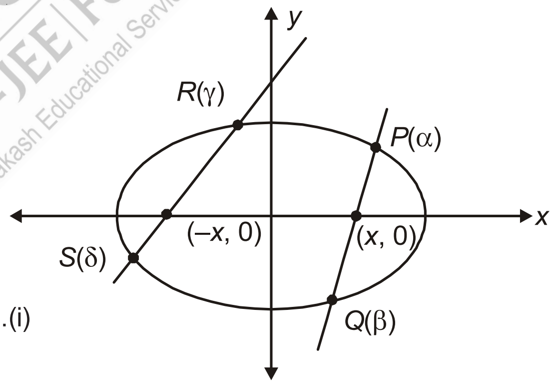
This director circle will cut the ellipse

$\frac{x^2}{50}+\frac{y^2}{20}=1$ at 4 points

10. If any two chords be drawn through two points on major axis of an ellipse equidistant from centre, then $\tan\left(\frac{\alpha}{2}\right)\tan\left(\frac{\beta}{2}\right)\tan\left(\frac{\gamma}{2}\right)\tan\left(\frac{\delta}{2}\right)$ = _______ , (where α, β, γ, δ are ecentric angles of extremities of chords)

Sol. Answer (1)

From figure



$$PQ = \frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$$

If passes through $(x_1, 0)$ so $\frac{x_1}{a}=\frac{\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)}$ …(i)

Similarly for RS, which passes through $(-x, 0)$

$$\frac{-x_1}{a}=\frac{\cos\left(\frac{\gamma-\delta}{2}\right)}{\cos\left(\frac{\gamma+\delta}{2}\right)} \quad \text{…(ii)}$$

From (i and (ii)

$$\frac{\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)}=-\frac{\cos\left(\frac{\gamma-\delta}{2}\right)}{\cos\left(\frac{\gamma+\delta}{2}\right)}$$

Using componendo and dividendo and solving we get

$$\frac{2\cos\frac{\alpha}{\beta}\cos\frac{\beta}{2}}{2\sin\frac{\alpha}{2}\sin\frac{\beta}{2}} = \frac{2\sin\frac{\gamma}{2}\sin\frac{\delta}{2}}{2\cos\frac{\gamma}{2}\cos\frac{\delta}{2}} \Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}\tan\frac{\delta}{2} = 1$$

11. If M_1 and M_2 are feet of perpendiculars from foci S_1 and S_2 of ellipse $\frac{x^2}{9}+\frac{y^2}{16}=1$ on the tangent at any point P an ellipse, then $\sqrt{(S_1M_1)(S_2M_2)}$ equals _______

Sol. Answer (4)

We know that product of perpendicular drawn from two foci S_1 and S_2 of an ellipse $\frac{x^2}{9}+\frac{y^2}{16}=1$ on the tangent at any point P on ellipse is equal to square of semi-minor axis

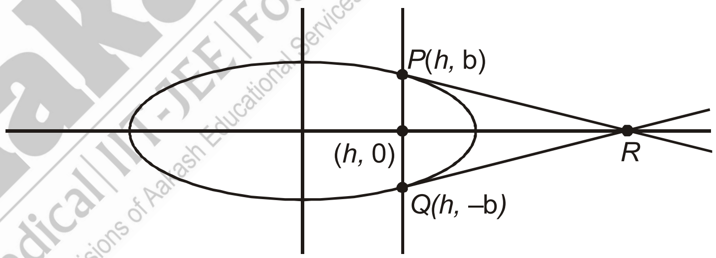
i.e., $\sqrt{(S_1M_1)(S_2M_2)} = \sqrt{16} = 4$

12. A vertical line passing through the point (h, 0) intersects the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ at the points P and Q. Let the tangents to the ellipse at P and Q meet at the point R. If $\Delta(h)$ = area of the triangle PQR, $\Delta_1 = \max\limits_{1/2\le h\le 1}\Delta(h)$ and $\Delta_2 = \min\limits_{1/2\le h\le 1}\Delta(h)$, then $\frac{8}{\sqrt{5}}\Delta_1 - 8\Delta_2 =$ **[JEE(Advanced) 2013]**

Sol. Answer (9)

$S \equiv \frac{x^2}{4}+\frac{y^2}{3}=1$

Let P and Q be (h, β) and (h, $-\beta$)



So, R is $S \equiv \frac{x^2}{4}+\frac{y^2}{3}=1$

Now, $\Delta = S \equiv \frac{x^2}{4}+\frac{y^2}{3}=1$

$= S \equiv \frac{x^2}{4}+\frac{y^2}{3}=1$

$= S \equiv \frac{x^2}{4}+\frac{y^2}{3}=1\text{s}$

So, $\frac{d\Delta}{dh} < 0$

i.e., Δ is decreasing

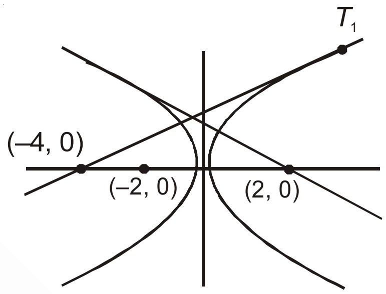
i.e., $\Delta_1 = \Delta\left(\frac{1}{2}\right) = \frac{15\sqrt{45}}{8}$

and $\Delta_2 = \Delta(1) = \frac{9}{2}$

Now $\frac{8}{\sqrt{5}}\Delta_1 - 8\Delta_2 = 9$

13. Suppose that the foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ are $(f_1, 0)$ and $(f_2, 0)$ where $f_1 > 0$ and $f_2 < 0$. Let P_1 and P_2 be two parabolas with a common vertex at (0, 0) and with foci at $(f_1, 0)$ and $(2f_2, 0)$, respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2, 0)$ and T_2 be a tangent to P_2 which passes through $(f_1, 0)$. If m_1 is the slope of T_1 and m_2 is the slope of T_2, then the value of $\left(\frac{1}{m_1^2}+m_2^2\right)$ is **[JEE(Advanced)-2015]**

Sol. Answer (4)



$f_1 \equiv (2, 0)$

$f_2 \equiv (-2, 0)$

Parabolas are

$y^2 = 8x$

$y^2 = -16x$

$T_1 : \; y = mx + \frac{2}{m}$

$0 = -4m + \frac{2}{m}$

$\Rightarrow \quad m^2 = \frac{1}{2}$

$T_2 : \; y = mx - \frac{4}{m}$

$2m - \frac{4}{m} = 0$

$\Rightarrow \quad m^2 = 2$

Now, $m_2^2 + \frac{1}{m_1^2} = 2 + 2 = 4$

14. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes and the equation of the ellipse is $x^2 + ky^2 = k$ then the value of k is ______.

Sol. Answer (2)

Let the equation of ellipse is

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a > b$ …(i)

$e_1 = \sqrt{1-\frac{b^2}{a^2}}$

$x^2 - y^2 = \frac{1}{2}$ …(ii)

$e_2 = \sqrt{1+\frac{1}{1}} = \sqrt{2}$

But $e_2 = \dfrac{1}{e_1}$

$$\Rightarrow \frac{1}{\sqrt{2}} = \sqrt{1-\frac{b^2}{a^2}} \Rightarrow \frac{1}{2} = 1-\frac{b^2}{a^2} \Rightarrow \frac{b^2}{a^2} = \frac{1}{2}$$

$$a = b\sqrt{2} \quad \ldots\text{(iii)}$$

by (i) $\dfrac{2x}{a^2} + \dfrac{2y}{b^2}\dfrac{dy}{dx} = 0$

$$\frac{dy}{dx} = \left(\frac{-\frac{x}{a^2}}{\frac{y}{b^2}}\right) = -\frac{b^2}{a^2}.\frac{x}{y} = -\frac{x}{2y} \quad \ldots\text{(iv)}$$

(ii) $2x - 2y\dfrac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{x}{y} \quad \ldots\text{(v)}$$

by but $\left(\dfrac{x}{y}\right)\left(-\dfrac{x}{2y}\right) = -1$

$\Rightarrow \quad x^2 = 2y^2 \quad \ldots$(vi)

by (ii) and (vi)

$$y^2 = \frac{1}{2}$$

$$x^2 = 1$$

Putting $x^2 = 1$, $y^2 = \dfrac{1}{2}$ in (i)

$$\frac{1}{a^2} + \frac{1}{2b^2} = 1 \quad \ldots\text{(vii)}$$

by (iii) and (vii)

$a^2 = 2$, $b^2 = 1$

Hence the equation of ellipse

$$\frac{x^2}{2} + \frac{y^2}{1} = 1,\ x^2 + 2y^2 = 2$$

$\Rightarrow \quad k = 2$

15. The equation of directrix of a hyperbola is $3x + 4y + 8 = 0$. The focus of the hyperbola is (1, 1). If eccentricity of the hyperbola is 2 and the length of conjugate axis is k then [k], where [ ] represents the greatest integer function, is equal to _____.

Sol. Answer (6)

The distance between focus and directrix is $= \left|\dfrac{3\times1+4\times1+8}{\sqrt{9+16}}\right| = 3$

e = eccentricity, $2a$ = transverse axis,

$2b$ = conjugate axis

Then $\left|\dfrac{a}{e} - ae\right| = 3$

$\Rightarrow \quad ae - \dfrac{a}{e} = 3 \qquad \Rightarrow a\left(2 - \dfrac{1}{2}\right) = 3 \qquad \Rightarrow a\left(\dfrac{3}{2}\right) = 3, \quad a = 2$

Between know that $e^2 = 1 + \dfrac{b^2}{a^2}$

$\Rightarrow 4 = 1 + \dfrac{b^2}{4} \qquad \Rightarrow b^2 = 12, \quad b = 2\sqrt{3}$

$k = 2b = 4\sqrt{3}$

$[k] = 6$

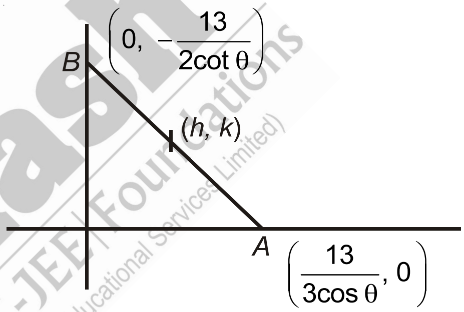
16. Consider the hyperbola $\dfrac{x^2}{9} - \dfrac{y^2}{4} = 1$. If the locus of middle point of segment of any normal intercepted between the coordinate axis is $36x^2 - 16y^2 = k$, then the value of $\dfrac{k}{169}$, is _____.

Sol. Answer (1)

Equation of normal

$$ax \cos\theta - by \cot\theta = a^2 + b^2$$

$$\Rightarrow \quad 3x\cos\theta - 2y\cot\theta = 13$$



at $y = 0$, $x = \dfrac{13}{3\cos\theta}$

at $x = 0$, $y = -\dfrac{13}{2\cot\theta}$

Let middle point of (h, k) is

$$h = \frac{13}{6\cos\theta}$$

$$k = -\frac{13}{4\cot\theta}$$

$$\Rightarrow \quad \cos\theta = \frac{13}{6h} \quad \Rightarrow \quad \sec\theta = \frac{6h}{13}$$

$$\Rightarrow \quad \cot\theta = -\frac{13}{4k} \quad \Rightarrow \quad \tan\theta = -\frac{4k}{13}$$

For locus of (h, k) we use

$$1 + \tan^2\theta = \sec^2\theta$$

$$\Rightarrow \quad 1 + \frac{16k^2}{169} = \frac{36h^2}{169}$$

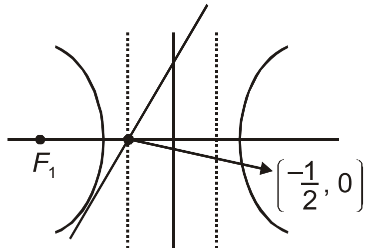
$$\Rightarrow \quad 36h^2 - 16y^2 = 169$$

$$\Rightarrow \quad k = 169 \quad \Rightarrow \quad \frac{k}{169} = 1$$

17. The line $2x + y = 1$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If this line passes through the point of intersection of the nearest directrix and the x-axis, then the eccentricity of the hyperbola is **[IIT-JEE 2010]**

Sol. Answer (2)

Clearly $\frac{a}{e} = \frac{1}{2}$ ...(i)



Also $y = -2x + 1$ is tangent to hyperbola

$\therefore\ 1 = 4a^2 - b^2$ ...(ii)

$\Rightarrow\ \frac{1}{a^2} = 4 - \left(e^2 - 1\right)$

$\Rightarrow\ \frac{4}{e^2} = 5 - e^2$

$\Rightarrow\ e^4 - 5e^2 + 4 = 0$

$\Rightarrow\ (e^2 - 4)(e^2 - 1) = 0$

$\Rightarrow\ e = 2, e = 1$

$e = 1$ gives the conic as parabola. But conic is given as hyperbola, hence

$e = 2$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : The locus of centroid of a triangle formed by three co-normal points on a parabola is the axis of parabola.

STATEMENT-2 : One of the angles between the parabolas $y^2 = 8x$ and $x^2 = 27y$ is $\tan^{-1}\left(\frac{9}{13}\right)$.

STATEMENT-3 : Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. The product of lengths of perpendiculars drawn from foci to a tangent is 4.

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (2)

Statement-1:

Statement is clearly true it is the standard result.

Statement-2:

The angle between $y^2 = ax$ and $x^2 = by$ is

$$\tan^{-1}\left(\frac{3a^{\frac{1}{3}}b^{\frac{1}{3}}}{2\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)}\right)$$

Hence $a = 8$, $b = 27$

Hence angle $= \tan^{-1}\left(\dfrac{3\times2\times3}{2(4+9)}\right) = \tan^{-1}\left(\dfrac{9}{13}\right)$

Hence statement-2 is also true.

Statement-3:

For the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$, the product of perpendicular is b^2, here $b^2 = 4$

Hence statement - 3 is true.

2. STATEMENT-1 : Let us consider three points A, B, C on a parabola. Tangents at A, B and C form a triangle PQR. If the area of triangle ABC is 10 then the area of triangle PQR is 5.

   STATEMENT-2 : If the foci of an ellipse are (3, 4) and (5, 12) and (0, 0) lies on the ellipse then the length of major axis of the ellipse is 9.

   STATEMENT-3 : If $k < AB$, where A, B are two fixed point in x-y plane and P is a variable point and $|PA - PB| = k$, the eccentricity of the conic is $\dfrac{AB}{k}$.

   (1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (1)

Statement-1:

The area of $\Delta ABC = 2 \times$ Area of ΔPQR

$\Rightarrow$ Area of $\Delta PQR = \dfrac{10}{2} = 5$

Hence statement-1 is true

Statement-2:

Let $S_1 \equiv (3, 4)$, $S_2 = (5, 12)$, $P = (0, 0)$

For an ellipse $PS_1 + PS_2$ = length of major axis

$PS_1 = \sqrt{(3-0)^2+(y-0)^2} = 5$

Similarly $PS_2 = 13$

Hence the length of major axis is 18.

Hence statement-2 is false

Statement-3:

$|PA - PB| = k$

If $k < AB$, then locus of P will be a hyperbola.

If length of transverse axis of hyperbola is $2a$ and eccentricity = e

Then $AB = 2ae$ ...(i)

$k = 2a$ ...(ii)

$\Rightarrow \dfrac{AB}{k} = e$

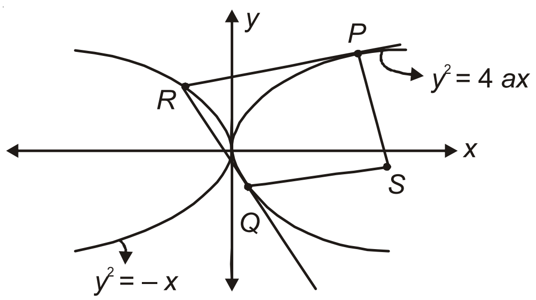
Hence statement-3 is also true.

SECTION - H

Aakash Challengers Questions

1. Tangents at points P and Q of parabola $y^2 = 4x$ intersect each other at point R on parabola $y^2 = -x$. The normal at P and Q intesect at right angle at S. Find the diameter of circumcircle of quadrilateral $PQRS$.

Sol. Let co-ordinates be $P(t_1^2, 2t_1)$, $Q(t_2^2, 2t_2)$



Co-ordinates of points of intersection R is $(t_1t_2, \{t_1 + t_2\})$

R lies on $y^2 = -x \Rightarrow (t_1 + t_2)^2 = -t_1t_2$ ...(i)

Co-ordinates of point of intersection of normal at P and Q

are $S(2 + \{t_1^2 + t_2^2 + t_1t_2\}, -t_1t_2\{t_1 + t_2\})$

P, Q, R, S are angular points, hence centre of circle is the point of intersection of PQ and RS

Co-ordinates of mid-point of PQ is

$$\left(\frac{t_1^2 + t_2^2}{2}, 2\frac{(t_1 + t_2)}{2}\right)$$

Co-ordinates of mid-point of RS is

$$\left(\frac{t_1t_2 + 2 - 2t_1t_2}{2}, \frac{(t_1 + t_2) - t_1t_2(t_1 + t_2)}{2}\right)$$

So,

$$\frac{\{t_1^2 + t_2^2\}}{2} = \frac{2 - t_1t_2}{2} \text{ and } \frac{2\{t_1 + t_2\}}{2} = \frac{(t_1 + t_2)[1 - t_1t_2]}{2}$$

$\Rightarrow$ $-3t_1t_2 = 2 - t_1t_2$ and $(t_1 + t_2)$ $[1 - t_1t_2 - 2] = 0$

$\Rightarrow$ $t_1t_2 = -1$ ...(ii)

$\Rightarrow$ From equation (i), $t_1 + t_2 = \pm 1$ ...(iii)

Hence, co-ordinates of centre are $\left(\frac{3}{2}, \pm 1\right)$ and co-ordinates of R are $(-1, \pm 1)$

$\therefore$ Radius of required circle is $\sqrt{\left(\frac{3}{2} + 2\right)^2 + (\pm 1 \mp 1)^2} = \frac{5}{2}$

$\Rightarrow$ Diameter = 5

2. The sides of triangle ABC are tangents to parabola $y^2 = 4ax$, let D, E, F be the points of contact of side BC, CA and AB respectively. If lines AD, BE and CF are concurrent at focus of parabola and $\angle ABC$ of ΔABC is $\frac{\pi}{k}$, then find the value of k.

Sol. Let the parameter of points D, E, F on the parabola be t_1, t_2, t_3 respectively. Tangents at E and F meet at A,

So the co-ordinates of A are $(at_2t_3, a(t_2 + t_3))$.

Now AD passes through focus, so

$$\frac{a\{t_2 + t_2\} - 0}{at_2t_3 - a} = \frac{2at_1 - 0}{at_1^2 - a}$$

$\Rightarrow$ $2t_1t_2t_3 - 2t_1 = \{t_2 + t_3\}t_1^2 - \{t_2 + t_3\}$

$t_1^3 - (t_1 + t_2 + t_3)t_1^2 - 3t_1 + 2t_1t_2t_3 + (t_1 + t_2 + t_3) = 0$

$\Rightarrow$ t_1, t_2, t_3 are roots of equation

$$t^3 - \{t_1 + t_2 + t_3\}t^2 - 3t + 2t_1t_2t_3 + (t_1 + t_2 + t_3) = 0 \qquad \text{...(i)}$$

$\Rightarrow$ $\Sigma t_1t_2 = -3$

And $t_1t_2t_3 = -2t_1t_2t_3 - \{t_1 + t_2 + t_3\}$

$$\Rightarrow \quad t_1t_2t_3 = -\frac{1}{3}\{t_1 + t_2 + t_3\}$$

So (i) becomes,

$$t^3 + 3(t_1t_2t_3)t^2 - 3t - t_1t_2t_3 = 0$$

$$\Rightarrow \quad -t_1t_2t_3 = \frac{3t - t^3}{1 - 3t^3}$$

$\because$ Slope of sides of triangle are

$$\frac{1}{t_1} = \tan\theta_1, \frac{1}{t_2} = \tan\theta_2, \frac{1}{t_3} = \tan\theta_3$$

So, $\cot\theta_1$, $\cot\theta_2$, $\cot\theta_3$ are roots of equation

$$\frac{3\cot\theta - \cot^3\theta}{1 - 3\cot^2\theta} = \cot\alpha, \text{ where } \cot\alpha = -t_1t_2t_3$$

$\therefore$ $\cot 3\theta = \cot\alpha \Rightarrow 3\theta = n\pi + \alpha$

$$\Rightarrow \quad \theta = \frac{n\pi}{3} + \frac{\alpha}{3}$$

$$\therefore \quad \theta_1 = \frac{\alpha}{3}, \theta_2 = \frac{\pi + \alpha}{3}, \theta_3 = \frac{2\pi + \alpha}{3}$$

$$\therefore \quad \theta_2 - \theta_1 = \theta_3 - \theta_2 = \frac{\pi}{3} = \frac{\pi}{k}$$

On comparison, we get

$k = 3$

3. If the tangent at P to the parabola $y^2 = 7x$ is parallel to $6y - x + 11 = 0$, then square of the distance of P from the vertex of the parabola is________.

Sol. Let $P(h, k)$ lie on $y^2 = 7x$

$k^2 = 7h$ … (i)

Equation of tangent at $P(h, k)$

$$yk = \frac{7}{2}(x + h)$$

$7x - 2ky + 7h = 0$ … (ii)

$6y - x + 11 = 0$ … (iii)

Line (ii) and (iii) are parallel

$\therefore \quad \frac{1}{6} = \frac{7}{2k} \quad \Rightarrow \quad k = 21$

$h = 63$

$P(63, 21)$

$OP^2 = (63)^2 + (21)^2 = 4410$

4. Find the mirror image of the parabola $y^2 = 4x$ in the tangent to the parabola at the point (1, 2) .

Sol. Any point on the given parabola is $(t^2, 2t)$

Tangent at (1, 2) is $2(x + x_1)$

$\Rightarrow 2y = 2(x + 1)$

$\Rightarrow x - y + 1 = 0$

Let image of $(t^2, 2t)$ is (h, k)

$$\therefore \frac{h - t^2}{1} = \frac{k - 2t}{-1} = \frac{-2(t^2 - 2t + 1)}{1 + 1} = -\left(t^2 - 2t + 1\right)$$

$\therefore h = t^2 - t^2 + 2t - 1 = 2t - 1$

and $k = 2t + t^2 - 2t + 1 = t^2 + 1$

$$\therefore t = \frac{h + 1}{2}$$

$$\therefore k = \frac{(h + 1)^2}{2} + 1 = \frac{(h + 1)^2 + 4}{4}$$

$\Rightarrow (h + 1)^2 - 4k + 4 = 0$

Locus of (h, k) is $(x + 1)^2 = 4(y - 1)$.

5. The locus of points such that two of three normals drawn from them to the parabola $y^2 = 4ax$ coincide is of the form $m\,ay^2 = n(x - 2a)^3$. The value of $\dfrac{m^{\frac{2}{3}}\left(m^{\frac{1}{3}} + n^{\frac{1}{2}}\right)}{\sqrt{2025}}$ is.

Sol. Let (h, k) be the point of intersection of three normals to the parabola $y^2 = 4ax$.

The equation of any normal to $y^2 = 4ax$ is

$$y = mx - 2am - am^3$$

If it passes through (h, k), then

$$k = mh - 2am - am^3$$

$$\Rightarrow am^3 + (2a - h)m + k = 0 \qquad \ldots(i)$$

If roots are m_1, m_2, m_3.

$\therefore\ m_1 + m_2 + m_3 = 0$

$$\sum m_1m_2 = \frac{(2a-h)}{a}$$

$$m_1m_2m_3 = -\frac{k}{a}$$

Given $m_1 = m_2$

$\therefore 2m_1 + m_3 = 0$

$\Rightarrow m_3 = -2m_1$

$$\therefore m_1 \times m_1 \times -2m_1 = -\frac{k}{a}$$

$$\Rightarrow m_1^3 = \frac{k}{2a}$$

Since m_1 is a root of equation (i), we get

$$am_1^3 + (2a - h)m_1 + k = 0$$

$$\Rightarrow a\left(\frac{k}{2a}\right) + (2a-h)\left(\frac{k}{2a}\right)^{1/3} + k = 0$$

$$\Rightarrow (2a-h)\left(\frac{k}{2a}\right)^{1/3} + \frac{3k}{2} = 0$$

$$\Rightarrow (2a-h)\left(\frac{k}{2a}\right)^{1/3} = -\frac{3k}{2}$$

$$\Rightarrow (2a-h)^3 \times \frac{k}{2a} = -\frac{27k^3}{8}$$

$\Rightarrow 27ak^2 = 4(h - 2a)^3$

Thus the latus of (h, k) is

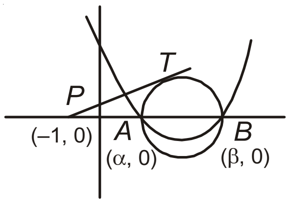
$$27ay^2 = 4(x - 2a)^3$$

$$\Rightarrow m = 27, n = 4$$

$$\therefore \frac{m^{\frac{2}{3}}\left(m^{\frac{1}{3}} + n^{\frac{1}{2}}\right)}{\sqrt{2025}} = \frac{9(3+2)}{\sqrt{2025}} = \frac{45}{45} = 1$$

6. A parabola $y = (a^2 + 1)x^2 + (b^2 + 1)x + (c^2 + 1)$ crosses the x-axis at $(\alpha, 0)$, $(\beta, 0)$, both are right to the origin. A circle also passes through these two points. The length of the tangent from $(-1, 0)$ to the circle is.

Sol. Using the concept of the power point of a circle



$PT^2 = PA.\ PB$

$= (\alpha + 1)(\beta + 1)$

$= \alpha\beta + (\alpha + \beta) + 1$

$$= \frac{c^2+1}{a^2+1} + \frac{b^2+1}{a^2+1} + 1$$

$$= \frac{c^2+1+b^2+1+a^2+1}{a^2+1}$$

$$= \frac{a^2+b^2+c^2+3}{a^2+1}$$

$$\Rightarrow PT = \sqrt{\frac{a^2+b^2+c^2+3}{a^2+1}}$$

7. Let P and Q be two points on the ellipse $x^2 + 4y^2 = 16$, whose eccentric angles are $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ respectively. The tangent at P and the normal at Q cuts the ellipse at M. The area of the triangle PQM is

(1) 1 sq. unit (2) 2 sq. units (3) $\frac{160}{17}$ sq. units (4) 4 sq. units

Sol. Answer (3)

Here the given ellipse is

$x^2 + 4y^2 = 16$

Co-ordinate of $P \equiv \left(4\cos\frac{\pi}{4}, 2\sin\frac{\pi}{4}\right) \equiv (2\sqrt{2}, \sqrt{2})$

Co-ordinate of $Q \equiv \left(4\cos\frac{3\pi}{4}, 2\sin\frac{3\pi}{4}\right) \equiv (-2\sqrt{2}, \sqrt{2})$

The equation of tangent t point P is *i.e.*, PR

$xx_1 + 4yy_1 = 16$

$\Rightarrow 2\sqrt{2}x + 4\sqrt{2}y = 16$

$\Rightarrow x + 2y = 4\sqrt{2}$

Now equation of the normal at point Q, *i.e.* QR

$ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$

$4x(-\sqrt{2}) - 2y(\sqrt{2}) = 16 - 4$

$\Rightarrow -4x\sqrt{2} - 2\sqrt{2}y = +12$

$\Rightarrow 2x + y = -3\sqrt{2}$

Now, solving, we get

$$x=-\frac{14}{17}\sqrt{2}$$

Putting this we get $y=-\frac{23\sqrt{2}}{17}$

Co-ordinate of M is $\left(\frac{-14\sqrt{2}}{17},-\frac{23\sqrt{2}}{17}\right)$

$$\text{So, area}=\frac{1}{2}\begin{vmatrix} 2\sqrt{2} & \sqrt{2} & 1\\ -2\sqrt{2} & \sqrt{2} & 1\\ -\frac{14\sqrt{2}}{17} & \frac{-23\sqrt{2}}{17} & 1\end{vmatrix}=\frac{160}{17}\ \text{units}$$

8. Let S and S' be the foci of an ellipse whose eccentricity is e. P is a variable point on the ellipse the locus of incentre of the triangle PSS' is an ellipse whose eccentricity is

(1) $\sqrt{\frac{1+e}{2e}}$ (2) $\sqrt{\frac{2e}{1+e}}$ (3) $\sqrt{\frac{e}{1+e}}$ (4) $\frac{e}{\sqrt{1+e}}$

Sol. Answer (2)

SP = focal distance of the point

$P = a - ae\cos\theta$

$S'P = a + ae\cos\theta$

$S'S = 2ae$

If (h, k) be the co-ordinate of the incentre of $\Delta PSS'$

$$h=\frac{2ae(a\cos\theta)+a(1-e\cos\theta)(-ae)+a(1+e\cos\theta)-ae}{2ae+a(1-e\cos\theta)+a(1+e\cos\theta)}$$

$\Rightarrow$ $h = ae\cos\theta$

and $$k=\frac{2ae(a\sin\theta)+a(1-e\cos\theta)(-ae)\cdot 0+a(1+e\cos\theta)}{2ae+a(1-e\cos\theta)+a(1+e\cos\theta)}$$

$$k=\frac{eb\sin\theta}{(e+1)}$$

Eliminating θ from (i) and (ii) we get

$$\frac{x^2}{a^2-e^2}+\frac{y^2}{\left(\frac{be}{e+1}\right)^2}=1$$

Which is an ellipse let e, be its eccentricity

$$\frac{b^2e^2}{(e+1)^2}=a^2e^2(1-e_1^2)$$

$$e_1^2=1-\frac{b^2}{a^2(e+1)^2}$$

$$\Rightarrow\quad e_1=\sqrt{\frac{2e}{e+1}}$$

9. There are exactly two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose distance from its centre is the same and is equal to $\sqrt{\frac{a^2+2b^2}{2}}$. The eccentricity of the ellipse is

(1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{2}}$ (3) $\frac{1}{4}$ (4) $\frac{1}{3}$

Sol. Answer (2)

Since there are exactly two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose distance from centre is same, the points would be either end points of the major axis or of the minor axis

But $\sqrt{\frac{a^2+2b^2}{2}}>b$

$\Rightarrow \sqrt{\frac{a^2+2b^2}{2}}=a$

$\Rightarrow a^2 = 2b^2$

$\Rightarrow e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{1}{2}}=\frac{1}{\sqrt{2}}$

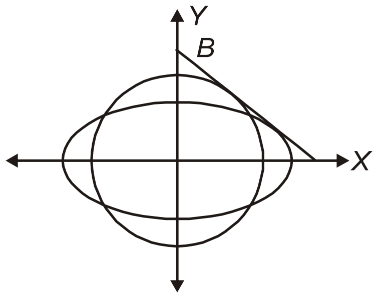
10. The area of the quadrilateral formed by the common tangents of the circle $x^2 + y^2 = c^2$ and the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $b < a$ is

(1) $\frac{c^2(a^2-b^2)}{\sqrt{(a^2-c^2)(c^2-b^2)}}$ (2) $\frac{2c^2(a^2-b^2)}{\sqrt{(a^2-c^2)(c^2-b^2)}}$

(3) $\frac{b^2(a^2-b^2)}{\sqrt{(a^2-c^2)(c^2-b^2)}}$ (4) $\frac{2b^2(a^2-c^2)}{\sqrt{(a^2-c^2)(c^2-b^2)}}$

Sol. Answer (2)

Let m be the slope of the common tangent AB of the given circle and ellipse in the first quadrant, intersecting x and y axes at A and B



Then $c\sqrt{1+m^2}=\sqrt{a^2m^2+b^2}$

$m=-\sqrt{\frac{c^2-b^2}{a^2-c^2}}$

$A\equiv\left(c\sqrt{\frac{a^2-b^2}{c^2-b^2}},0\right)$, $B\equiv\left(0,c\sqrt{\frac{a^2-b^2}{a^2-c^2}}\right)$

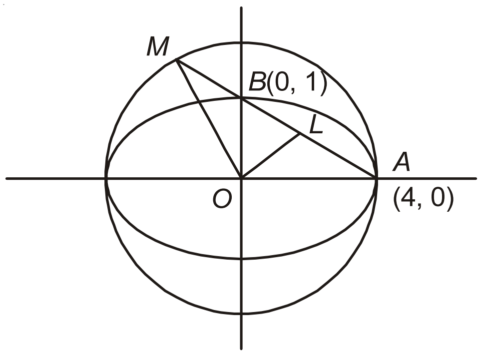
Due to symmetry the quadrilateral formed will be a rhombus having diagonals along the axes of co-ordinates of length

$2c\sqrt{\frac{a^2-b^2}{c^2-b^2}}$ and $2c\sqrt{\frac{a^2-b^2}{a^2-c^2}}$

Hence required area $=\frac{2c^2(a^2-b^2)}{\sqrt{a^2-c^2}\sqrt{c^2-b^2}}$

11. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 16y^2 = 16$ meets its auxiliary circle of the point M. Then the area of the triangle with vertices of A, M and the origin O is __________.

Sol. $E : \dfrac{x^2}{16} + \dfrac{y^2}{1} = 0$



Auxiliary circle : $x^2 + y^2 = 16$

$AM : x + 4y - 4 = 0$

$OL = \dfrac{4}{\sqrt{17}}$

Length of AM $= 2\,MN$

$$= 2\sqrt{OM^2 - ON^2}$$

$$= 2\sqrt{16 - \frac{16}{17}}$$

$$= 2\sqrt{16\left(1 - \frac{1}{17}\right)}$$

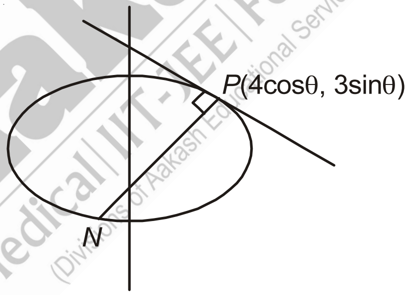
$$= \frac{32}{\sqrt{17}}$$

$$\text{ar}(\Delta OAM) = \frac{1}{2} \times \frac{32}{\sqrt{17}} \times \frac{4}{17} = \frac{64}{17} \text{ sq.units}$$

12. If the tangents and normal to an ellipse $9x^2 + 16y^2 = 144$ at a point cut off intercept a_1, a_2 on x-axis and b_1, b_2 on y-axis. Then the value of $a_1a_2 + b_1b_2 + 1000$ is __________.

Sol. $E : 9x^2 + 16y^2 - 144$

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$



$T : \dfrac{x}{4}\cos\theta + \dfrac{y}{3}\sin\theta = 1$

$\therefore \quad a_1 = 4\sin\theta$, $b_1 = 3\text{cosec}\theta$

$$PN : \frac{x}{\frac{7\cos\theta}{4}} + \frac{y}{\frac{-7\sin\theta}{3}} = 1$$

$$a_2 = \frac{7}{4}\cos\theta,\ b_2 = -\frac{7\sin\theta}{3}$$

$$a_1a_2 + b_1b_2 = \frac{4}{\cos\theta} \times \frac{7}{4}\cos\theta - \frac{3}{\sin\theta} \times \frac{7\sin\theta}{3} = 7 - 7 = 0$$

$\Rightarrow \quad a_1a_2 + b_1b_2 + 1000 = 1000$

13. The locus of the point of intersection of tangents to an ellipse drawn at the extremities of a normal chord __________.

Sol. Let (h, k) be the point of intersection of tangents, so that the equation of chord of contact is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1 \qquad \ldots\text{(i)}$$

Since (i) is a normal chord of the ellipse at any point θ, so

$ax\sec\theta - by\text{cosec}\theta = a^2 - b^2 \qquad \ldots$(ii)

From (i) & (ii), we get

$$\frac{h}{a^3\sec\theta}=\frac{k}{-b^3\text{cosec}\theta}=\frac{1}{a^2-b^2}$$

$$\Rightarrow \quad \cos\theta=\frac{a^3}{h}\times\left(\frac{1}{a^2-b^2}\right),\ \sin\theta=-\frac{b^3}{k}\times\frac{1}{a^2-b^2}$$

$$\Rightarrow \quad \left(\frac{a^6}{h^2}+\frac{b^6}{k^2}\right)=(a^2-b^2)^2$$

Locus of (h, k) is $\frac{a^6}{x^2}+\frac{b^6}{y^2}=(a^2-b^2)^2$

14. Let P be a point on the hyperbola $x^2 - y^2 = a^2$ where a is a parameter, such that P is nearest to the line $y = 2x$. The locus of P is

(1) $y = 2x$ (2) $2y = x$ (3) $y = x$ (4) $y + x = 0$

Sol. Answer (2)

Any point on the hyperbola is ($a\sec\theta$, $a\cos\theta$). The point will be nearest to $y = 2x$, if the tangent at this point is parallel to $y = 2$.

Slope is $\text{cosec}\theta$

$$\Rightarrow \quad \text{cosec}\theta=2$$

$$\theta=\frac{\pi}{6}$$

$$P\equiv\left(a\sec\frac{x}{6},a\tan\frac{x}{6}\right)\ \equiv\ \left(\frac{2a}{\sqrt{3}},\frac{a}{\sqrt{3}}\right)\ \Rightarrow k=\frac{h}{2}\ \Rightarrow\ 2y-x=0$$

15. Tangents are drawn from points on a tangent of the hyperbola $x^2 - y^2 = a^2$ to the parabola $y^2 = 4ax$. If all the chords of contact pass through Q, the locus of point Q for different tangents on the hyperbola is a/an

(1) Circle (2) Ellipse (3) Parabola (4) Hyperbola

Sol. Answer (2)

Tangent at a point ($a\sec\theta$, $a\tan\theta$) is $x\sec\theta - y\tan\theta = a$ …(i)

Any point on (1) will be of the form $\left(t,\frac{t\sec\theta-a}{\tan\theta}\right)$.

Equation of chord of contact w.r.t. the parabola $y^2 = 4ax$ will be

$$y\left(\frac{t\sec\theta-a}{\tan\theta}\right)-2a(x+t)=0$$

$$\Rightarrow \quad \left(\frac{-ay}{\tan\theta}-2ax\right)+t\left(\frac{y\sec\theta}{\tan\theta}-2a\right)=0$$

They represent a family of liens each member of which passes through the point of intersection of straight lines $\frac{-ay}{\tan\theta}-2ax=0$ and $y\frac{\sec\theta}{\tan\theta}-2a=0$.

$$\Rightarrow y=2a\sin\theta;\ x=-a\cos\theta.$$

$$\Rightarrow Q\equiv(-a\cos\theta,\ 2a\sin\theta)$$

$$\alpha=-a\cos\theta,\ \beta=2a\sin\theta$$

$$\Rightarrow \quad \frac{x^2}{a^2}+\frac{y^2}{4a^2}=1 \text{ which is an ellipse}$$

16. Let $F(x) = (1 + b^2)x^2 + 2bx + 1$. The minimum value of $F(x)$ is the reciprocal of the square of the ecentricity of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$. Then the equation of the common tangents to the curves $\frac{x^2}{16} + \frac{y^2}{9/16} = 1$ and $x^2 + y^2 = b^2$ is

Sol. Given $F(x) = (1+b^2)x^2 + 2bx + 1$

$\Rightarrow F'(x) = 2(1+b^2)x + 2b = 0$

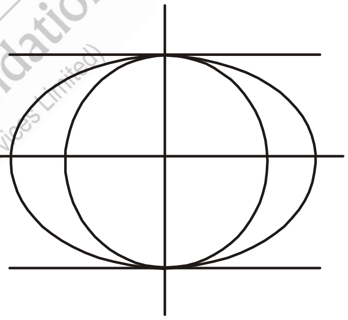
$\Rightarrow x = -\frac{b}{1+b^2}$

$$\therefore F\left(-\frac{b}{1+b^2}\right) = (1+b^2)\times\frac{b^2}{\left(1+b^2\right)^2} - \frac{2b^2}{(1+b^2)} + 1$$

$$= \frac{b^2}{\left(1+b^2\right)} - \frac{2b^2}{(1+b^2)} + 1$$

$$= \frac{b^2 - 2b^2 + 1 + b^2}{1+b^2}$$

$$= \frac{1}{1+b^2}$$



Also, $e = \sqrt{1+\frac{9}{16}} = \sqrt{\frac{25}{16}}$

$\Rightarrow \frac{1}{1+b^2} = \frac{16}{25}$

$\Rightarrow 1 + b^2 = \frac{16}{25}$

$\Rightarrow b^2 = \frac{16}{25} - 1 = \frac{9}{16}.$

$\Rightarrow b = \pm\frac{3}{4}$

Equation of the common tangents are

$\Rightarrow y = \pm\frac{3}{4}.$

$\Rightarrow 4y - 3 = 0, 4y + 3 = 0$

17. If a, b, c, d are positive real numbers such that $a + b + c + d = \lambda$ and $(a + b)(c + d) = \mu$ where $\lambda = e^2$ to the curve $x^2 - y^2 = \lambda$ and μ is the length of the *L.R.* to the parabola $y^2 = \mu x$, then the range of μ is

Sol. Given curves are

$$x^2 - y^2 = \lambda \Rightarrow e = \sqrt{1+\frac{1}{1}} = \sqrt{2} \Rightarrow \lambda^2 = 2$$

$$y^2 = \mu x \Rightarrow \text{L.R.} = 4.\frac{\mu}{4} = \mu$$

Now, $\dfrac{(a+b)+(c+d)}{2} \geq \sqrt{(a+b)(c+d)}$

$\Rightarrow \dfrac{\lambda}{2} \geq \sqrt{(a+b)(c+d)}$

$\Rightarrow \sqrt{(a+b)(c+d)} \leq \dfrac{2}{2} = 1$

$\Rightarrow (a+b)(c+d) \leq 1$

$\Rightarrow \mu \leq 1$

Since *a, b, c* and *d* are all positive, so

$0 < \mu \leq 1$

$\Rightarrow \mu \in (0, 1]$

18. If PN be the ordinate of a point P on the hyperbola $\dfrac{x^2}{(97)^2} - \dfrac{y^2}{(79)^2} = 1$ and the tangent at P meets the transverse axis in T, O is the origin, then $ON.OT$ is equal to __________.

Sol. Let $P(a \sec\theta, b \tan)$

Equation of tangent at P

$\dfrac{x\sec\theta}{a} - \dfrac{y\tan\theta}{b} = 1$

Put $y = 0$

$\therefore \quad x = \dfrac{a}{\sec\theta}$

$OT = \dfrac{a}{\sec\theta}$

$ON = a \sec\theta$

$\therefore \quad OT\,.\,ON = a^2 = (97)^2 = 9409$

19. If $f(x) = ax^3 + bx^2 + cx + d$, ($a, b, c, d \in Q$) and roots of $f(x) = 0$ are eccentricities of a parabola and a rectangular hyperbola, then $a + b + c + d + 2010$ is

Sol. As we know that the eccentricities of a parabola and a rectangular hyperbola are 1 and $\sqrt{2}$ respectively. Also, irrational roots occur in conjugate pairs.

$\therefore$ The roots of $f(x) = 0$ are $b\sqrt{2}$ and $-\sqrt{2}$

$\therefore \quad f(x) = (x-1)(x-\sqrt{2})(x+\sqrt{2}) = x^3 - x^2 - 2x + 2$

$\Rightarrow \quad ax^3 + bx^2 + cx + d = x^3 - x^2 - 2x + 2$

$\Rightarrow \quad a = 1, b = -1, c = -2, d = 2$

$\Rightarrow \quad a + b + c + d + 2010 = (1 - 1 - 2 + 2) + 2010 = 2010$

20. If $f(x) = ax^2 + bx + c$, where $a, b, c \in Q$ and roots of $f(x) = 0$ is the eccentricity of a hyperbola $3x^2 - y^2 = 3$ and $g(x) = ax^3 + bx^2 + cx + d$, where ($a, b, c, d \in Q$) and roots of $g(x) = 0$ is the eccentricity of $8x^2 - y^2 = 8$, then the value of $\dfrac{15a + 3b + d + 2010}{2010}$ is

Sol. $H_1 : 3x^2 - y^2 = 3$

$\Rightarrow \quad \dfrac{x^2}{1} - \dfrac{y^2}{3} = 1$

$\therefore\ e_1 = \sqrt{1+\frac{3}{1}} = 2$

$H_2 : 8x^2 - y^2 = 8$

$\Rightarrow\ \frac{x^2}{1} - \frac{y^2}{8} = 1 \qquad \therefore\ e_2 = \sqrt{1+\frac{8}{1}} = 3$

Thus, $f(2) = 0 \Rightarrow 4a + 2b + c = 0$ ...(1)

and $g(3) = 0 \Rightarrow 27a + 9b + 3c + d = 0$ ...(2)

From (1) & (2), we get

$27a + 9b + 3(-4a - 2b) + d = 0$

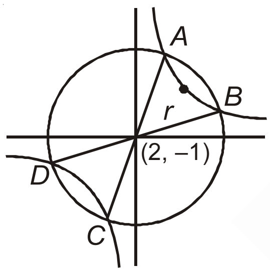
$\Rightarrow\ 27a + 9b - 12a - 6b + d = 0$

$\Rightarrow\ 15a + 3b + d = 0$

From $\frac{15a+3b+d+2010}{2010} = \frac{0+2010}{2010} = 1$

21. Let C be a curve, which is locus of point of intersection of lines $x = 2 + \lambda$ and $\lambda y = 4 - \lambda$. A circle $(x - 2)^2 + (y + 1)^2 = 25$ intersect the curve at four points A, B, C and D. If O is the centre of the curve C, then $OA^2 + OB^2 + OC^2 + OD^2$ is

Sol.



Given, $x = 2 + \lambda \Rightarrow x - 2 = \lambda$

Also $\lambda y = 4 - \lambda \Rightarrow y = \frac{4}{\lambda} - 1 \Rightarrow y + 1 = \frac{4}{\lambda}$

$\therefore\ (x-2)(y+1) = \lambda \times \frac{4}{\lambda} = 4$

$\Rightarrow\ (x-2)(y+1) = 4$

$\Rightarrow\ XY = 4,\ X = x - 2,\ Y = y + 1$

Also, $(x-2)^2 + (y+1)^2 = 25$

$\Rightarrow\ X^2 + Y^2 = 25,\ X = x - 2,\ Y = y + 1$

$\therefore\ OA^2 + OB^2 + OC^2 + OD^2 = 4 \times r^2 = 4 \times 25 = 100$

22. If all the points of the circle $x^2 + y^2 - 8x - 8y = 20$ and the rectangular hyperbola $xy + 4x + 4y = 40$ satisfy the equation $|x + y| = A$, then the value of $2A^3 + 12$ is

Sol. Given $x^2 + y^2 - 8(x + y) = 20$ ...(1)

and $xy + 4(x + y) = 40$ ...(2)

Now, (1) × 1 and (2) × 2 and adding, we get

$x^2 + y^2 - 8(x + y) + 2xy + 8(x + y) = 20 + 80$

$\Rightarrow \quad x^2 + y^2 + 2xy = 100$

$\Rightarrow \quad (x + y)^2 = 10^2$

$\Rightarrow \quad |(x + y)| = 10$

$\Rightarrow \quad A = 10$

Now, $2A^3 + 12 = 2 \times 10^3 + 12 = 2000 + 12 = 2012$

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